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F54 CONTROLLER

Operation Manual



INSTRUCTIONS:

We sincerely thank you for trusting and choosing QS Technology's controller products. At QS Technology, we take pride in offering a wide range of controllers featuring modern designs and user-friendly interfaces. In particular, our F54 controller is designed with flexibility in mind, making it suitable for various applications. The F54 not only meets your current operational needs but also expands your machine’s capabilities thanks to its integrated PLC processing module and Macro functionality.

This user manual provides an overview of the features and usage of the F54 controller, making operation easier than ever before.

QS Technology Co., Ltd. We are proud to be a pioneer in developing, applying, and commercializing CNC controller products "Made by Vietnam." After many years of development, QS Technology-branded products have increasingly gained trust and interest from both domestic and international customers. In addition, with the strengths and experience accumulated over years of working in machine manufacturing, automation systems, and electronic circuit boards, we are confident in mastering the technology and fulfilling our customers’ expectations.

WEBSITE

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CONTENT

PART 1:CONTROLLER OVERVIEW 1

1.1 Overview 1

1.2 Dimension 2

1.3 Software structure 3

PART 2: WIRING – CONNECTION - SETTING 4

2.1 Controller I/O ports 4

2.2 Controller power supply 5

2.2.1 Powering instructions 5

2.3 Input devices connection 6

2.3.1 Sensor Input wiring 7

2.3.1.1 High-level input signal wiring 8

2.3.1.2 Low-level input signal wiring 8

2.3.2 Connecting X,Y,Z,A homing sensor 9

2.3.2.1 [3-wire] homing sensor wiring diagram 9

2.3.2.2 [2-wire] homing sensor wiring diagram (NPN signal) 9

2.3.2.3 [2-wire] homing sensor wiring diagram (PNP signal) 10

2.3.2.4 Homing switch wiring diagram 10

2.3.3 Automatic tool setter connection 12

2.3.3.1 [3-wire] Automatic tool setter wiring diagram 13

2.3.3.2 [4-wire] Automatic tool setter wiring diagram (NPN signal) 13

2.3.3.3 [4-wire] Automatic tool setter wiring diagarm (PNP signal) 14

2.3.3.4 [6-wire] Automatic tool setter wiring diagram (NPN signal) 14

2.3.3.5 [6-wire] Automatic tool setter wiring diagram (PNP signal) 15

2.3.4 Alarm signal for axes (X,Y,Z,A), spindle 16

2.3.4.1 Servo alarm wiring diagram (NPN type) 19

2.3.4.2 Servo alarm wiring diagram (PNP type) 20

2.3.4.3 Spindle alarm wiring diagram (NPN, PNP) 21

2.3.5 MPG Handle 23

2.3.5.1 MPG wiring diagram (4-wire type – differential output signal) 24

2.3.5.2 MPG wiring diagram (6-wire type – differential output signal) 25

QS TECHNOLOGY

2.3.5.3 MPG wiring diagram(4-wire type – NPN output signal) 26
2.3.5.4 MPG wiring diagram (4-wire type – PNP output signal) 27
2.3.5.5 MPG wiring diagram (6-wire – NPN output) 28
2.3.5.6 MPG wiring diagram (6-wire – PNP output) 29
2.3.5.7 MPG pendant with EMG button 30
2.3.6 STARTbutton wiring diagram with NPN – PNP signal 31
2.3.7 STOP button wiring diagram with NPN – NPN signal 31
2.3.8 EMG button wiring diagram (NPN – PNP signal) 32
2.3.8.1 EMG button wiring diagram (NPN type) 32
2.3.8.2 EMG button wiring diagram (PNP type) 32
2.4 Output signal wiring diagram 32
2.4.1 Y40 output signal (S-ON) 34
2.4.2 Y41 output signal (SP-FW) 35
2.4.3 Y42 output signal (M8) 36
2.4.4 Y43 output signal (Air-Pro) 37
2.4.5 Y44 output signal (Start light) 38
2.4.6 Y45 output signal (Alarm light) 39
2.4.7 Analog signal wiring diagram for the inverter 40
2.4.7.1 Analog signal connection diagram to the inverter using 0–10V voltage 41
2.4.7.2 Analog signal connection diagram to the inverter using 0–20mA current 41
2.5 I/O link board connection 42
2.5.1 I/O link 32_V1.1_0722 board wiring diagram 42
2.5.1.1 I/O Link 32_V1.1_0722 board wiring diagram (LOW INPUT) 43
2.5.1.2 Link 32_V1.1_0722 board wiring diagram (HIGH INPUT) 44
2.5.1.3 I/O link 32_V1.1_0722 board output signal 45
2.5.2 I/O Link 08_RLTR board wiring diagram 46
2.5.2.1 I/O link 08_RLTR board wiring diagram (LOW INPUT) 47
2.5.2.2 I/O link 08_RLTR board wiring diagram (HIGH INPUT) 48
2.5.2.3 I/O link 08_RLTR board output signal 49
2.6 Axis connection (X,Y,Z,A) 50
2.6.1 X-Y axis wiring diagram (Pul/Dir) 52
QS TECHNOLOGY

2.6.2 Z-A axis wiring diagram (Pul/Dir) 52
2.6.3 Encoder feedback signal for phase Z from the servo drive of axes (X, Y, Z, A) 53
2.6.4 Axis configuration..... 54
2.6.4.1 Enable/Disabe axis (X,Y,Z,A)..... 54
2.6.4.2 Motor rotation direction for axes (X,Y,Z,A)..... 55
2.6.4.3 Software limit for axes (X,Y,Z,A) 56
2.6.4.4 Axes direction when moving to home position..... 57
2.6.4.5 Axis movement type (X,Y,Z,A) 59
2.6.4.6 Pulse generation type for axes (X,Y,Z,A) 60
2.6.4.7 Electronic gear box setting..... 62
2.6.4.8 Axis acceleration in JOG RAPID (G00) 65
2.6.4.9 Axis acceleraion for machining (G01)..... 66
2.6.4.10 Axis acceleration in JOG mode 67
2.6.4.11 Machine speed in JOG RAPID mode..... 68
2.6.4.12 Axis max speed in JOG mode 69
2.6.4.13 Axis max speed in G00 program..... 70
2.6.4.14 Axis max speed in G01 program..... 71
2.6.4.15 Axis homing type..... 72
2.6.4.16 Homing Feedrate 1st 74
2.6.4.17 Homing Feedrate 2nd 75
2.6.4.18 Homing Feedrate 3rd..... 76
2.6.4.19 Machine travel limit setting (Soft Limit) 77
2.6.5 Automatic tool setter..... 80
2.6.7 Backlash compensation – With load (G01) 86
PART 3 SPECIAL FUNCTIONS 87
3.1 Auto home 87
3.2 Workpiece homing 89
3.3 Resume the program when sudden power failure occurs 92
3.4 Start program from any line 93
3.5 Resume the program from the line where it was previously stopped 95
PART 4 OPERATION INTERFACE 97

QS TECHNOLOGY

4.1 Main screen 97
4.2 Coordinate sreen 98
4.2.1 Switching coordinate system screen 100
4.2.2 Divide the specified axis coordinates in the relative coordinates (REL) by 2 101
4.2.3 Set the specified axis coordinates in the relative coordinates (REL) to position 0 102
4.3 Program screen 103
4.3.1 Copy data from USB to the controller 105
4.3.2 Execute program 106
4.3.3 Go to any command line 108
3.3.4 Search and replace object in the program 110
3.3.5 Delete command line 113
4.3.6 Copy and Paste 115
4.4 Offset screen 117
4.4.1 Manual data input 119
4.4.2 Assign the coordinates and auxiliary coordinates to the specified coordinate system 120
4.4.3 Quickly determine the center coordinates for a 3-point or 4-point workpiece 122
4.5 Monitor screen 124
4.5.1 Toolpath simulation 126
4.5.2 MDI mode 127
4.5.3 Working history 128
4.6 Maintenance screen 129
4.7 Diagnostic screen 130
4.8 Paramters setting 131
4.9 PLC screen 134
3.9.1 Searching for an object in the program 136
3.9.2 PLC data loading 137
4.10 System data 140
4.10.1 Firmware update 142
4.10.2 Date & Time setting 145

4.11 Transfer data via Wi-Fi network 147
4.11.1 Software installation 148
4.11.2 Connect device to wi-fi network 149
4.11.3 Transfer data via wi-fi network 153
4.11.4 Monitor the operational status of the controller 158
PART 5 PARAMETERS TABLE 160
5.1 Axis parameters 160
5.2 Spindle parameters 162
5.3 General parameters 163

QS TECHNOLOGY

PART 1:CONTROLLER OVERVIEW

1.1 Overview

Specification:

Model	F54
Dimension	220x140x30 mm
Case material	Aluminum anod
Monitor size	5 Inches
Axis control	4 axis
Input/Output	16/6 (Max.Optional 256/256)



Figure 1. Front view of F54 controller

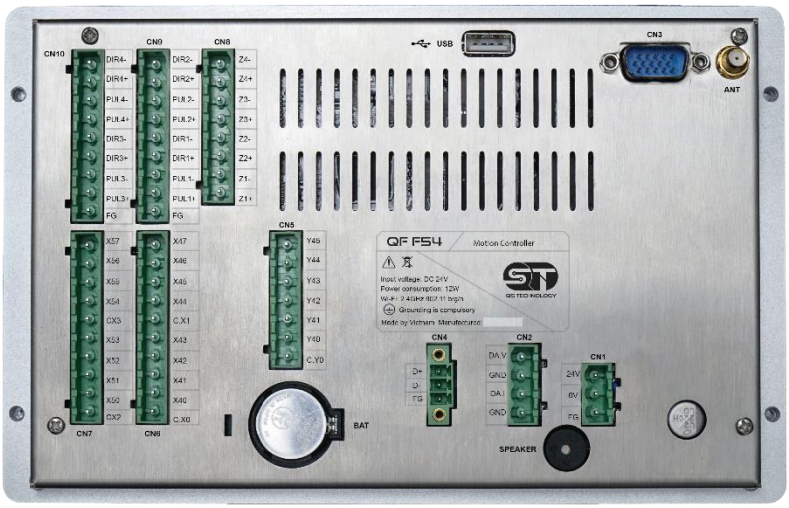


Figure 2. Back view of F54 controller

1.2 Dimension

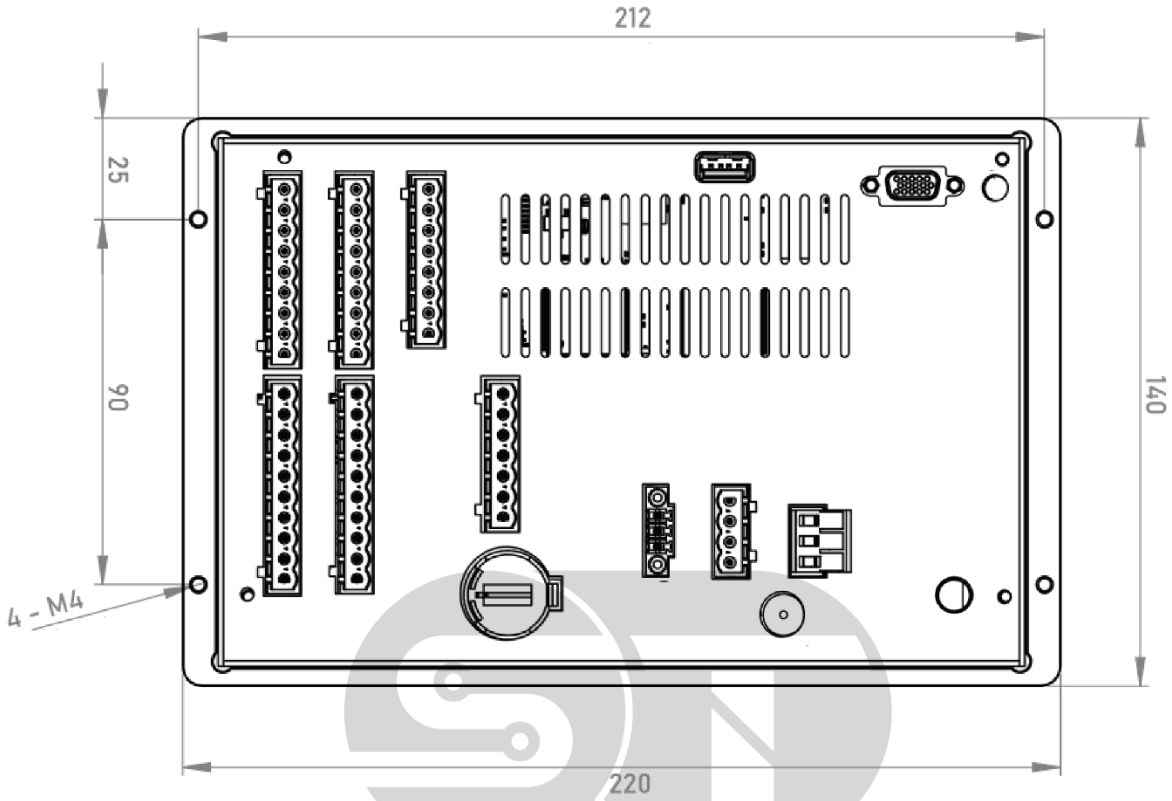


Figure 3. F54 controller dimension drawing

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1.3 Software structure

Model: F54

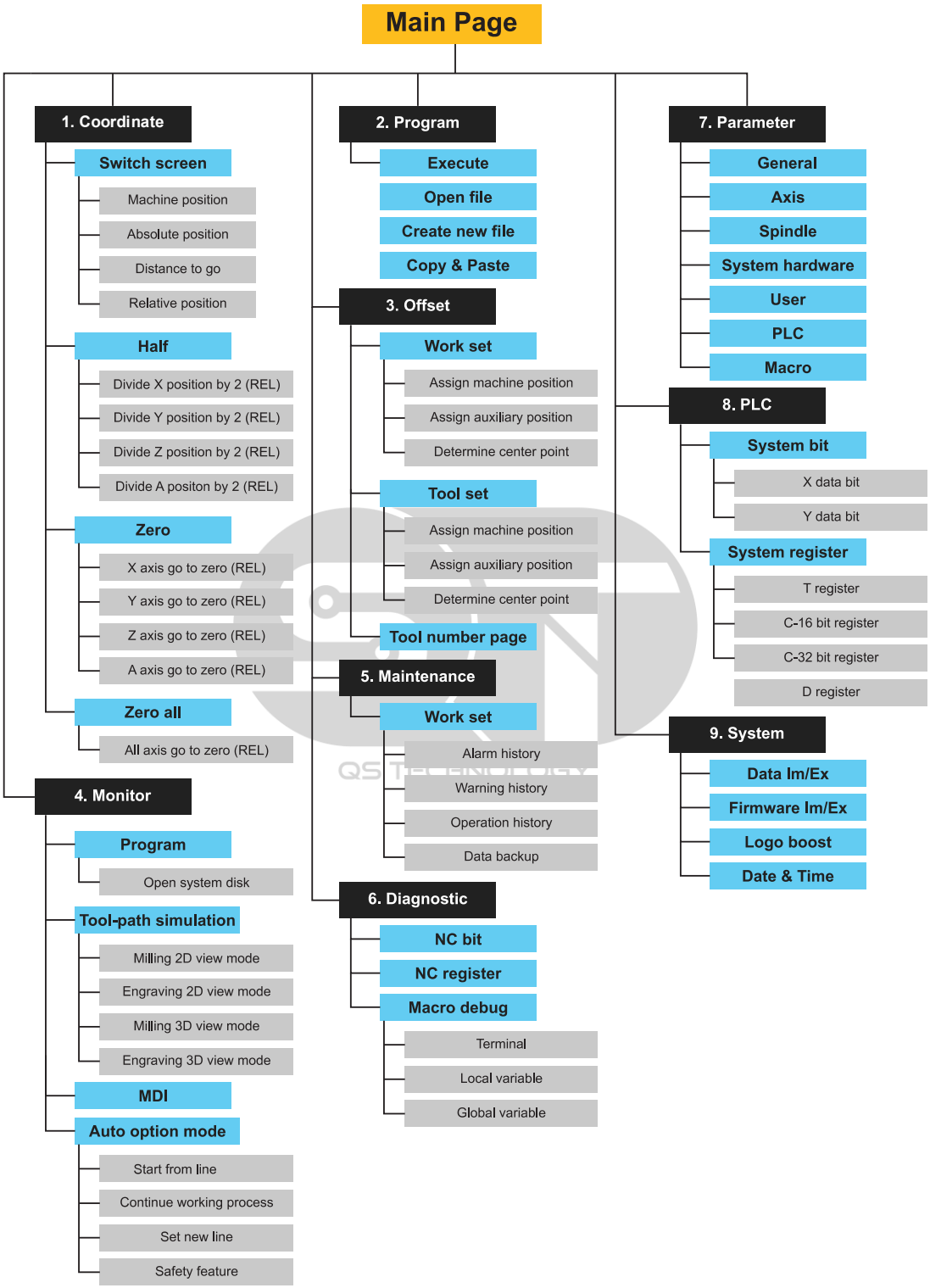


Figure 4. F54 controller software structure

PART 2: WIRING – CONNECTION - SETTING

2.1 Controller I/O ports

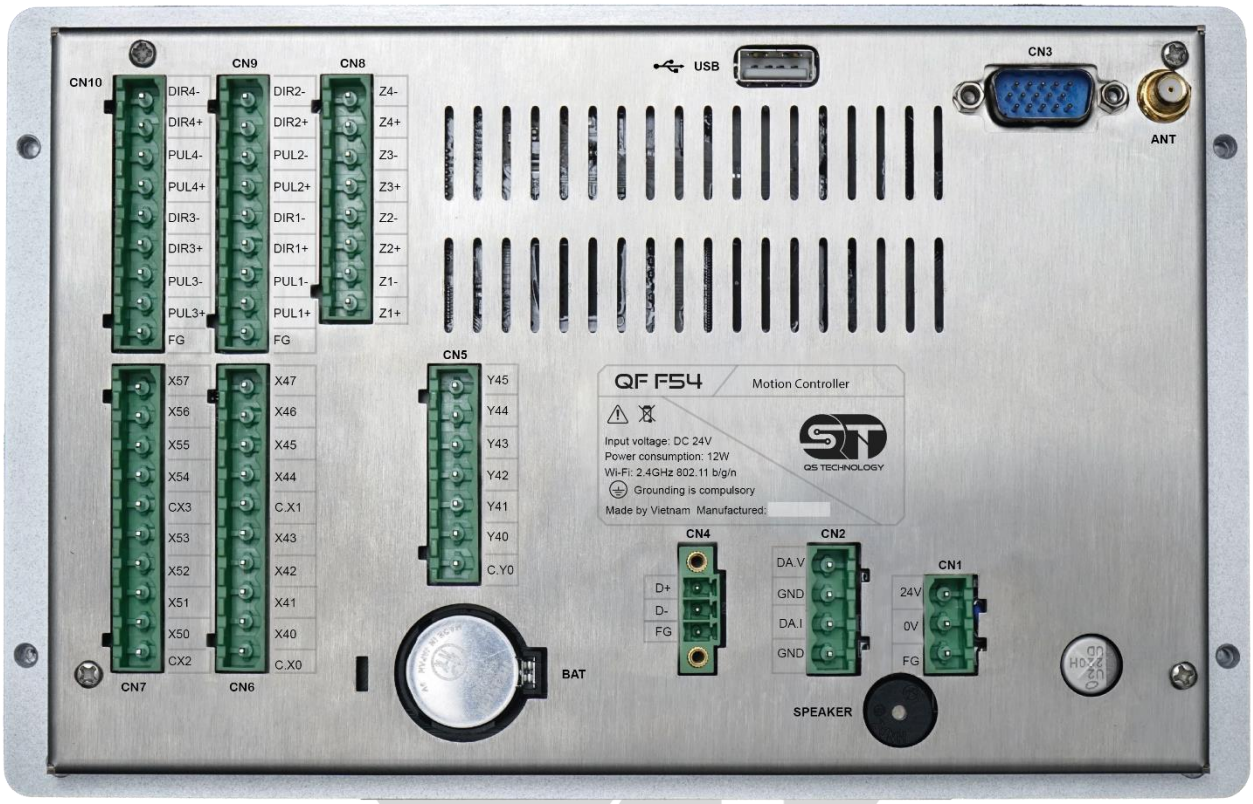


Figure 5. F54 controller I/O ports

Port	Functions
CN1	Power supply
CN2	Analog output signal port for spindle speed control
CN3	MPG Handle
CN4	I/O expansion module connection
CN5	Output signal
CN6	Input port (1)
CN7	Input port (2)
CN8	Feedback Encoder Z Phase Signal Port for axes (X, Y, Z, A)
CN9	Pul/Dir Signal Connection Port for X, T axis servo
CN10	Pul/Dir Signal Connection Port for Z, A axis servo

2.2 Controller power supply



Figure 6. Commonly used power supply type

Note:

- Using pulse-type power supply is better than transformer
- Input voltage must be connected to the FG pin to prevent noise signal for controller

2.2.1 Powering instructions

The power supply for the controller will be connected to CN1 port on controller



Figure 7. CN1 power supply port

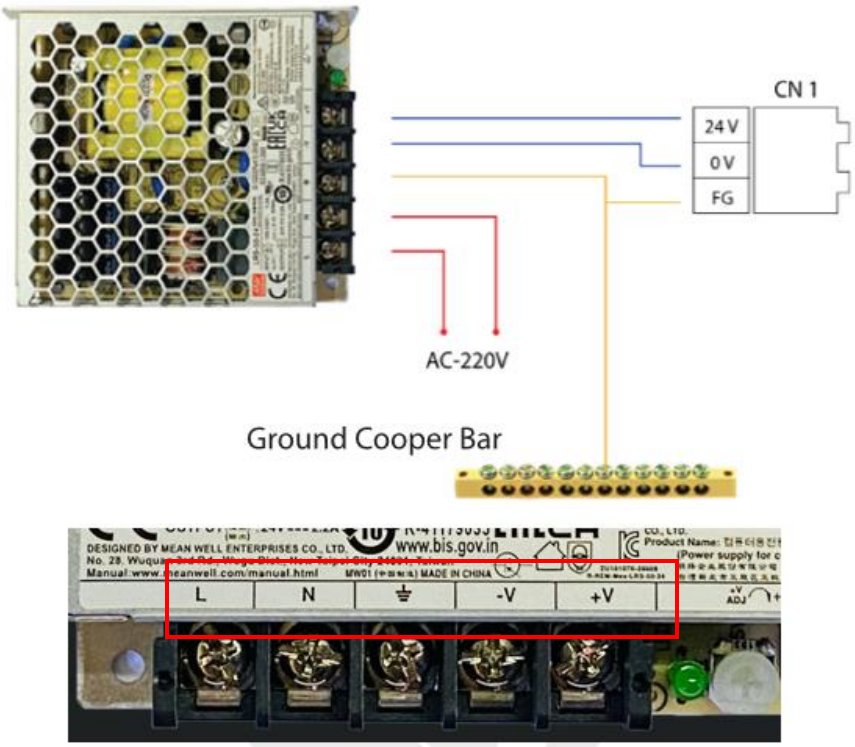


Figure 8. MeanWell 24-2.2A power supply terminal

Installer should carefully check the power terminals on the device before connecting to the controller to avoid causing damage to the equipment.

2.3 Input devices connection

Input devices such as sensors, Start/Stop/EMG button, automatic tool setter, ... will be connected to CN6 and CN7 port of controller

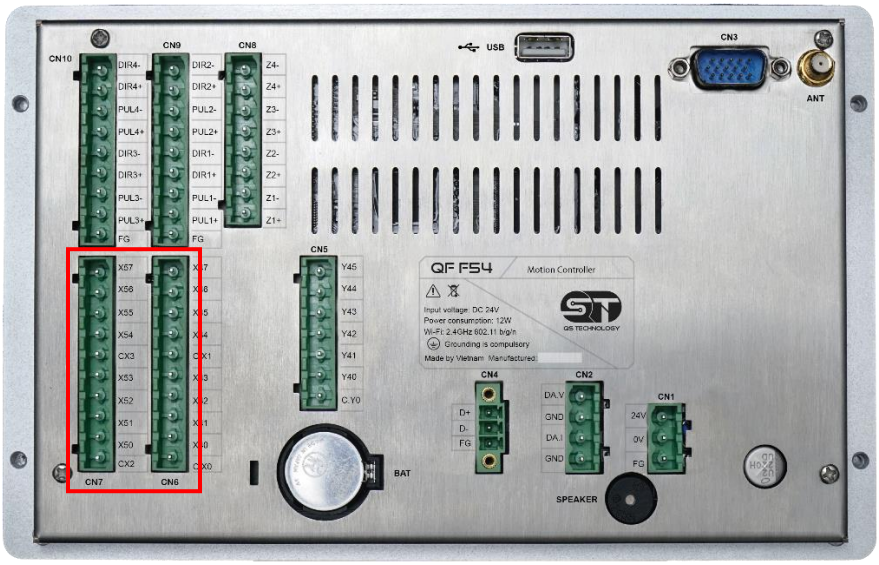


Figure 9. Input devices connection port on F54 controller

Input devices connection address table on F54 controller

Add	Name	Function
X40	Home X	X axis homing sensor
X41	Home Y	Y axis homing sensor
X42	Home Z	Z axis homing sensor
X43	Home A	A axis homing sensor
X44	Automatic tool setter	Automatic tool setter
X45	Overtravel protection signal of (Automatic tool setter)	Overtravel limit switch of automatic tool setter
X46	Start button	Start button signal
X47	Stop button	Stop button signal
X50	EMG button	Emergency button signal
X51	Alarm X	X-axis servo alarm
X52	Alarm Y	Y-axis servo alarm
X53	Alarm Z	Z-axis servo alarm
X54	Alarm A	A-axis servo alarm
X55	Alarm SP	Spindle alarm
X56	Alarm M8 Coolant	Overtemperature alarm for the cooling water pump motor
X57	Empty port	Empty signal

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2.3.1 Sensor Input wiring

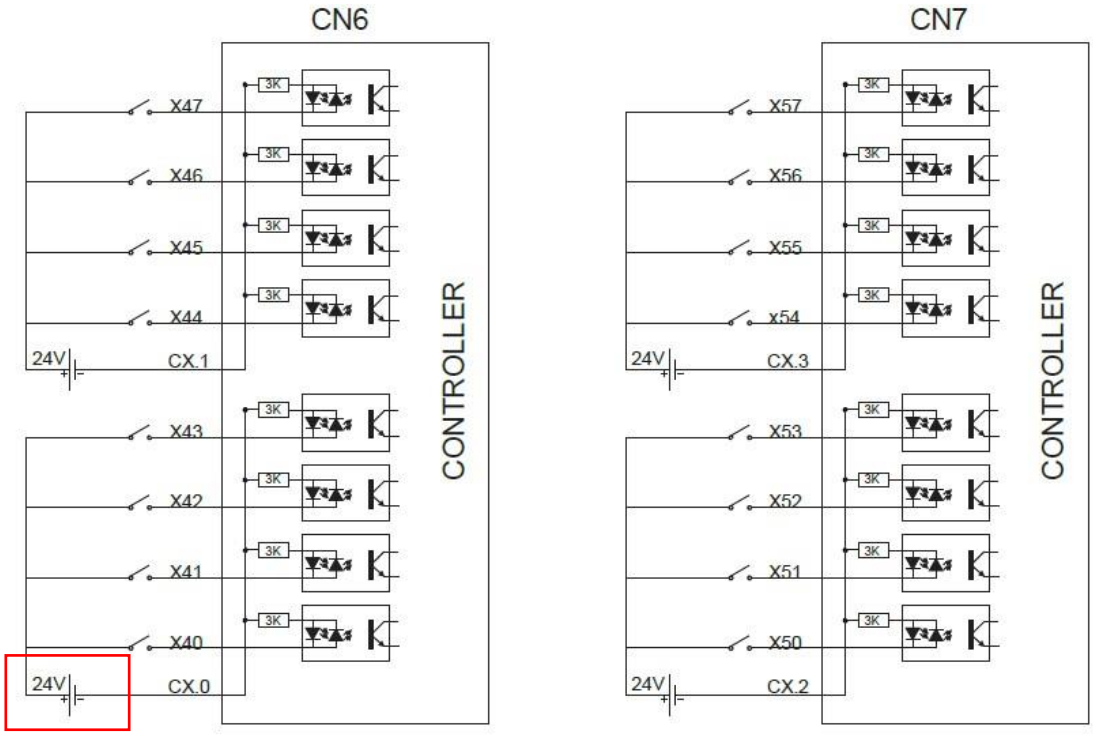
There are two common sensor wiring types:

- High-level input signal (PNP)
- Low-level input signal (NPN)

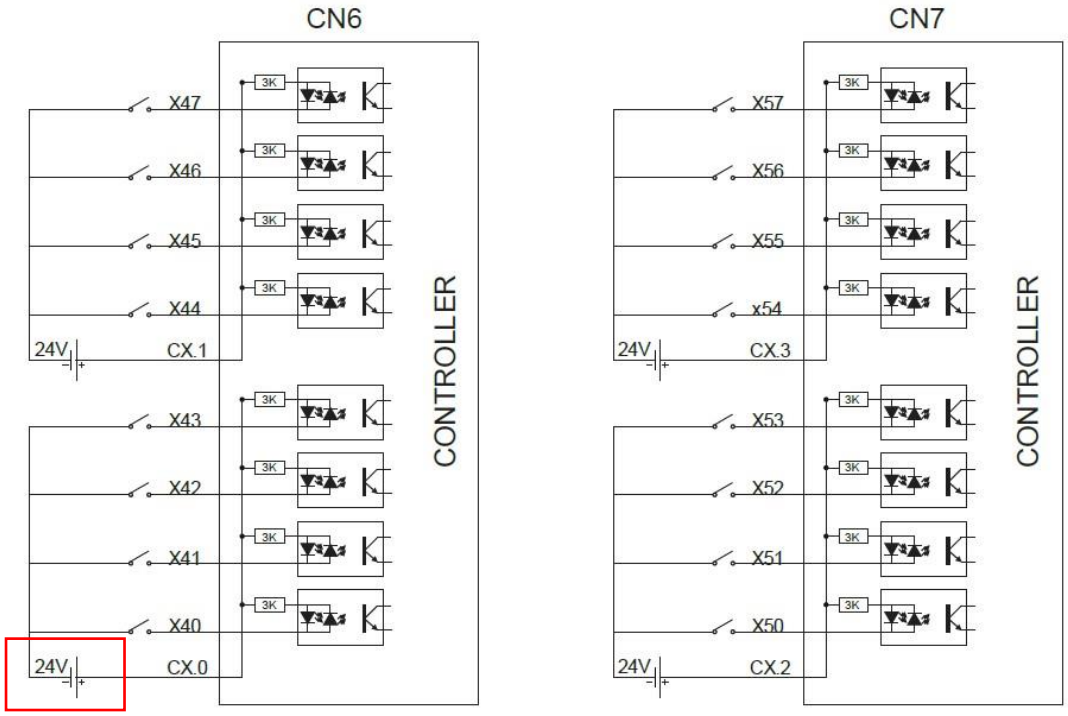
Notes:

- **High-level input (PNP):** This is when the CX terminal is connected to 0V and a sensor with a high-level (PNP) output signal is used. This signal is fed into the input terminal of the controller
- **Low-level input (NPN):** This is when the CX terminal is connected to 24V and a sensor with a low-level (NPN) output signal is used. This signal is fed into the input terminal of the controller
- Terminals CX.0, CX.1, CX.2, and CX.3 are separate COM terminals for input signals
- All input signal voltages must be of the same type
- **DO NOT** connect both high-level and low-level input signals to the machine at the same time

2.3.1.1 High-level input signal wiring



2.3.1.2 Low-level input signal wiring



2.3.2 Connecting X,Y,Z,A homing sensor

Before installing homing sensors for axes (X,Y,Z,A). Installer must identify the sensor type, wire colors, and the function of the signal wires. Connections should be made according to the sensor wiring diagram below.

2.3.2.1 [3-wire] homing sensor wiring diagram

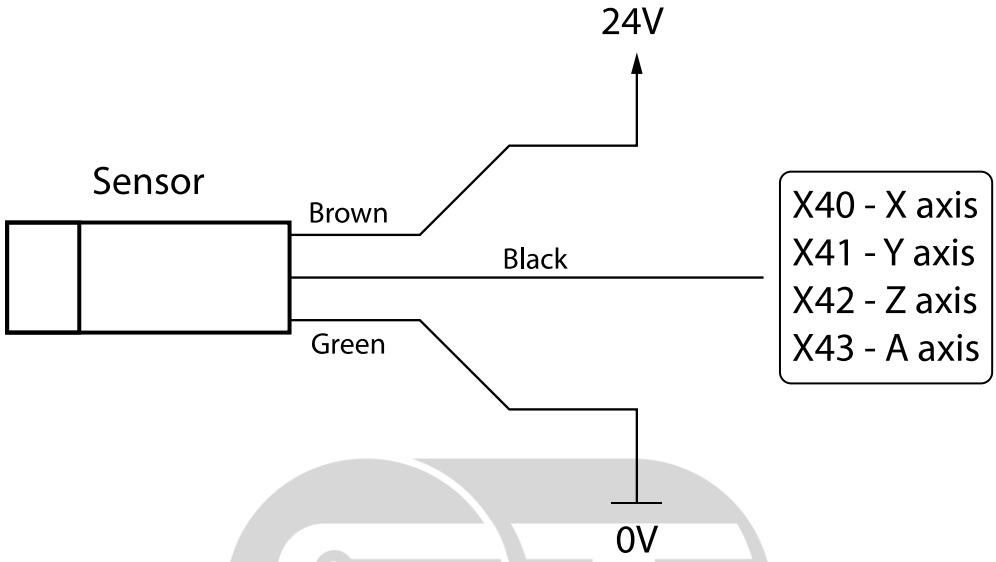


Figure 10. 3-wire homing sensor wiring diagram for axes (X,Y,Z,A)

2.3.2.2 [2-wire] homing sensor wiring diagram (NPN signal)

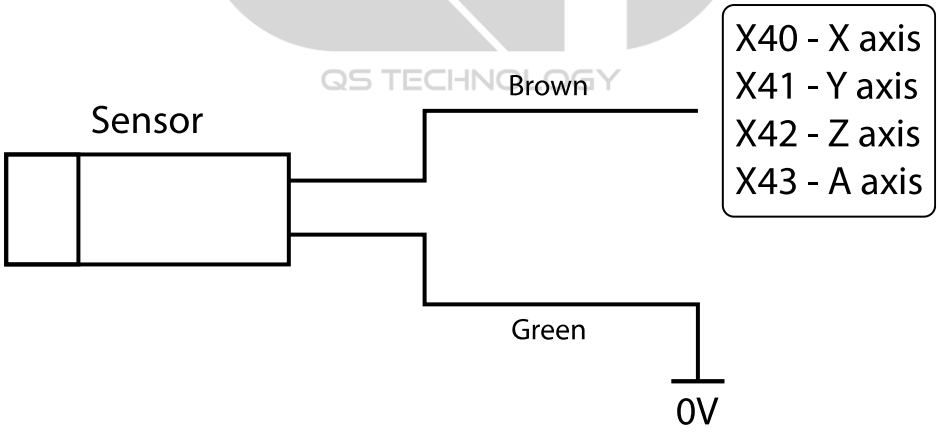


Figure 11. 2-wire homing sensor wiring diagram for axes (X,Y,Z,A) – NPN signal

2.3.2.3 [2-wire] homing sensor wiring diagram (PNP signal)

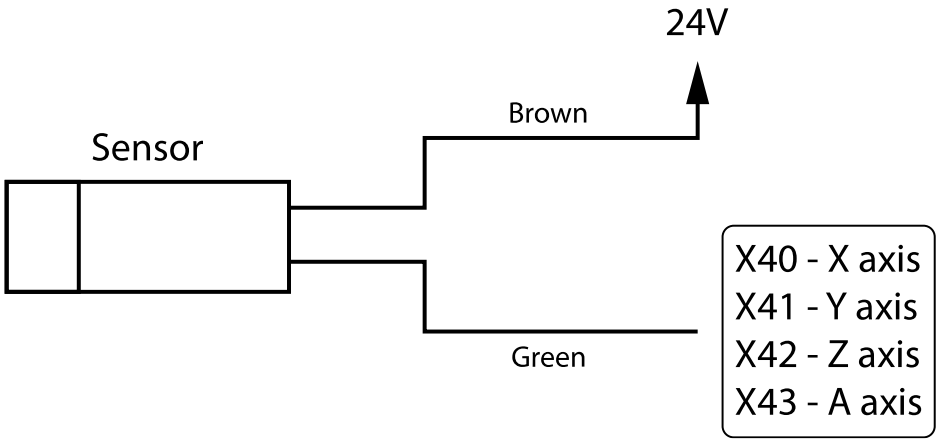


Figure 12. 2-wire homing sensor wiring diagram for axes (X,Y,Z,A) – PNP signal

2.3.2.4 Homing switch wiring diagram

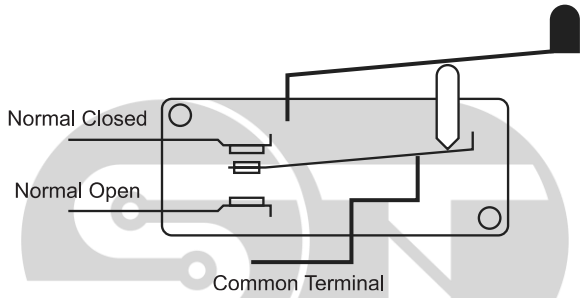


Figure 13. Homing switch operation structure

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Homing switch wiring diagram for NPN, PNP signal

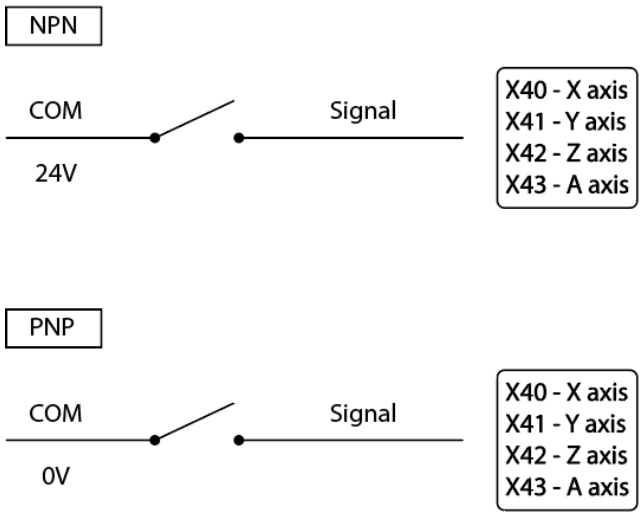


Figure 14. Homing switch wiring diagram for (NPN,PNP signal)

Note:

After connecting the homing sensor for the axes (X,Y,Z,A). The installer will proceed to configure the following setting on the F54 controller:

Step 1: On the main sreen of the controller, use the arrow buttons to switch to the right and select the PARAMETER button.

Step 2: From PARAMETER screen, select USER

Step 3: Proceed to configure the status parameters for axes homing sensor (X,Y,Z,A) at parameter M1920, M1921, M1922, M1923.

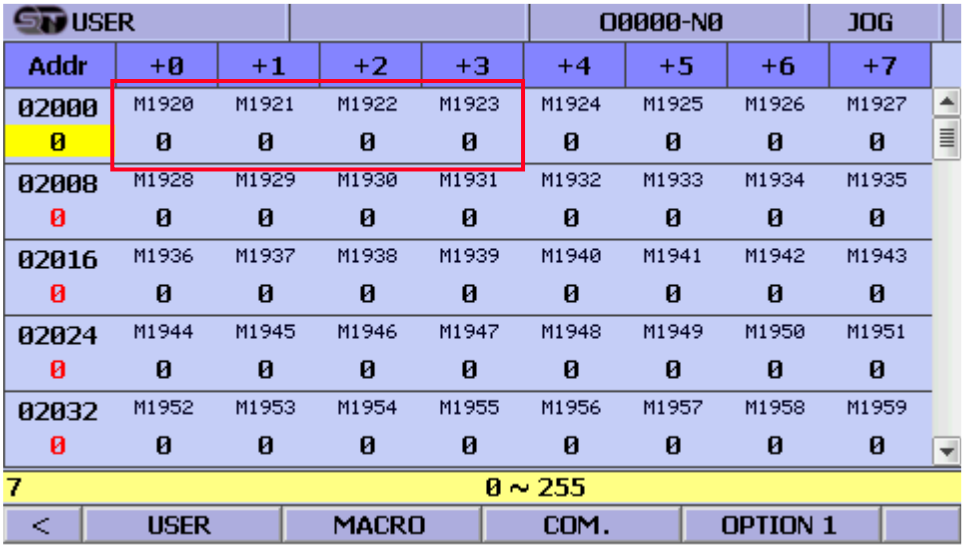


Figure 15. Homing sensor status configuration parameters for axes (X,Y,Z,A)

How to configure the parameter from 0 to 1:

- From the PARAMETER screen, use the arrows keys to select the parameter to configure.
- Then enter number 1 and press OK. A prompt will appear on the screen asking for the user password.
- Enter “1415” and press OK
- Next, enter number 1 again and press OK.
- Press RESET button to save the configured parameters.

Parameter stauts description:

- **M1920:** Homing sensor status configuration parameter for X axis
  - M1920 = 0 : Normally closed sensor (NC)
  - M1920 = 1 : Normally open sensor (NO)
- **M1921:** Homing sensor status configuration parameter for Y axis
  - M1921 = 0 : Normally closed sensor (NC
  - M1921 = 1 : Normally open sensor (NO)
- **M1922:** Homing sensor status configuration parameter for Z axis
  - M1922 = 0 : Normally closed sensor (NC
  - M1922 = 1 : Normally open sensor (NO)
- **M1923:** Homing sensor status configuration parameter for A axis
  - M1923 = 0 : Normally closed sensor (NC
  - M1923 = 1 : Normally open sensor (NO)

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2.3.3 Automatic tool setter connection

Before installing the automatic tool setter, installers must identify the type of tool setter, wire colors and the functions of the signal wires. Connections should be made according to the wiring diagram below.



Figure 16. Automatic tool setter deivce

2.3.3.1 [3-wire] Automatic tool setter wiring diagram

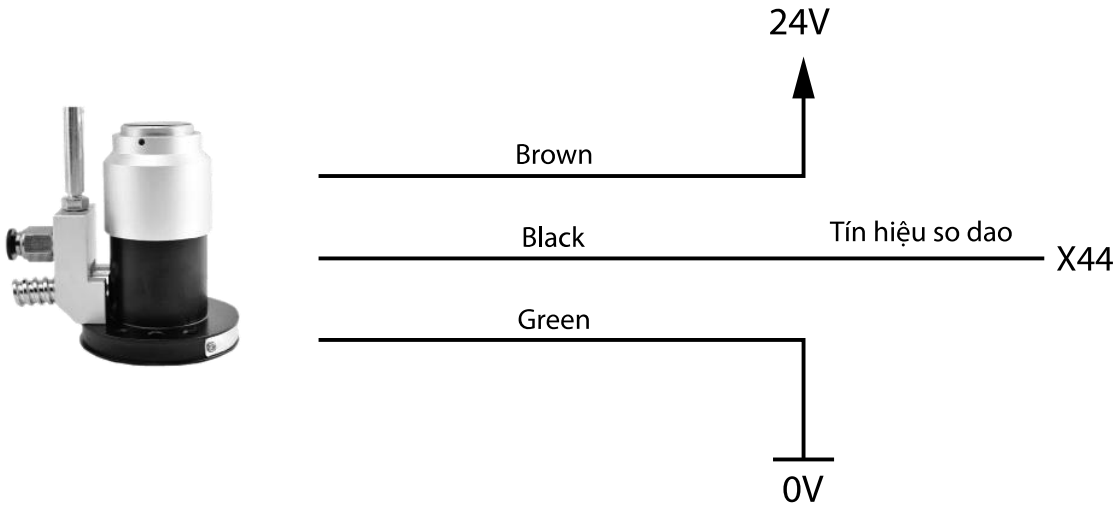


Figure 17. Automatic tool setter wiring diagram (3-wire)

2.3.3.2 [4-wire] Automatic tool setter wiring diagram (NPN signal)

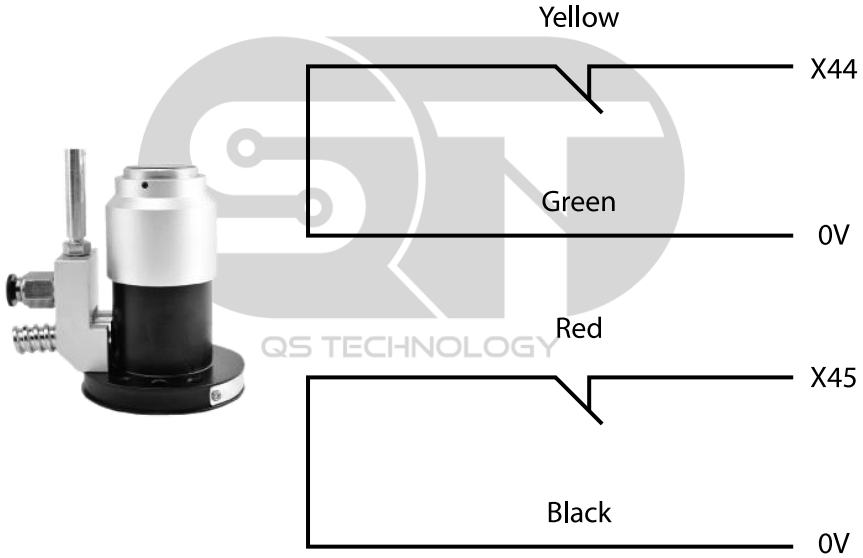


Figure 18. Automatic tool setter wiring diagram (4-wire, NPN signal)

2.3.3.3 [4-wire] Automatic tool setter wiring diagarm (PNP signal)

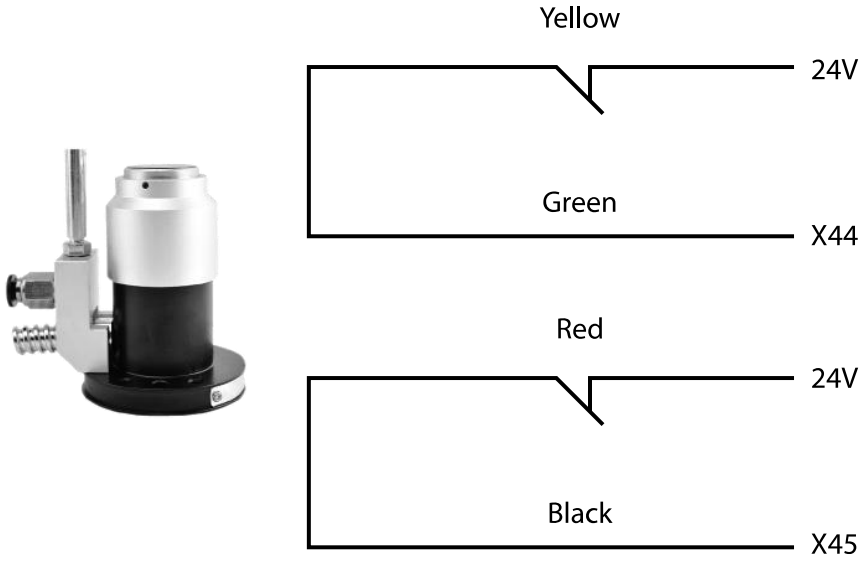


Figure 19.Automatic tool setter wiring diagram (4-wire, PNP signal)

2.3.3.4 [6-wire] Automatic tool setter wiring diagram (NPN signal)

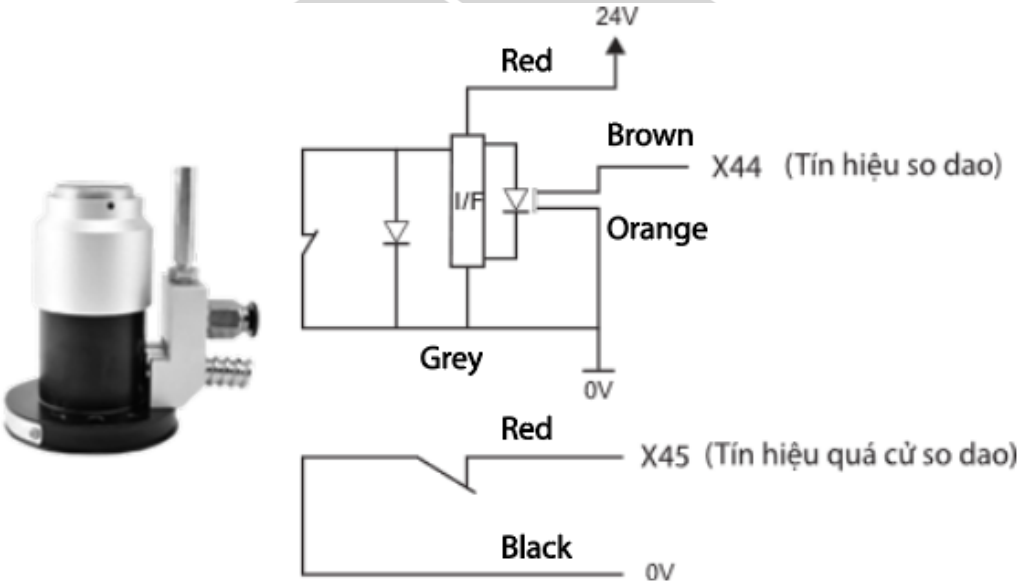


Figure 20. Automatic tool setter wiring diagram (6-wire, NPN signal)

2.3.3.5 [6-wire] Automatic tool setter wiring diagram (PNP signal)

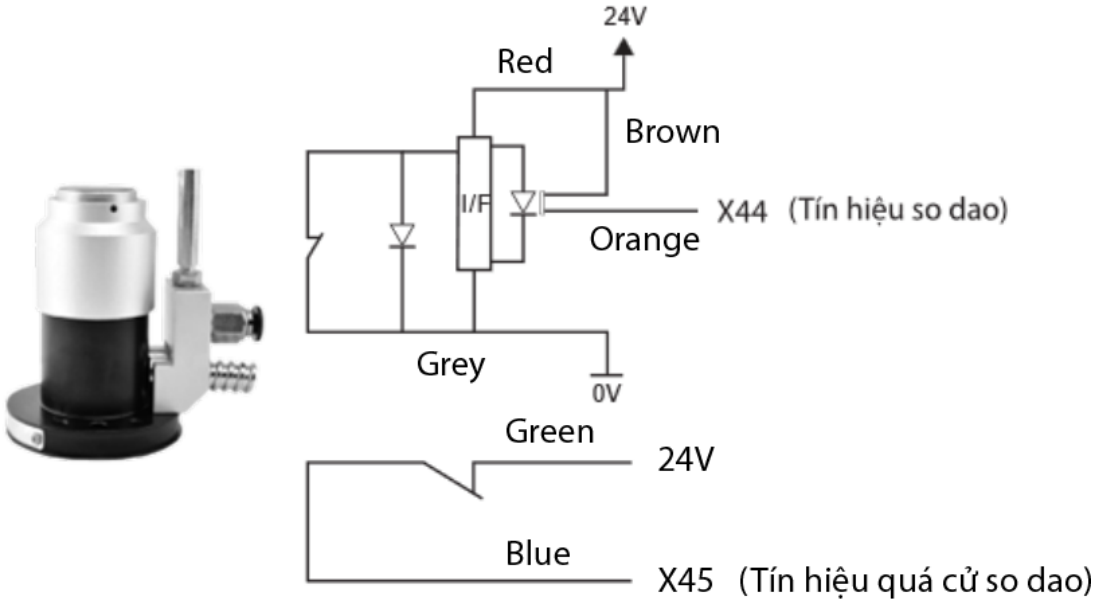


Figure 21. Automatic tool setter wiring diagram (6-wire,PNP signal)

Note:

After connecting the automatic tool setter to the machine, the installer will proceed to configure the following parameters on F54 : M1925, M1926, M1958.

Step 1: On the main screen of the controller, use the arrow buttons to switch to the right and select PARATER button.

Step 2: From PARAMETER screen, select USER

Step 3: Proceed to configure the stauts parameter of automatic tool setter device: M1925, M1926, M1958.

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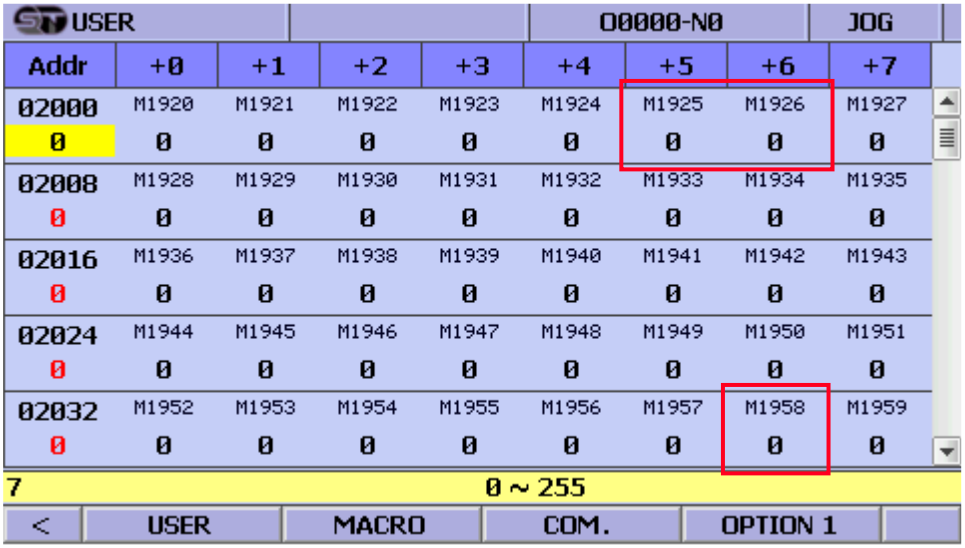


Figure 22. Automatic tool setter stauts parameter configuration

How to configure the parameter from 0 to 1:

- From the PARAMETER screen, use the arrows keys to select the parameter to configure.
- Then enter number 1 and press OK. A prompt will appear on the screen asking for the user password.
- Enter “1415” and press OK
- Next, enter number 1 again and press OK.
- Press RESET button to save the configured parameters.

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Parameter stauts description:

- **M1925:** Automatic tool setter touch head status parameter configuration
  - M1925 = 0 : Normally closed sensor (NC)
  - M1925 = 1 : Normally open sensor (NO)
- **M1926:** Status parameter configuration for automatic tool setter touch head travel limit
  - M1926 = 0 : Normally closed sensor (NC)
  - M1926 = 1 : Normally open sensor (NO)
- **M1958:** Disable/Enable travel limit of automatic tool setter touch head
  - M1958 = 0 : Disable travel limit feature
  - M1958 = 1 : Enable travel limit feature

2.3.4 Alarm signal for axes (X,Y,Z,A), spindle

Note: Before connecting the Alarm(+) and Alarm(-) signal. The installer must check the wiring document of the corresponding servo drive

Example: Servo SGDV Yaskawa *(Reference documentation from Yaskawa)*

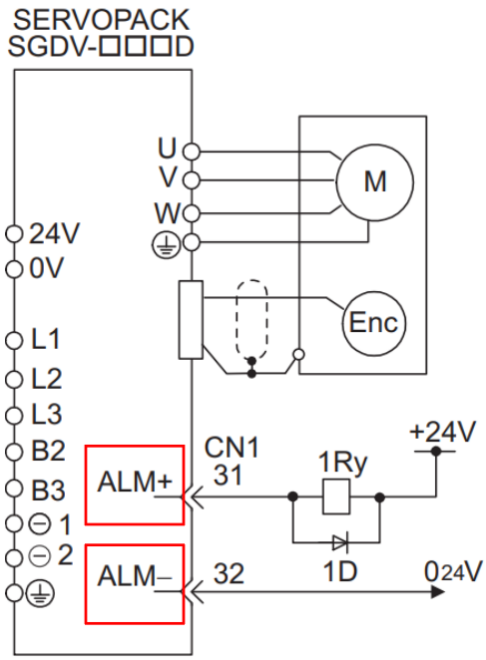


Figure 23.Servo alarm wiring diagram of Yaskawa SGDV servo drive (Reference documentation)

Note (1): After connecting alarm signal. Installer needs to configure the following (M1937, M1938, M1939, M1940, M1941) to enable servo alarm feature

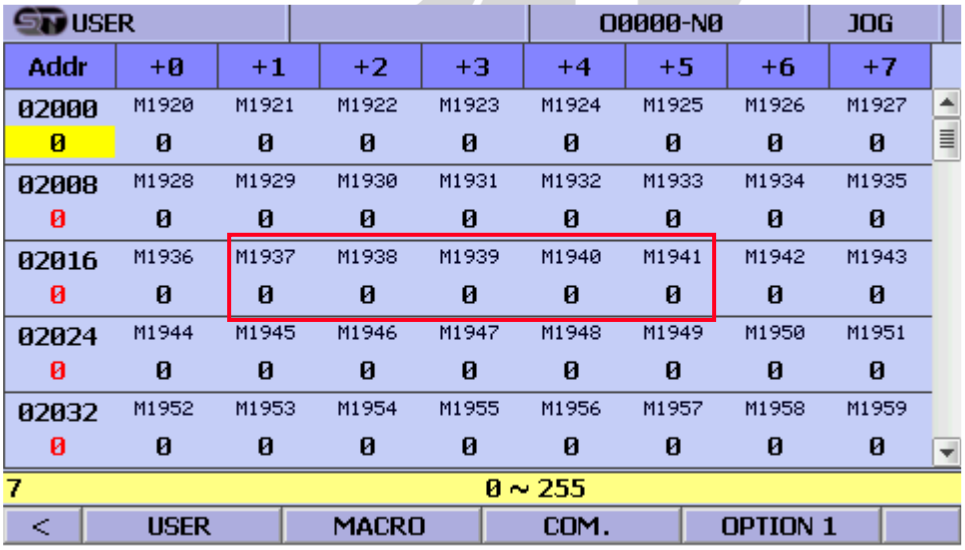


Figure 24. Spindle, servo alarm function parameters

How to configure the parameter from 0 to 1:

- From the PARAMETER screen, use the arrows keys to select the parameter to configure.

- Then enter number 1 and press OK. A prompt will appear on the screen asking for the user password.
- Enter “1415” and press OK
- Next, enter number 1 again and press OK.
- Press RESET button to save the configured parameters.

Parameter status description:

- **M1937:** Enable/Disable servo alarm of X-axis
  - M1937 = 0 : Disable
  - M1937= 1 : Enable
- **M1938:** Enable/Disable servo alarm of Y-axis
  - M1938 = 0 : Disable
  - M1938 = 1 : Enable
- **M1939:** Enable/Disable servo alarm of Z-axis
  - M1939 = 0 : Disable
  - M1939 = 1 : Enable
- **M1940:** Enable/Disable servo alarm of A-axis
  - M1940 = 0 : Disable
  - M1940 = 1 : Enable
- **M1941:** Enable/Disable servo alarm of spindle
  - M1941 = 0 : Disable
  - M1941 = 1 : Enable

Note (2): After enableing servo alarm function for axes (X,Y,Z,A), instailler must configure the alarm signal status at following parameter: (M1932, M1933, M1934, M1935, M1961)

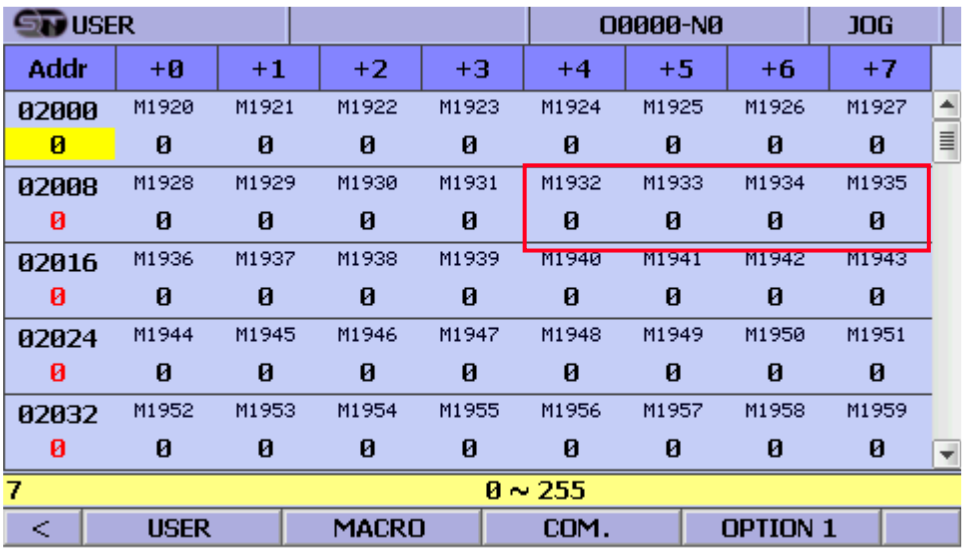


Figure 25. Axis,Spindle alarm status signal parameters

How to configure the parameter from 0 to 1:

- From the PARAMETER screen, use the arrows keys to select the parameter to configure.
- Then enter number 1 and press OK. A prompt will appear on the screen asking for the user password.
- Enter "1415" and press OK
- Next, enter number 1 again and press OK.
- Press RESET button to save the configured parameters.

Parameter status description:

- **M1932:** Servo alarm status of X axis
  - M1932 = 0 : Normally closed alarm signal (NC)
  - M1932= 1 : Normally open alarm signal (NO)
- **M1933:** Servo alarm status of Y axis
  - M1933 = 0 : Normally closed alarm signal (NC)
  - M1933 = 1 : Normally open alarm signal (NO)
- **M1934:** Servo alarm status of Z axis
  - M1934 = 0 : Normally closed alarm signal (NC)
  - M1934 = 1 : Normally open alarm signal (NO)
- **M1935:** Servo alarm status of A axis
  - M1935 = 0 : Normally closed alarm signal (NC)
  - M1935 = 1 : Normally open alarm signal (NO)
- **M1961:** Servo alarm status of spindle
  - M1961 = 0 : Normally closed alarm signal (NC)
  - M1961 = 1 : Normally open alarm signal (NO)

QS TECHNOLOGY

2.3.4.1 Servo alarm wiring diagram (NPN type)

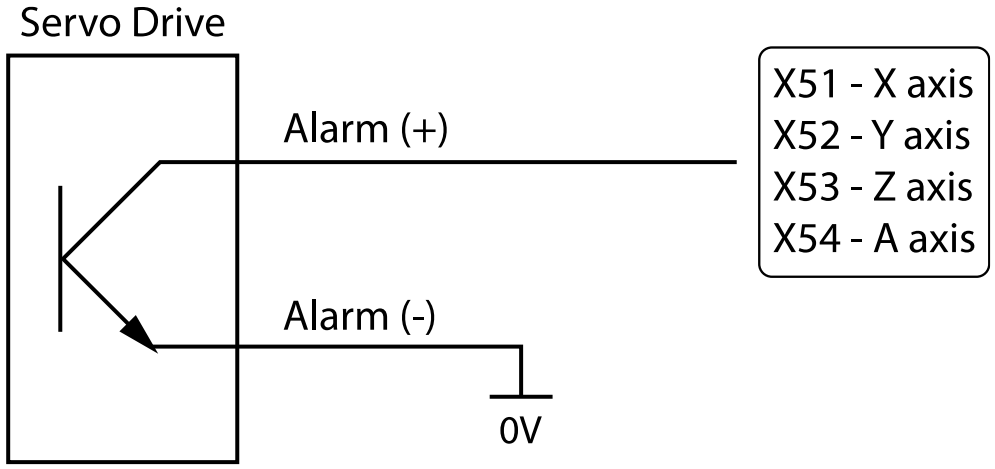


Figure 26. Servo alarm wiring diagram (NPN type)

2.3.4.2 Servo alarm wiring diagram (PNP type)

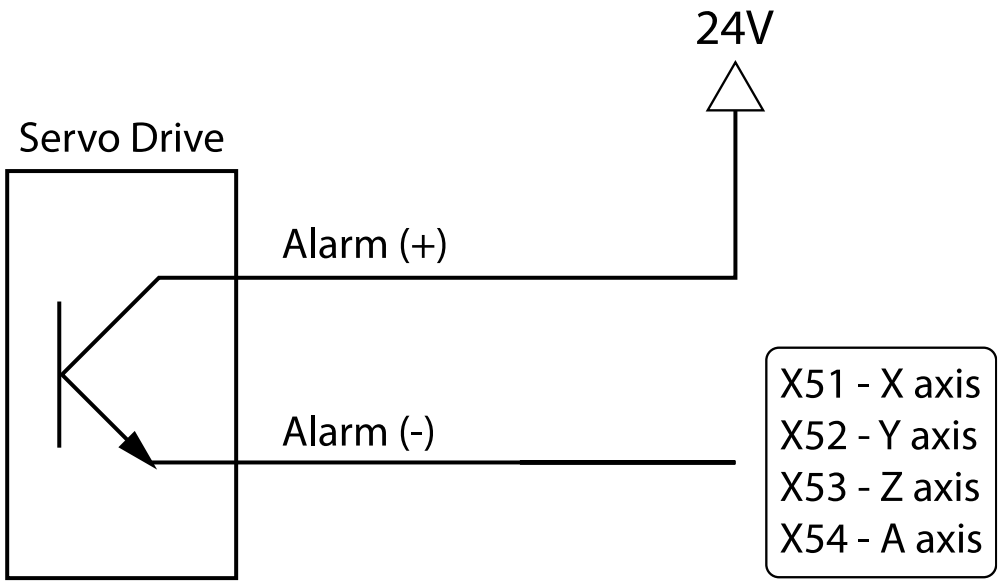


Figure 27. Servo alarm wiring diagram (PNP type)

QS TECHNOLOGY

2.3.4.3 Spindle alarm wiring diagram (NPN, PNP)

Note: Before connecting the alarm signal from the inverter to controller. The installer should check the wiring documentation of corresponding inverter model.

Example: INVT – GD350A *(Reference documentation from INVT)*

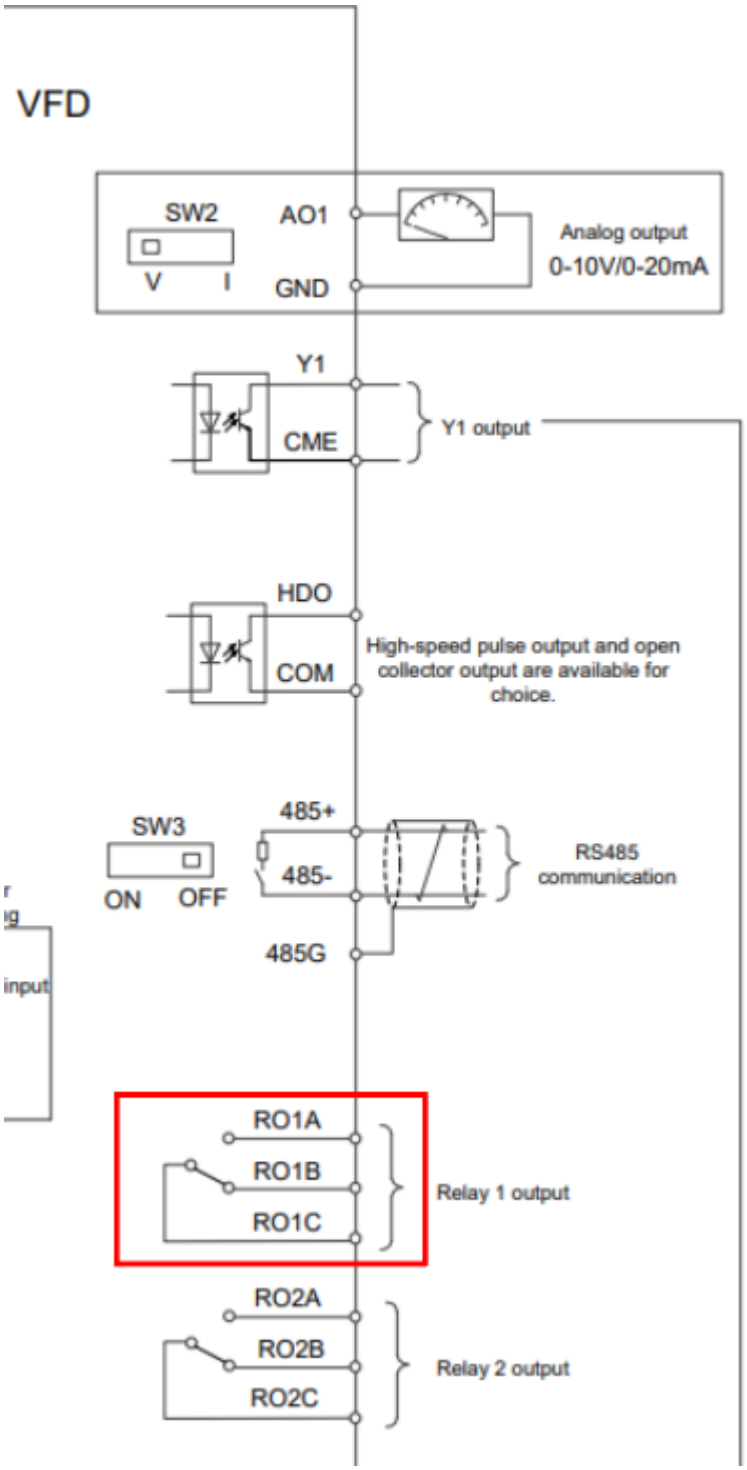


Figure 28. Spindle alarm wiring diagram from INVT inverter to controller

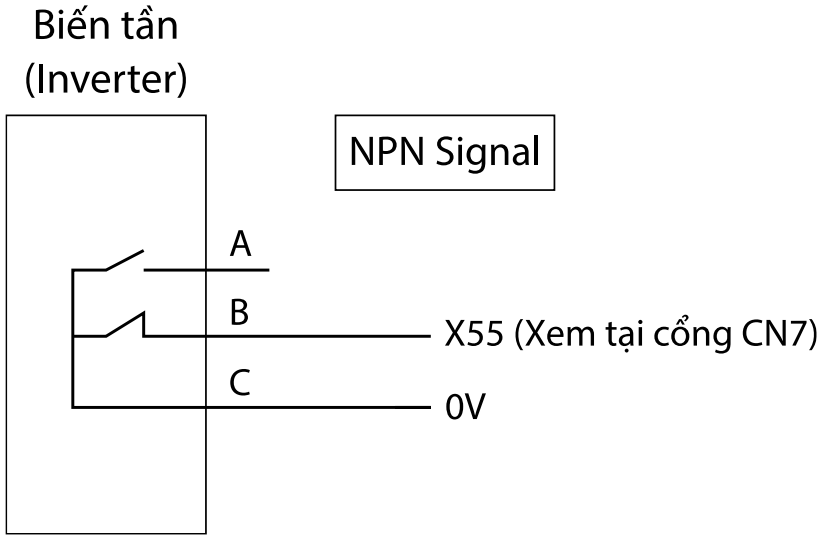


Figure 29. Spindle wiring diagram (NPN type)

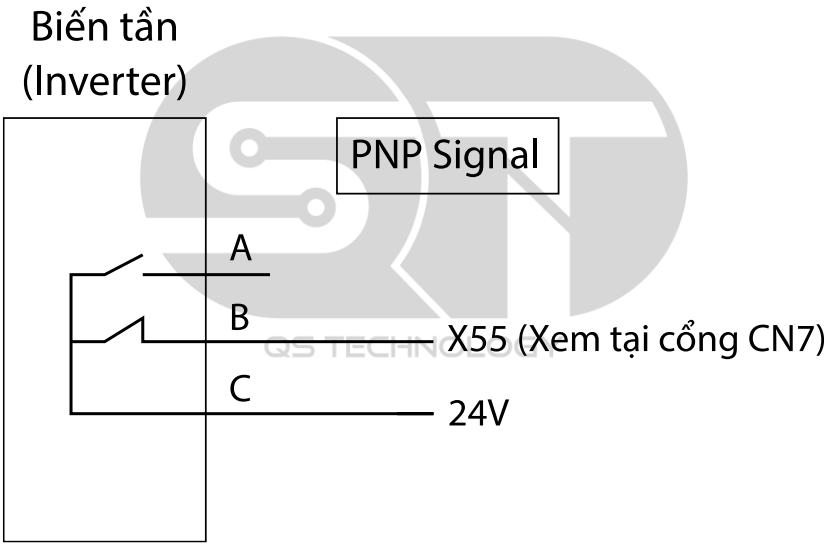
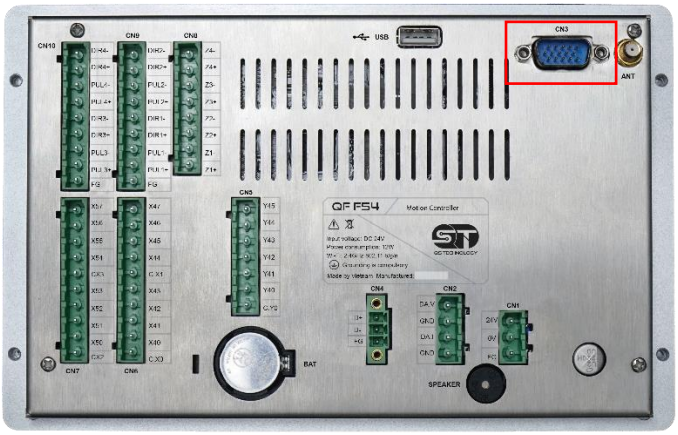


Figure 30. Spindle wiring diagram (PNP type)

2.3.5 MPG Handle

Note: Before connecting the MPG handle to the controller, the installer must identify the type of MPG handle wire colors, signal type and the corresponding wiring diagram



The signal from the MPG handle will be connected to the controller via the CN3 port on the controller.



Figure 31. Commonly used MPG handle type

2.3.5.1 MPG wiring diagram (4-wire type – differential output signal)

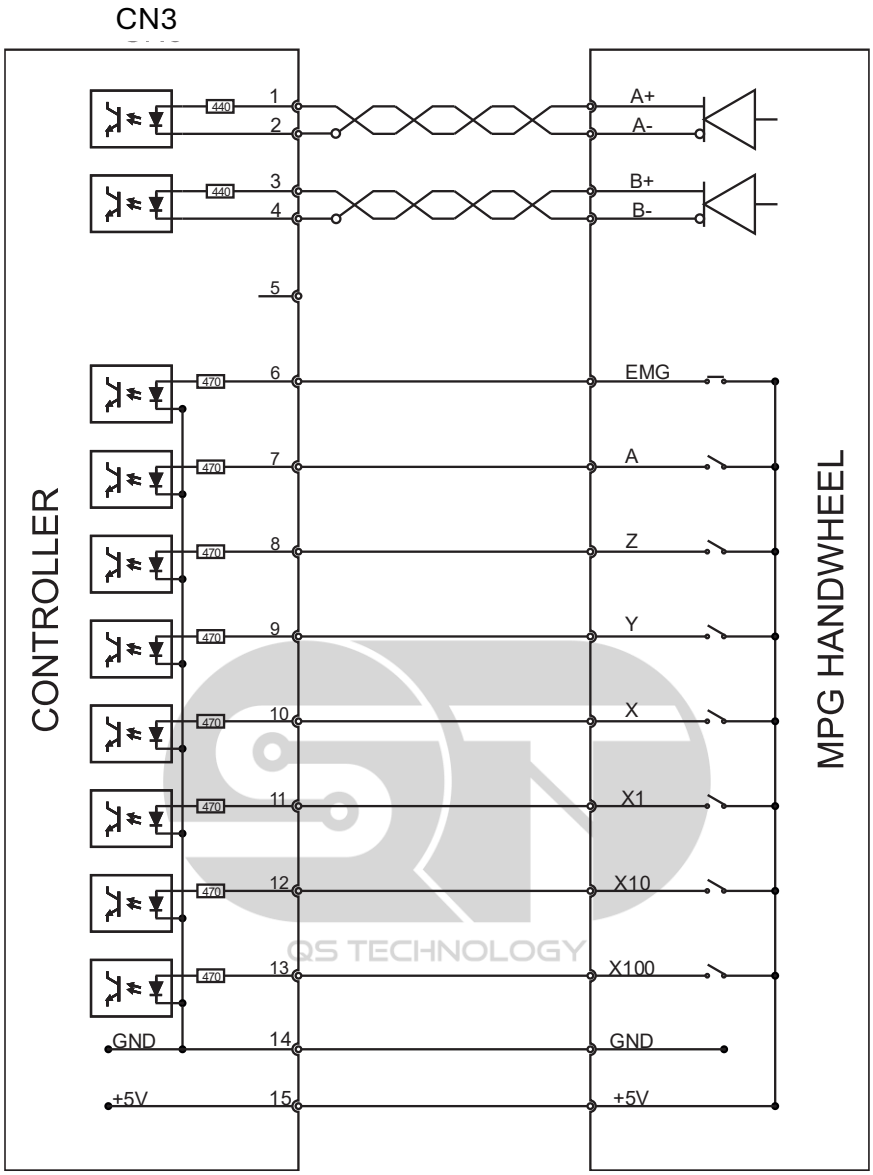


Figure 32. MPG wiring diagram (4-wire type – differential output signal)

2.3.5.2 MPG wiring diagram (6-wire type – differential output signal)

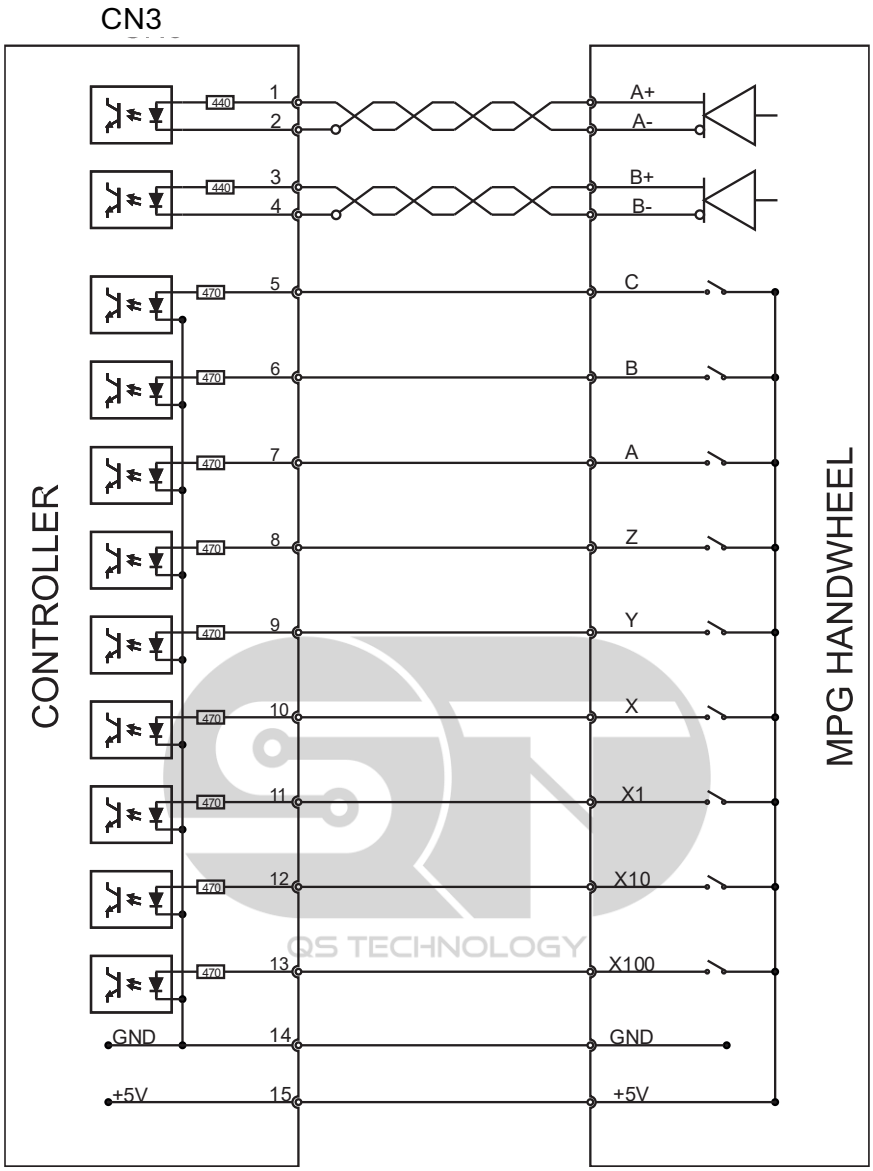


Figure 33. MPG wiring diagram (6-wire type – differential output signal)

2.3.5.3 MPG wiring diagram(4-wire type – NPN output signal)

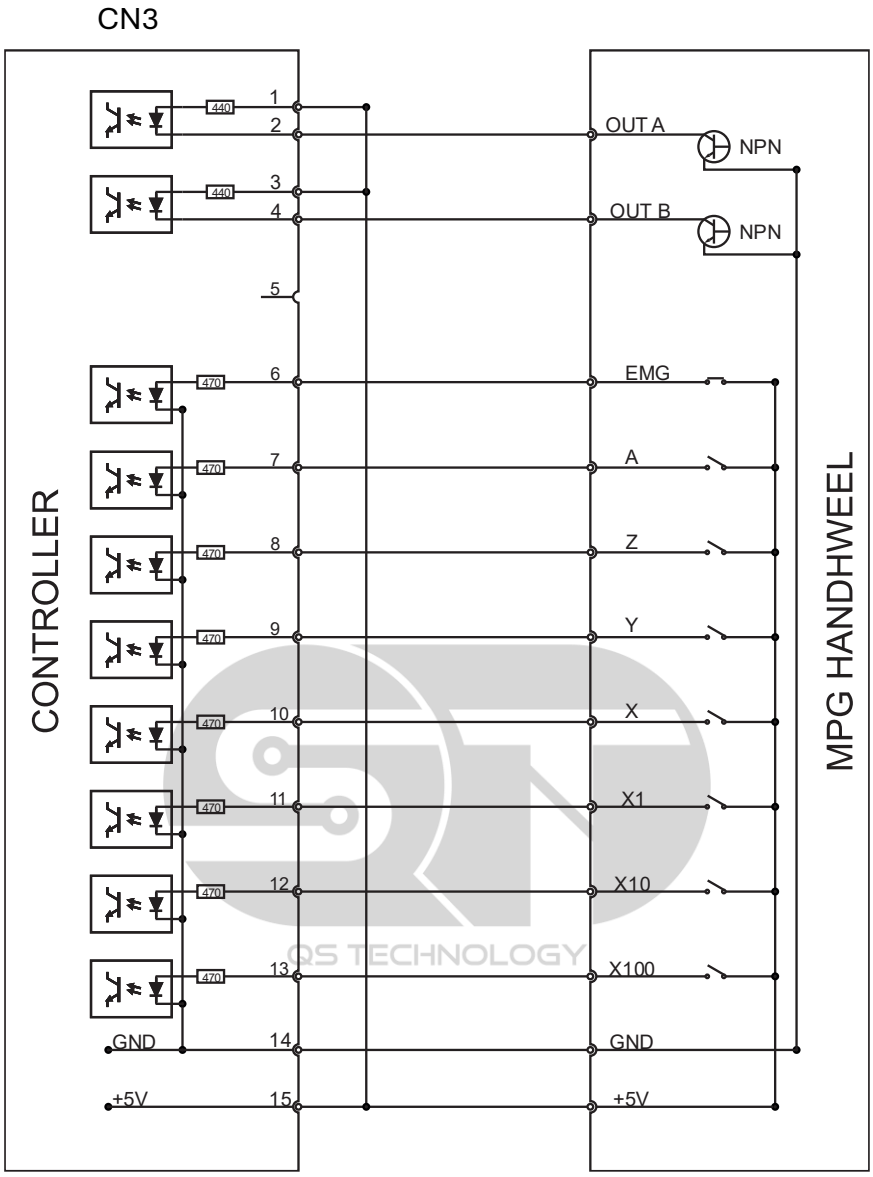


Figure 34. MPG wiring diagram (4-wire type – NPN output signal)

2.3.5.4 MPG wiring diagram (4-wire type – PNP output signal)

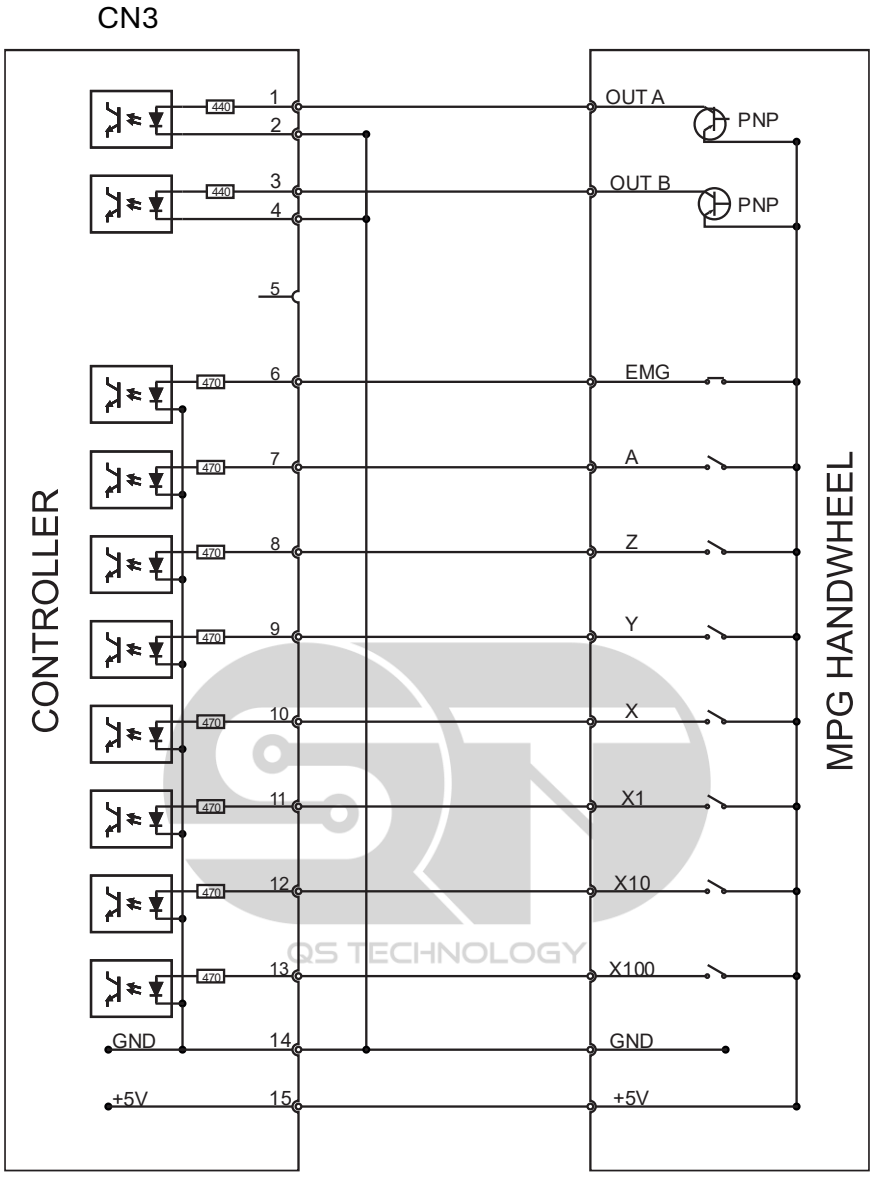


Figure 35. MPG wiring diagram (4-wire type – PNP signal)

2.3.5.5 MPG wiring diagram (6-wire – NPN output)

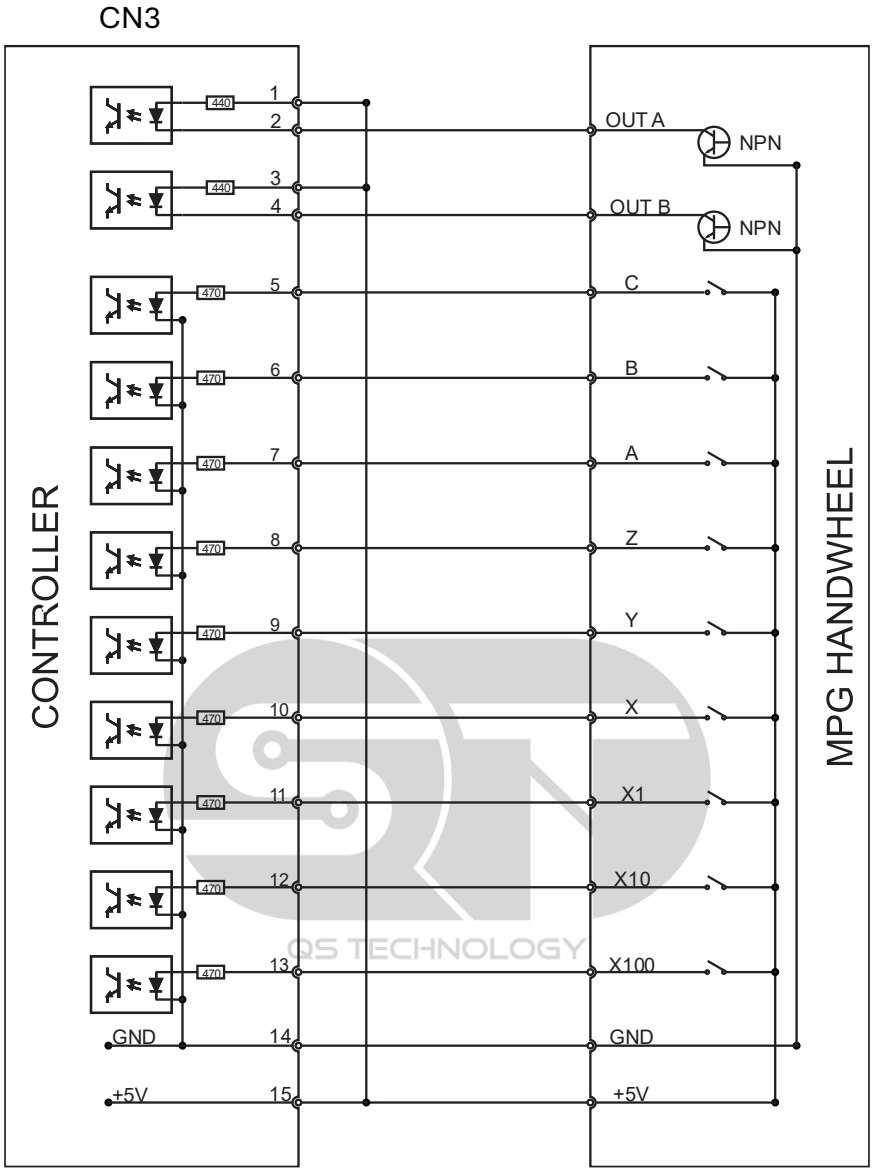


Figure 36. MPG wiring diagram (6-wire – NPN output signal)

2.3.5.6 MPG wiring diagram (6-wire – PNP output)

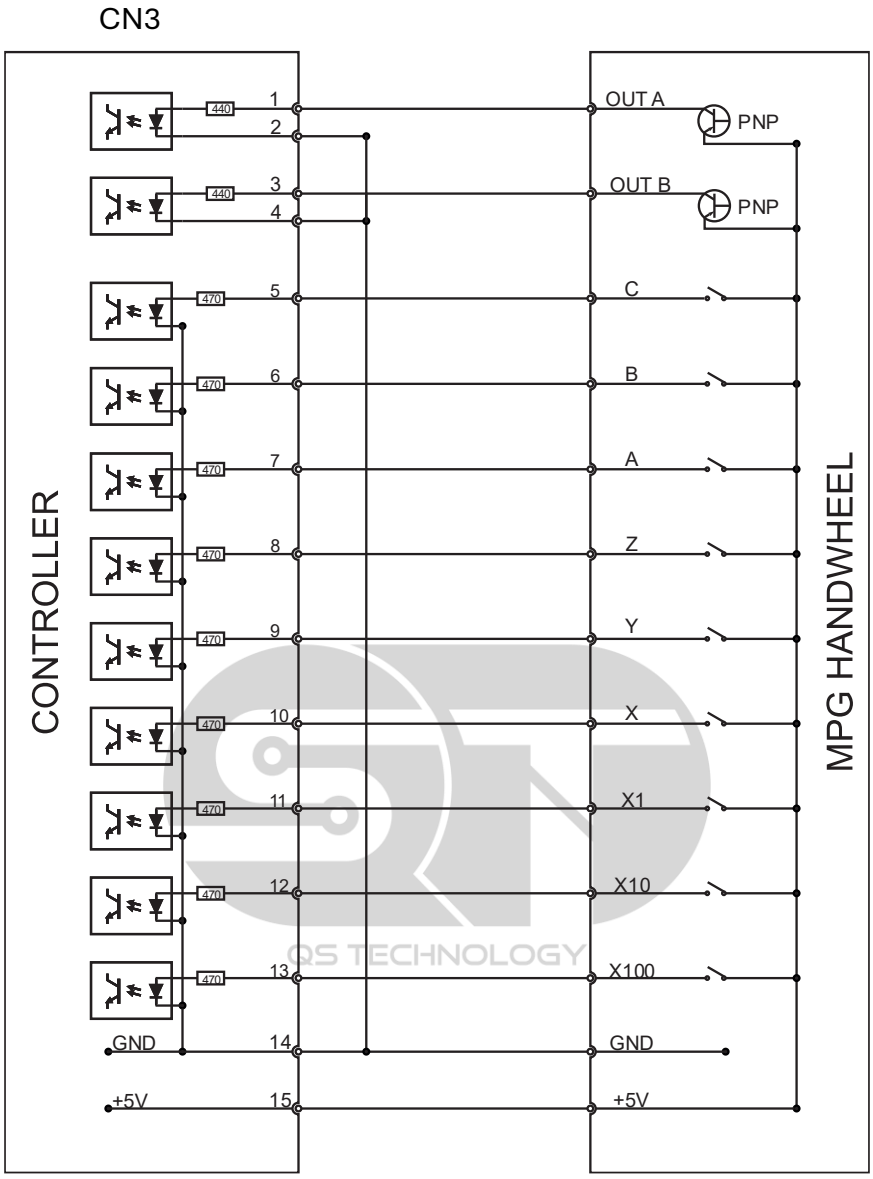


Figure 37. MPG wiring diagram (6-wire – PNP output signal)

2.3.5.7 MPG pendant with EMG button

Controller – CN3 Tay quay MPG

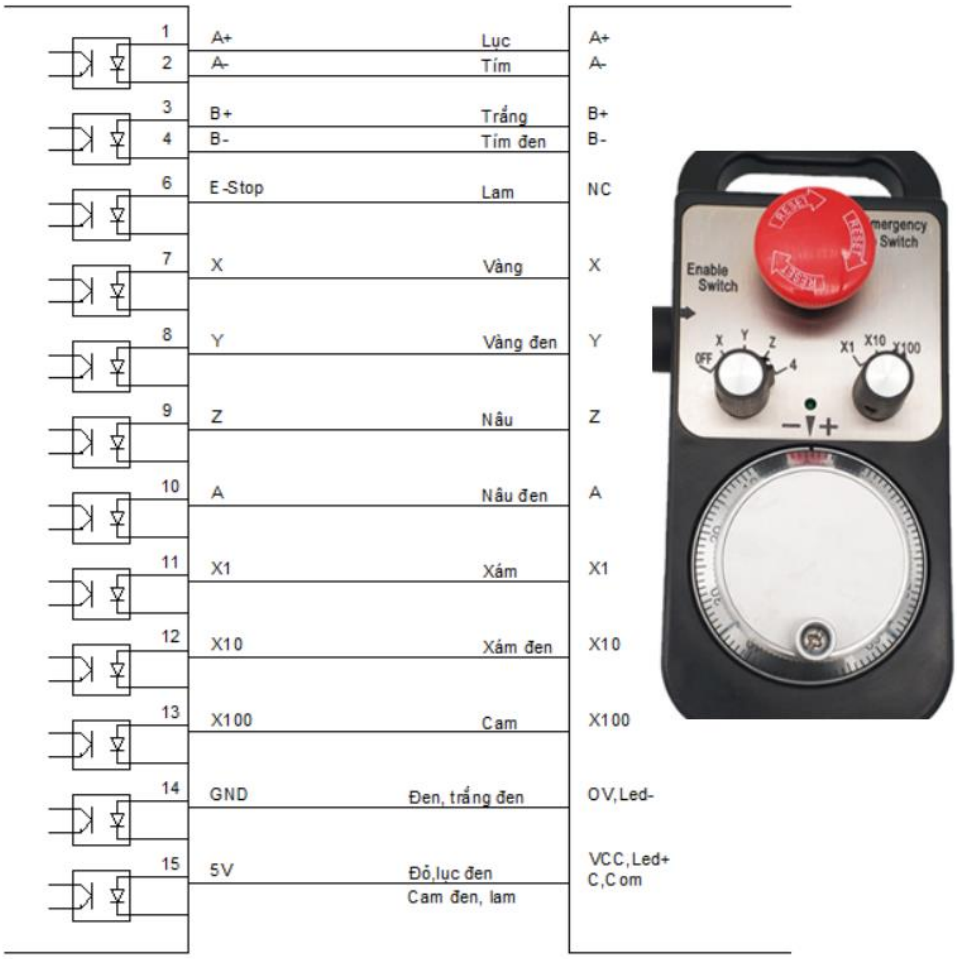


Figure 38. MPG wiring diagram (with EMG button) to F54 controller

Note: For MPG handles with an integrated EMG button, after wiring and connecting to the controller via the CN3 port, the installer must configure parameter M1963 to enable the EMG error detection function on the MPG handle

How to configure the parameter from 0 to 1:

- From the PARAMETER screen, use the arrows keys to select the parameter to configure.
- Then enter number 1 and press OK. A prompt will appear on the screen asking for the user password.
- Enter “1415” and press OK
- Next, enter number 1 again and press OK.

- Press RESET button to save the configured parameters.

Parameter stauts description:

- **M1963:** Enable/Disable alarm signal of EMG button on MPG pendant
  - M1963 = 0 : Disable
  - M1963= 1 : Enable

2.3.6 STARTbutton wiring diagram with NPN – PNP signal

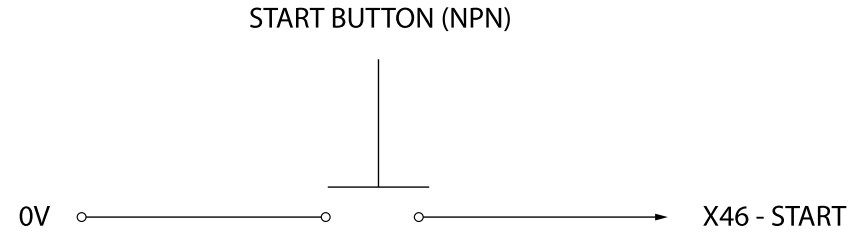


Figure 39. START button wiring diagram (NPN type)

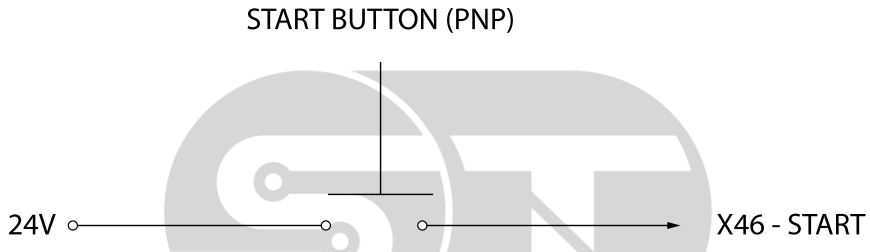


Figure 40. START button wiring diagram (PNP type)

2.3.7 STOP button wiring diagram with NPN – NPN signal

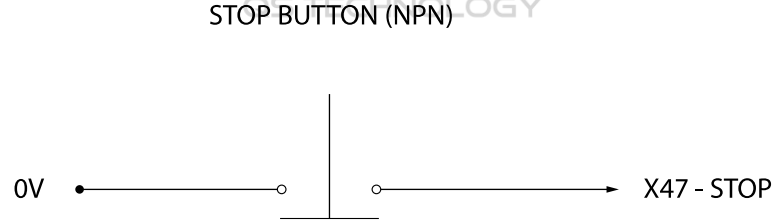


Figure 41. STOP button wiring diagram (NPN type)

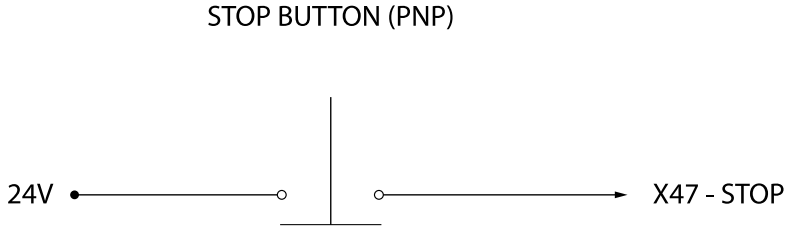


Figure 42. STOP button wiring diagram (PNP type)

2.3.8 EMG button wiring diagram (NPN – PNP signal)

2.3.8.1 EMG button wiring diagram (NPN type)

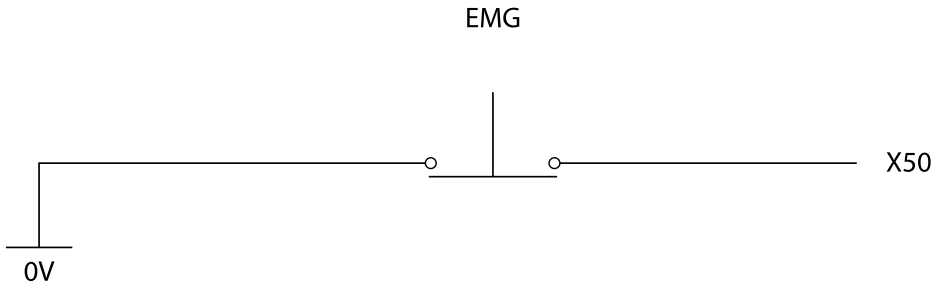


Figure 43. EMG button wiring diagram (NPN type)

2.3.8.2 EMG button wiring diagram (PNP type)

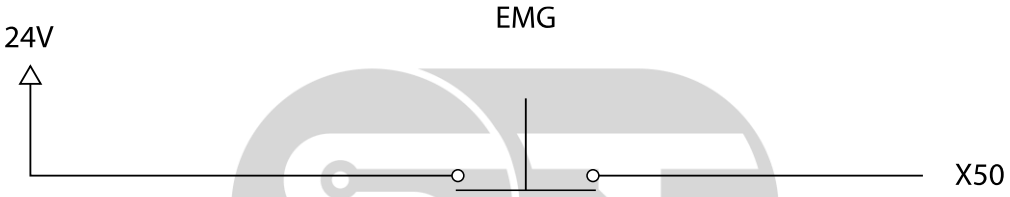


Figure 44. EMG button wiring diagram (PNP type)

2.4 Output signal wiring diagram

The functions of the output signals on the F54 controller are programmed in the controller's sample PLC ladder. The installer can modify the functions of these outputs by reprogramming the controller's PLC ladder.

Note: Before reprogramming the output signal functions, the installer should back up the sample PLC ladder program to allow for restoration if needed.

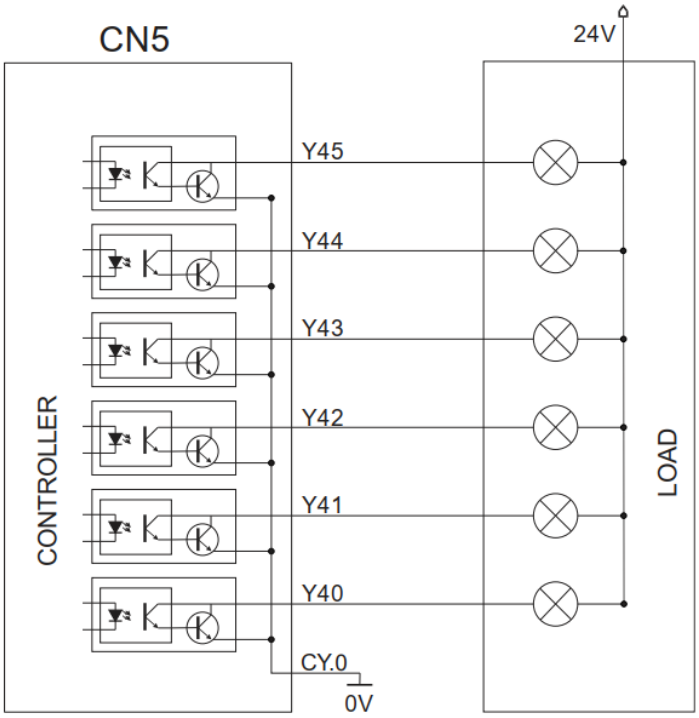


Figure 45. Output signal on F54 controller

Note:

- CY.0 signal pin must be connected to 0V
- The output signal of the F54 controller is a low-level signal (Low output – NPN)
- The output signal should not be directly connected to the load device. To ensure safety for the equipment, it is recommended that the installer connect the output signal to an intermediate relay before connecting it to the load device.

Output signal functions table

Add	Name	Function
Y40	S-ON	Điều khiển servo chạy
Y41	SP-FW	Điều khiển trục chính quay theo chiều thuận
Y42	M8	Bật motor nước tưới nguội
Y43	Air-Pro	Nở vale khí nén vệ sinh thiết bị so dao
Y44	Start light	Đèn báo trạng thái máy đang chạy chương trình
Y45	Alarm light	Đèn báo trạng thái máy đang có lỗi

2.4.1 Y40 output signal (S-ON)

Note: The following steps are an example performed with the Sigma V servo drive by Yaskawa. From the Y40 signal pin on the CN5 port of the controller, the installer will connect the Servo-ON signal through an intermediate relay.

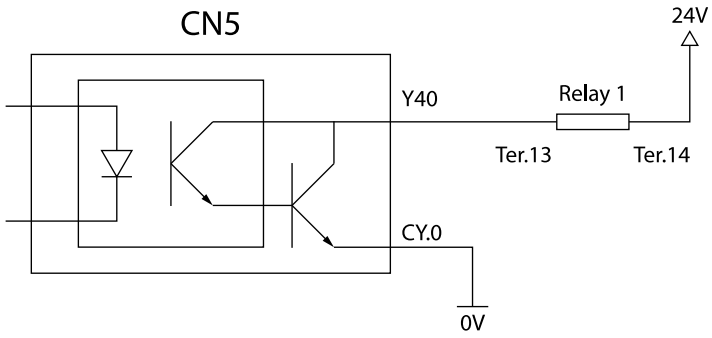


Figure 46. Y40 signal wiring diagram to relay 1(Ter.13 – Ter.14)

After connecting the Y40 signal pin to contacts 13 and 14 of the intermediate relay (Relay 1), continue by connecting 0V to contacts 9 and 5, and pin 40 on the SGDV – Yaskawa servo drive

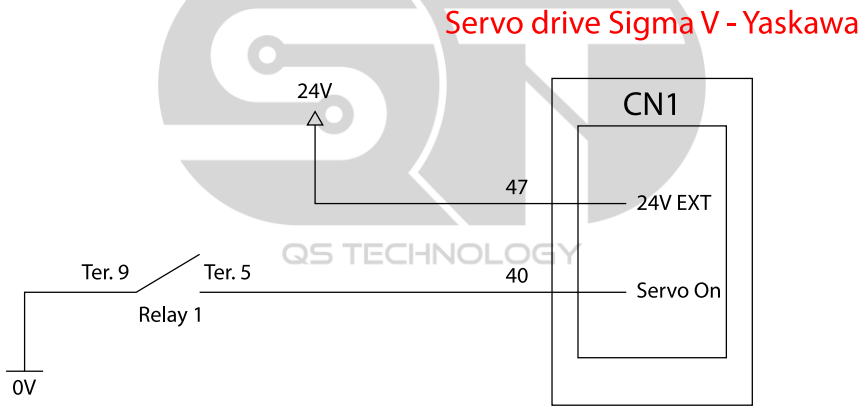


Figure 47. Ter.9 – Ter.5 terminal wiring diagram to Servo Drive Sigma V - Yaskawa

Note:

Ter.(5,9,13,14): Terminals on the intermediate relay

2.4.2 Y41 output signal (SP-FW)

Note: The following steps are an example performed with the INVT inverter.

From the Y41 signal pin on the CN5 port of the controller, the installer will connect the spindle control signal through an intermediate relay.

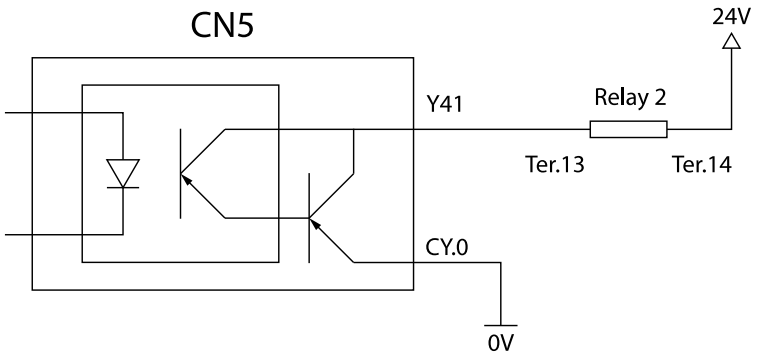


Figure 48. Y41 signal wiring diagram to relay 2(Ter.13 – Ter.14)

After connecting the Y41 signal pin to contacts 13 and 14 of the intermediate relay (Relay 2), continue by connecting contacts 9 and 5, and the S1 and COM pins on the INVT inverter. When the S1 and COM pins on the inverter are connected, the spindle will run in the forward direction

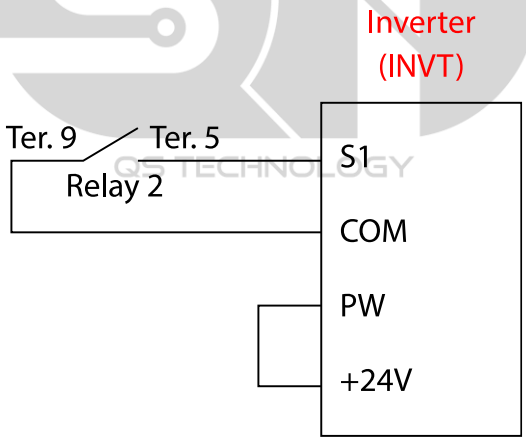


Figure 49. Ter.9 – Ter.5 terminal wiring diagram to INVT inverter

Note:

Ter.(5,9,13,14): Terminals on the intermediate relay

2.4.3 Y42 output signal (M8)

Note: The following steps are an example performed with a 220V-3 phase water pump motor

Từ chân tín hiệu Y42 trên cổng CN5 của bộ điều khiển, người lắp đặt sẽ đấu tín hiệu kích điều khiển trục chính thông qua 1 relay trung gian.

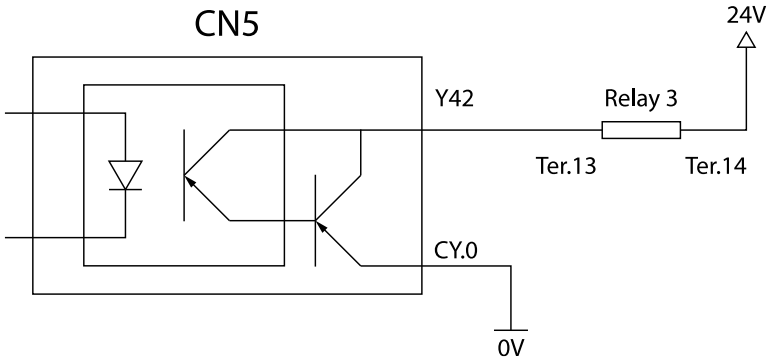


Figure 50. Y42 signal wiring diagram to relay 3(Ter.13 – Ter.14)

From the Y42 signal pin on the CN5 port of the controller, the installer will connect the spindle control signal through an intermediate relay

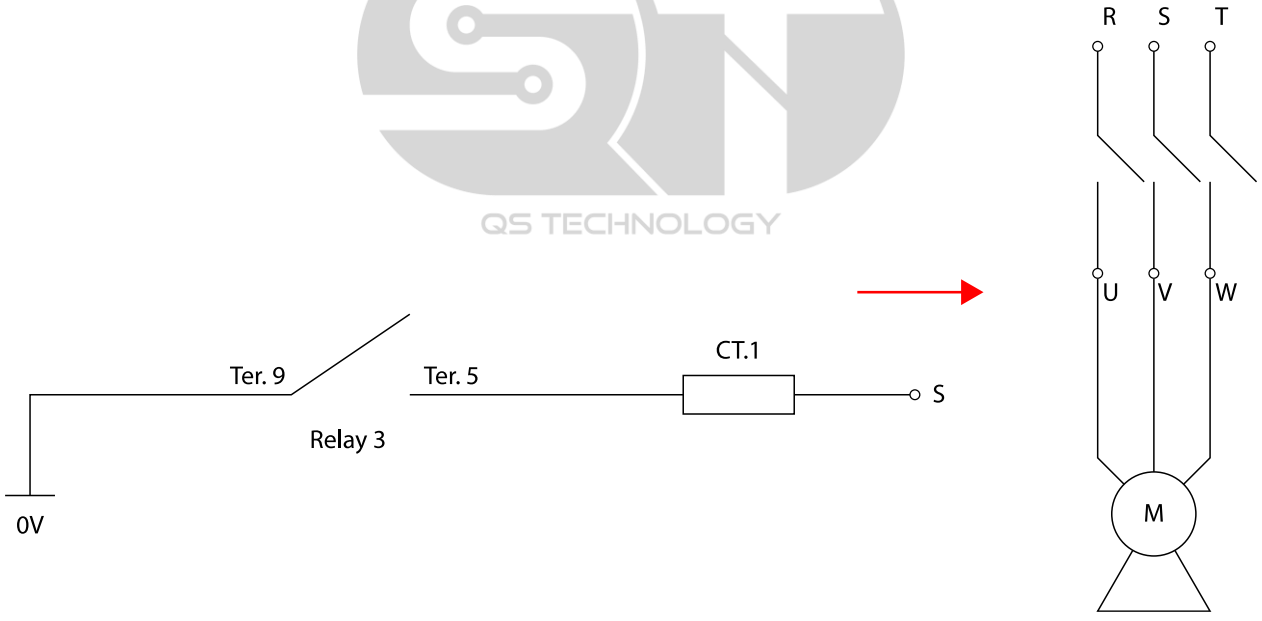


Figure 51. Ter.9 – Ter.5 terminal wiring diagram to control 3 phase motor

For a 1-phase 220V water pump motor, the installer will use the contacts on the intermediate relay to control the 1-phase 200V motor. Connect 0V to contacts 9 and 5 of the relay and then connect to the motor.

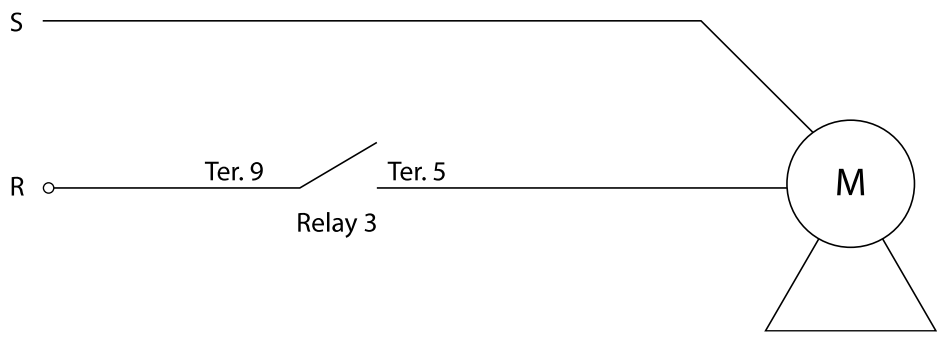


Figure 52. Ter.9 – Ter.5 terminal wiring diagram to control 1 phase - 220V motor

Note:

Ter.(5,9,13,14): Terminals on the intermediate relay

CT.1: Contactor 1

2.4.4 Y43 output signal (Air-Pro)

From the Y43 signal pin on the CN5 port of the controller, the installer will connect the spindle control signal through an intermediate relay..

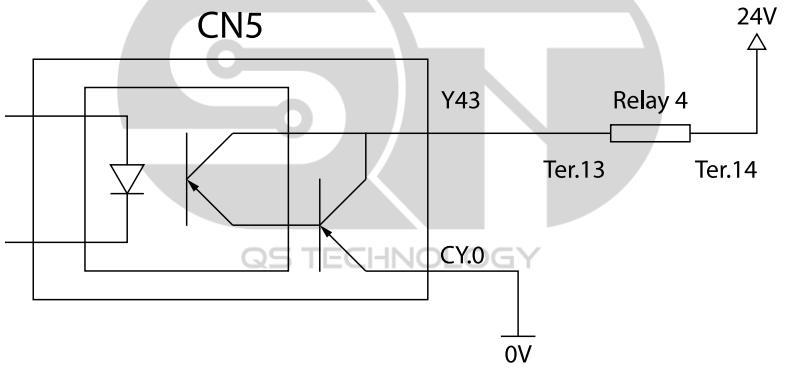


Figure 53. Y43 signal wiring diagram to relay 4(Ter.13 – Ter.14)

Connect 0V to contacts 9 and 5 of the relay and the compressed valve.

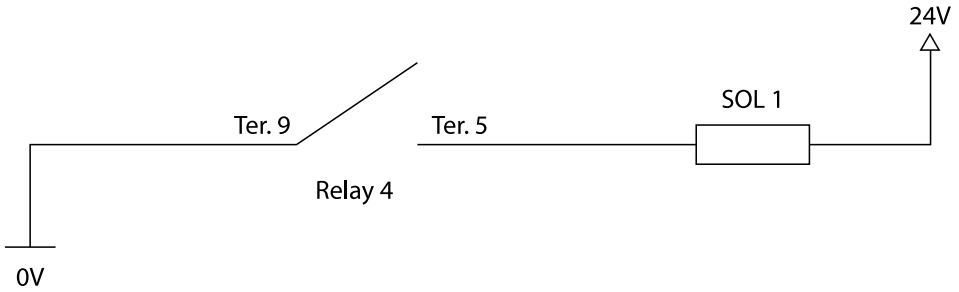


Figure 54. Ter.9 – Ter.5 terminal wiring diagram to control compressed valve

2.4.5 Y44 output signal (Start light)

From the Y44 signal pin on the CN5 port of the controller, the installer will connect the spindle control signal through an intermediate relay.

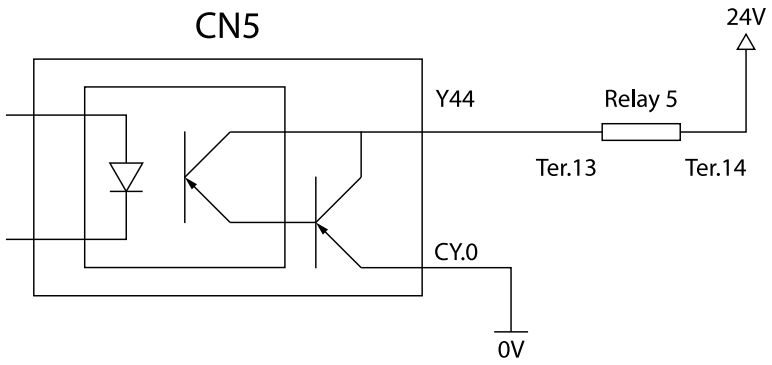


Figure 55. Y44 signal wiring diagram to relay 5(Ter.13 – Ter.14)

Connect the signal using DC voltage

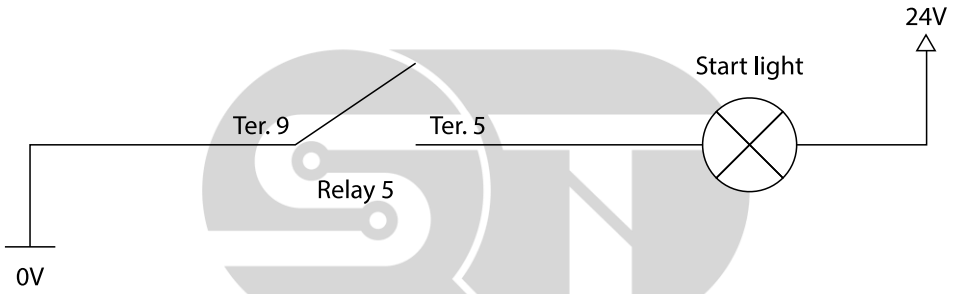


Figure 56 Ter.9 – Ter.5 terminal wiring diagram to control START light using DC voltage

QS TECHNOLOGY

Connect the signal using AC voltage

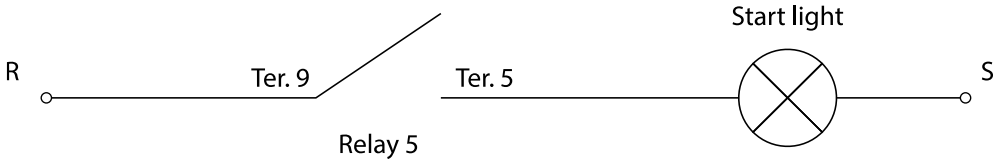


Figure 57. Ter.9 – Ter.5 terminal wiring diagram to control START light using AC voltage

2.4.6 Y45 output signal (Alarm light)

From the Y45 signal pin on the CN5 port of the controller, the installer will connect the spindle control signal through an intermediate relay.

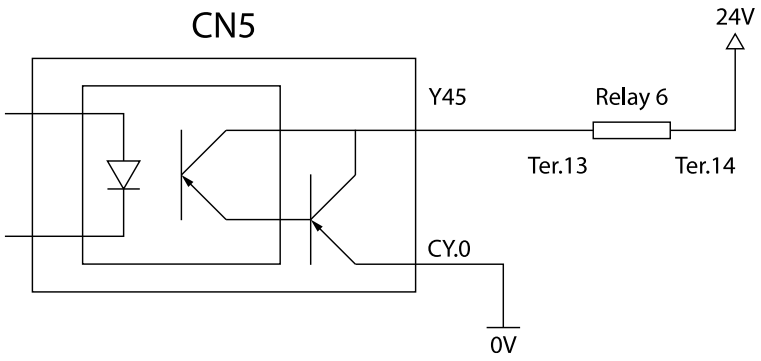


Figure 58. Y45 signal wiring diagram to relay6(Ter.13 – Ter.14)

Connect the signal using DC voltage

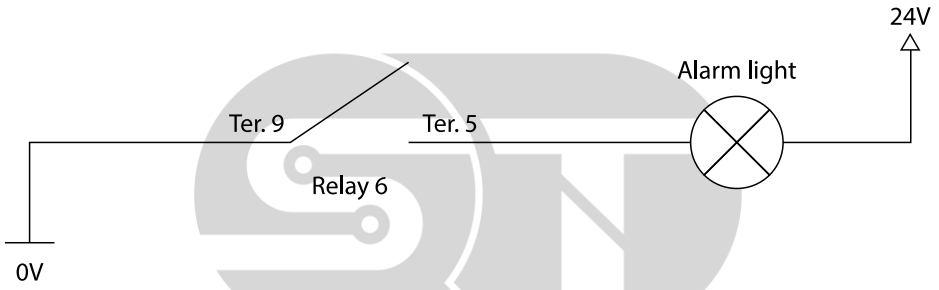


Figure 59. Ter.9 – Ter.5 terminal wiring diagram to control ALARM light using DC voltage

QS TECHNOLOGY

Connect the signal using AC voltage

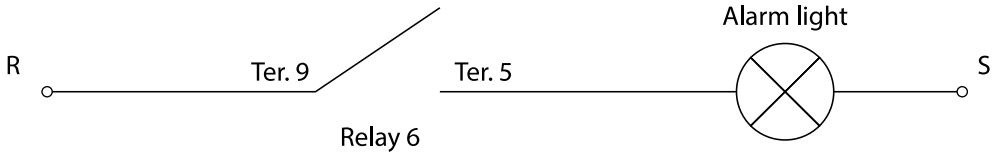


Figure 60. Ter.9 – Ter.5 terminal wiring diagram to control ALARM light using AC voltage

2.4.7 Analog signal wiring diagram for the inverter

Note: The purpose of wiring the analog signal from the controller to the inverter is to control the spindle speed. Before wiring, the installer must identify the type of inverter, its model, and the corresponding user manual.

Note: The following diagrams are examples based on the INVT GD350A inverter

Reference documentation: INVT – GD350A inverter wiring diagram

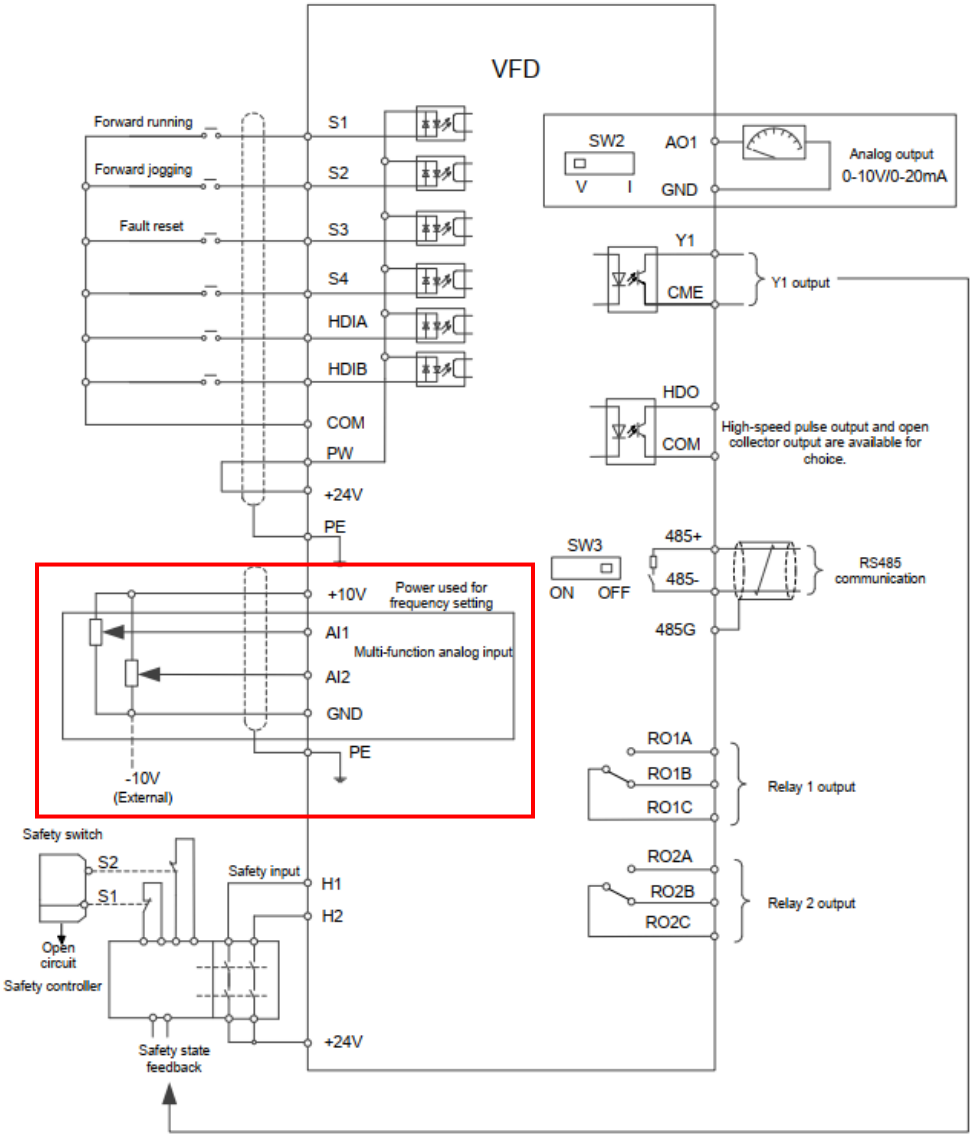


Figure 61. Analog signal wiring diagram of INVT-GD350A inverter

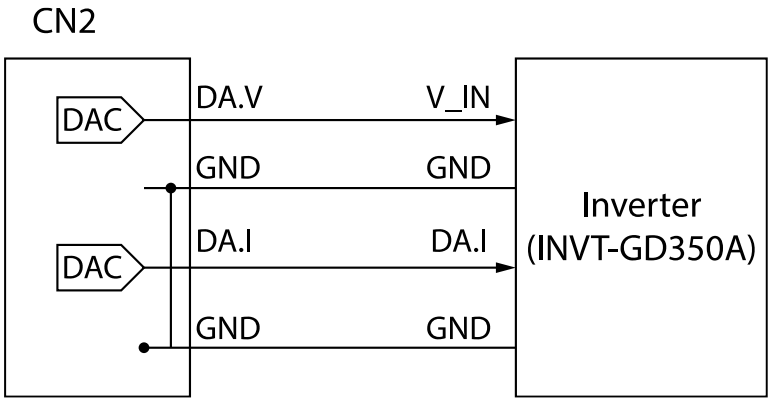


Figure 62. Analog signal wiring diagram from the F54 controller to the inverter for spindle speed control

2.4.7.1 Analog signal connection diagram to the inverter using 0–10V voltage

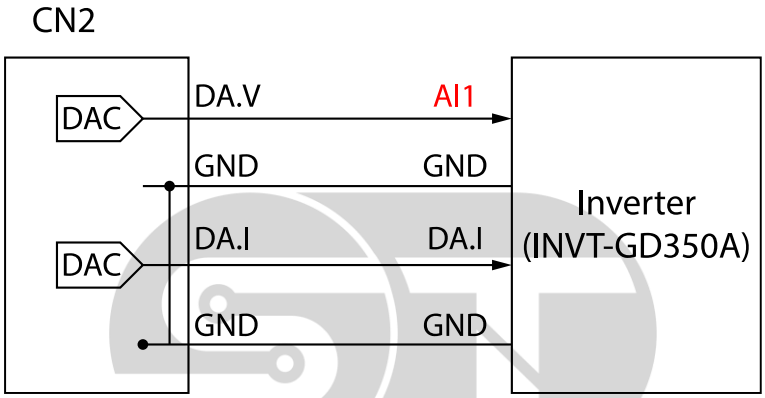


Figure 63. Analog signal wiring diagram to the inverter using 0–10V voltage

2.4.7.2 Analog signal connection diagram to the inverter using 0–20mA current

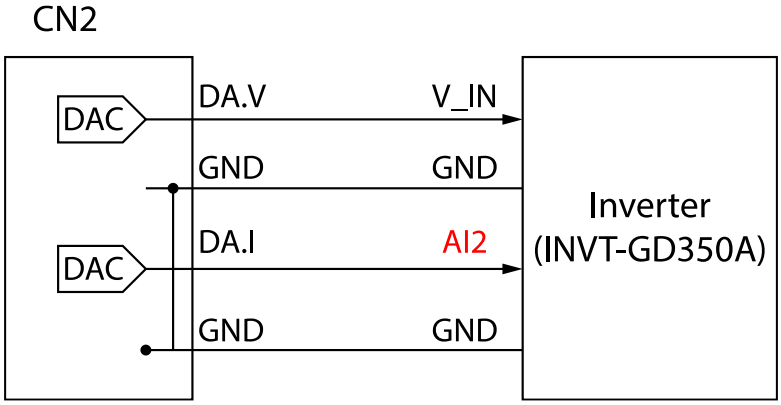


Figure 64. Analog signal wiring diagram to the inverter using 0–20mA current

2.5 I/O link board connection

The connection of an additional I/O link board will provide the controller with more input/output signal ports, allowing the controller to manage more devices

2.5.1 I/O link 32_V1.1_0722 board wiring diagram

Specification :

- Model: I/O link 32_V1.1_0722
- Input port: 32
- Output port: 32
- COM port: 8
- RS485 port: 2
- Voltage input: 12 – 24VDC

Note: I/O link/RS485 cable (recommend to use shield wire type to against noise signal)

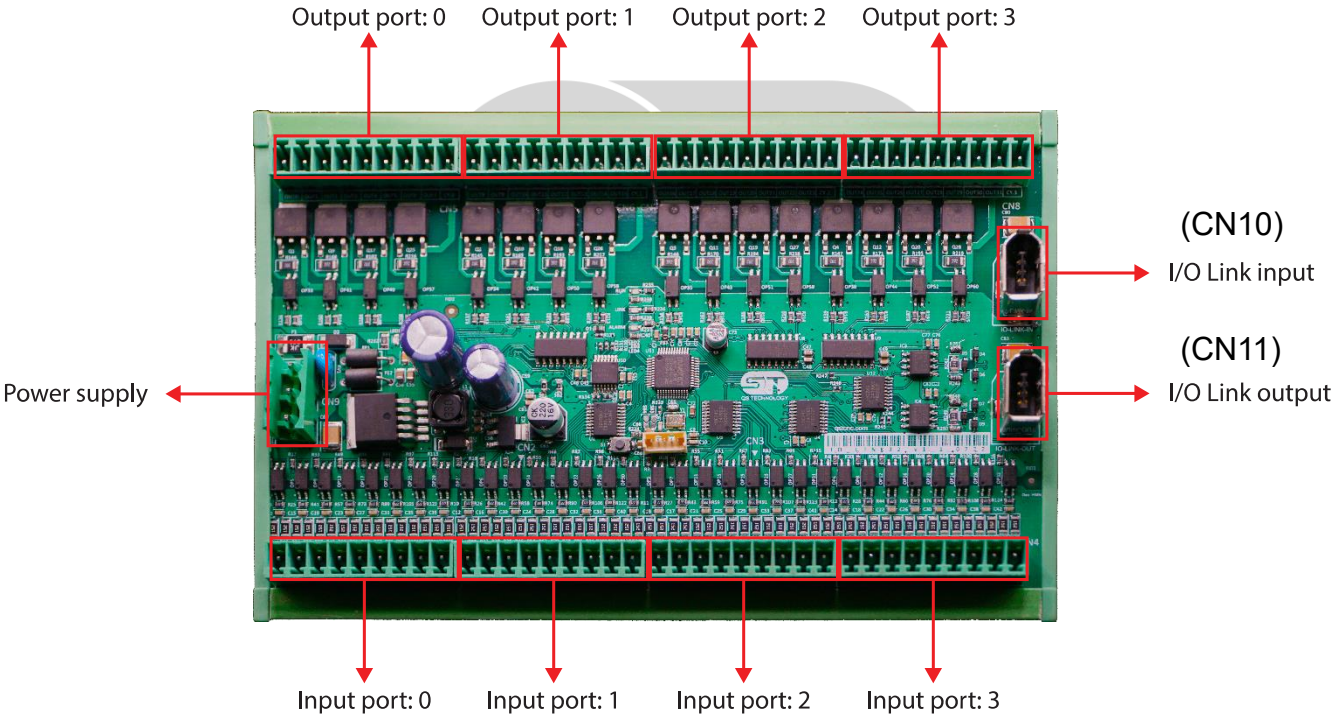


Figure 65. Board I/O link 32_V1.1_0722

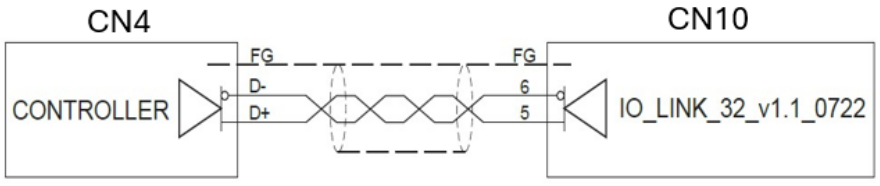
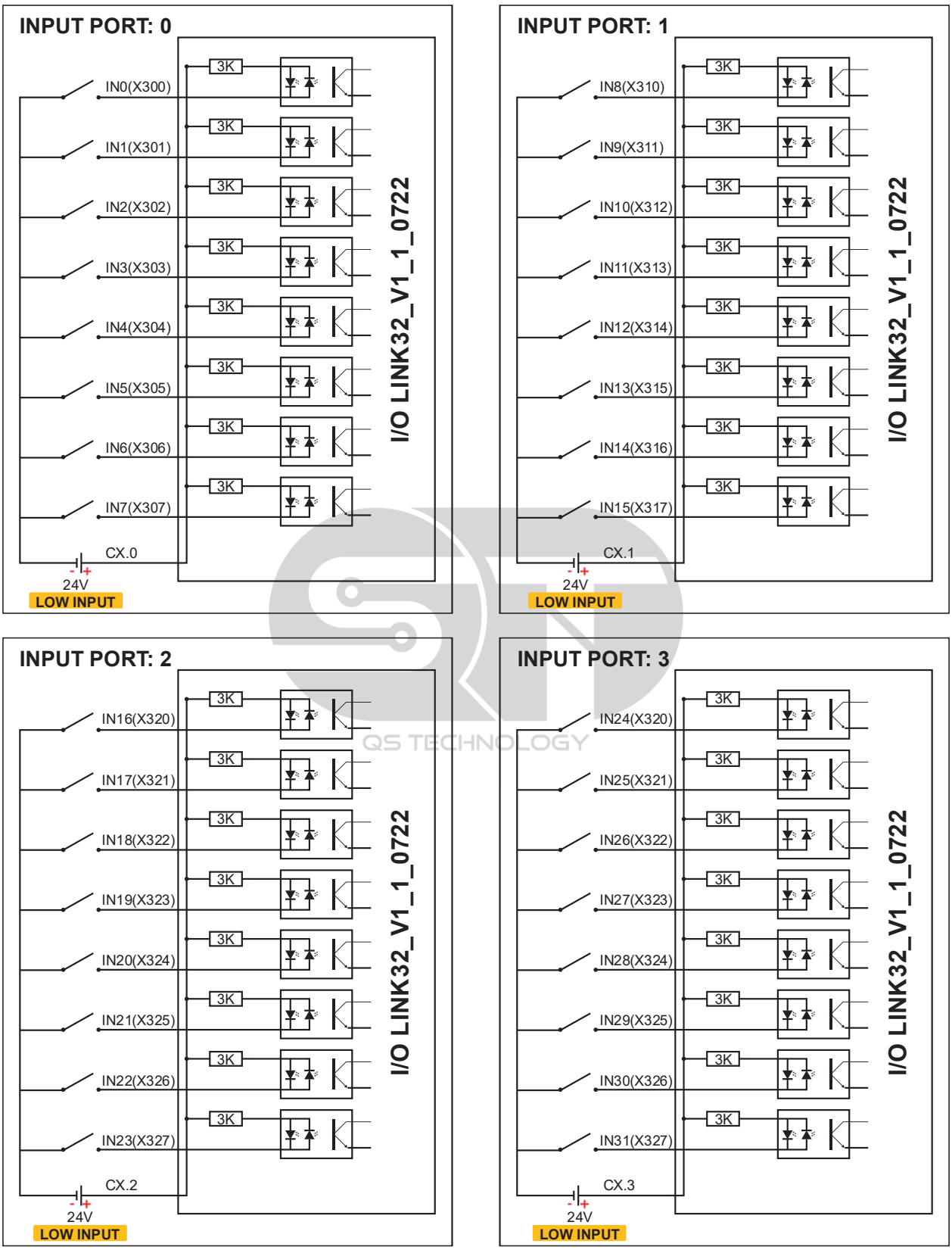


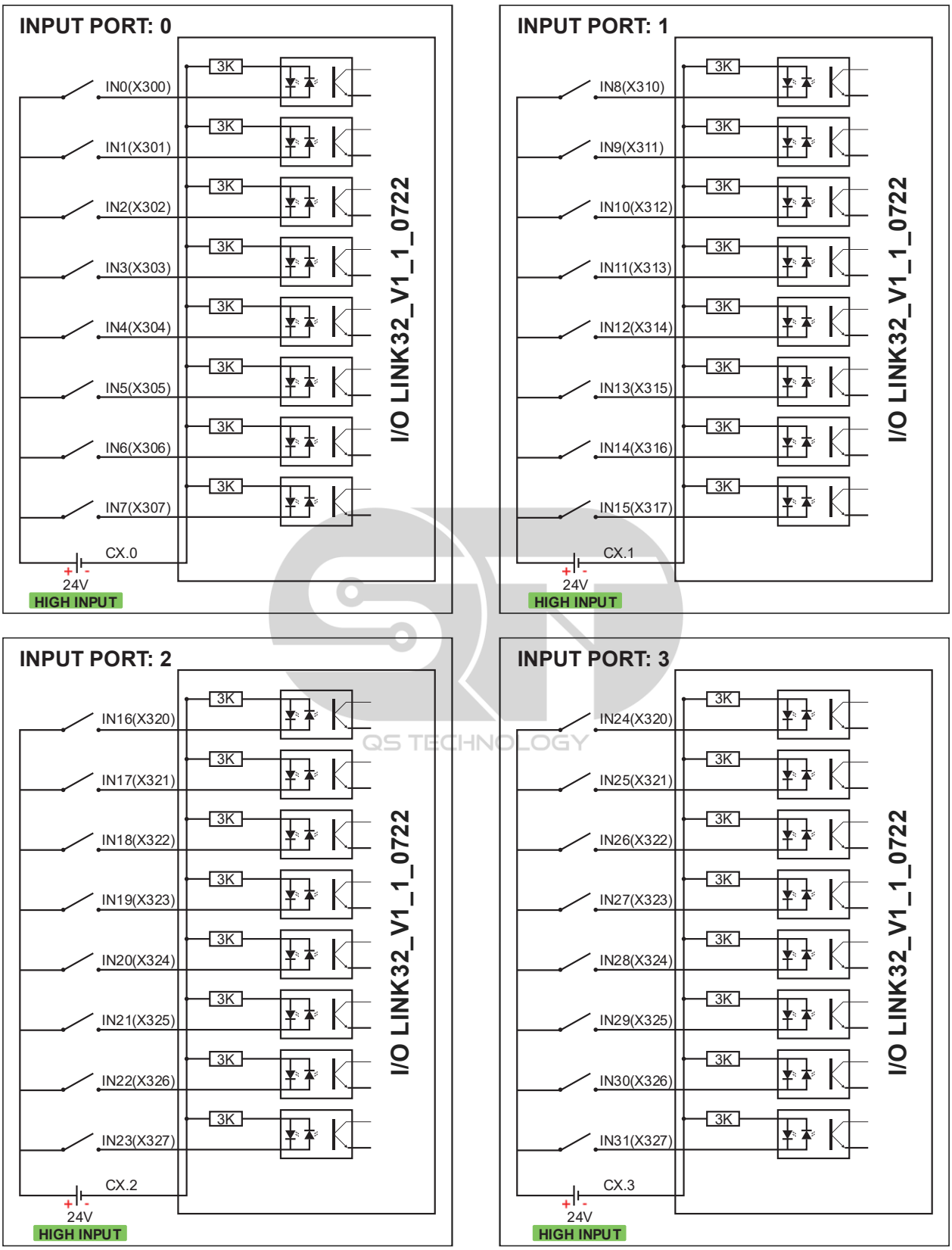
Figure 66. I/O link 32_V1.1_0722 wiring diagram to F54 controller

2.5.1.1 I/O Link 32_V1.1_0722 board wiring diagram (LOW INPUT)

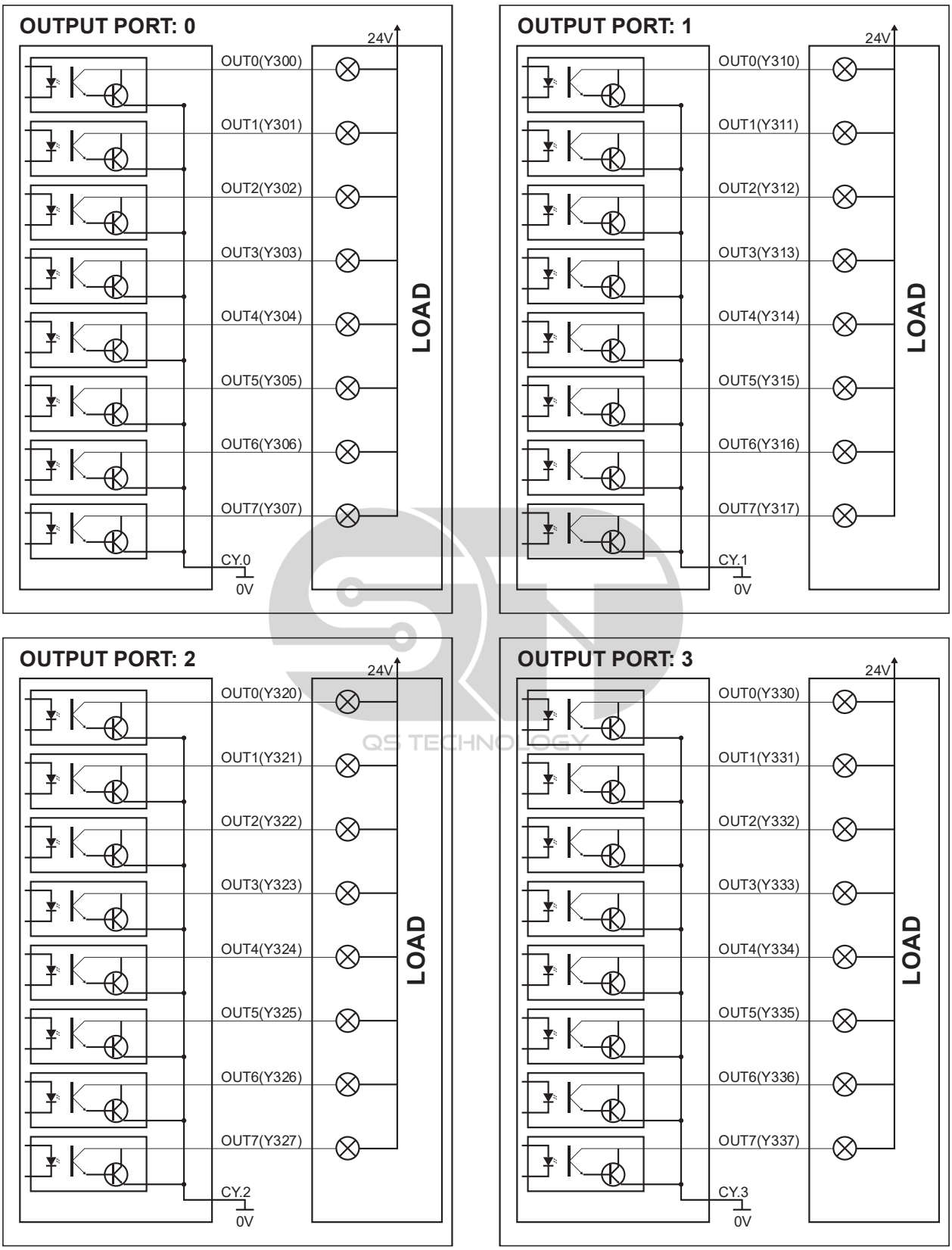


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2.5.1.2 Link 32_V1.1_0722 board wiring diagram (HIGH INPUT)



2.5.1.3 I/O link 32_V1.1_0722 board output signal



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2.5.2 I/O Link 08_RLTR board wiring diagram

Specification:

- Model: I/O link 08_RLTR
- Input port: 8
- Output port: 8
- COM port: 3 (Input: 1 – Output: 2)
- RS485 port: 1
- Voltage input: 12 – 24VDC

Note: I/O link/RS485 cable (recommend to use shield wire type to against noise signal)

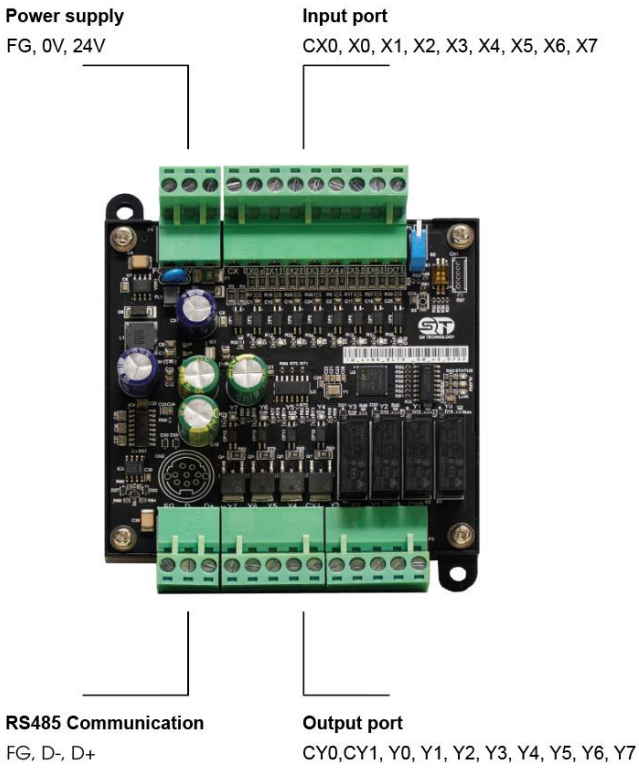


Figure 67. Board I/O link 08_RLTR

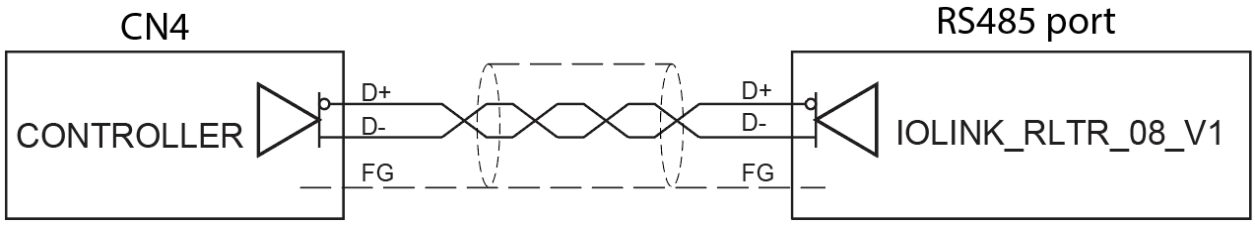


Figure 68. I/O link 08_RLTR board wiring diagram to F54 controller

2.5.2.1 I/O link 08_RLTR board wiring diagram (LOW INPUT)

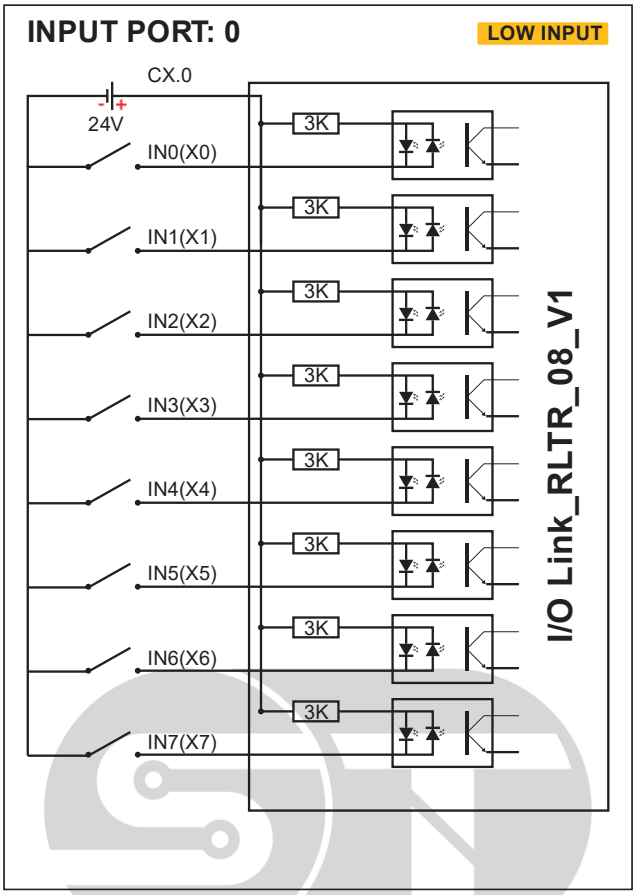


Figure 69. I/O Link 08_RLTR low input wiring diagram

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2.5.2.2 I/O link 08_RLTR board wiring diagram (HIGH INPUT)

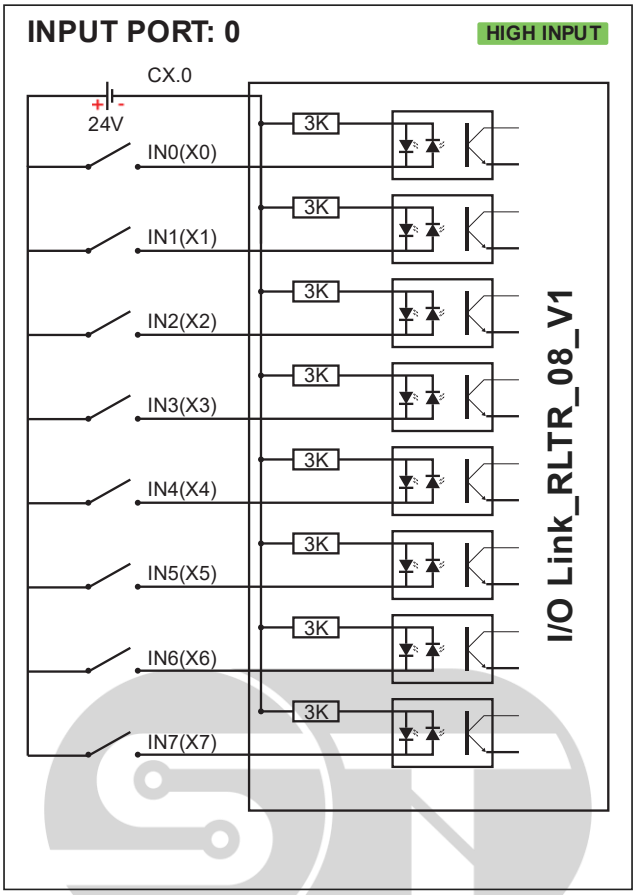


Figure 70. I/O Link 08_RLTR high input wiring diagram

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2.5.2.3 I/O link 08_RLTR board output signal

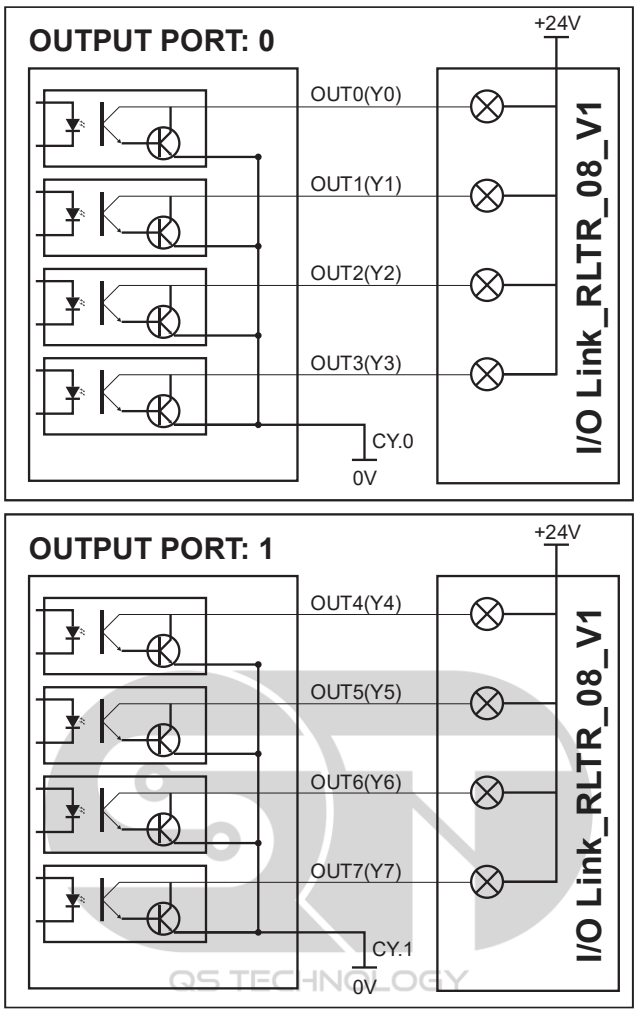


Figure 71. I/O link 08_RLTR output board

2.6 Axis connection (X,Y,Z,A)

Note: Before connecting the servo to the moving axes, the installer must identify the type of servo, model, and the corresponding product documentation.

Note(1): The servo control cable must be of the type with a shielded jacket and should be welded to the connector's casing (refer to the diagram below).



Figure 72. Shielded cable type, soldered to the connector shell for against noise

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Reference documentation: Pul/Dir wiring diagram of Servopack SGDV Sigma V Yaskawa

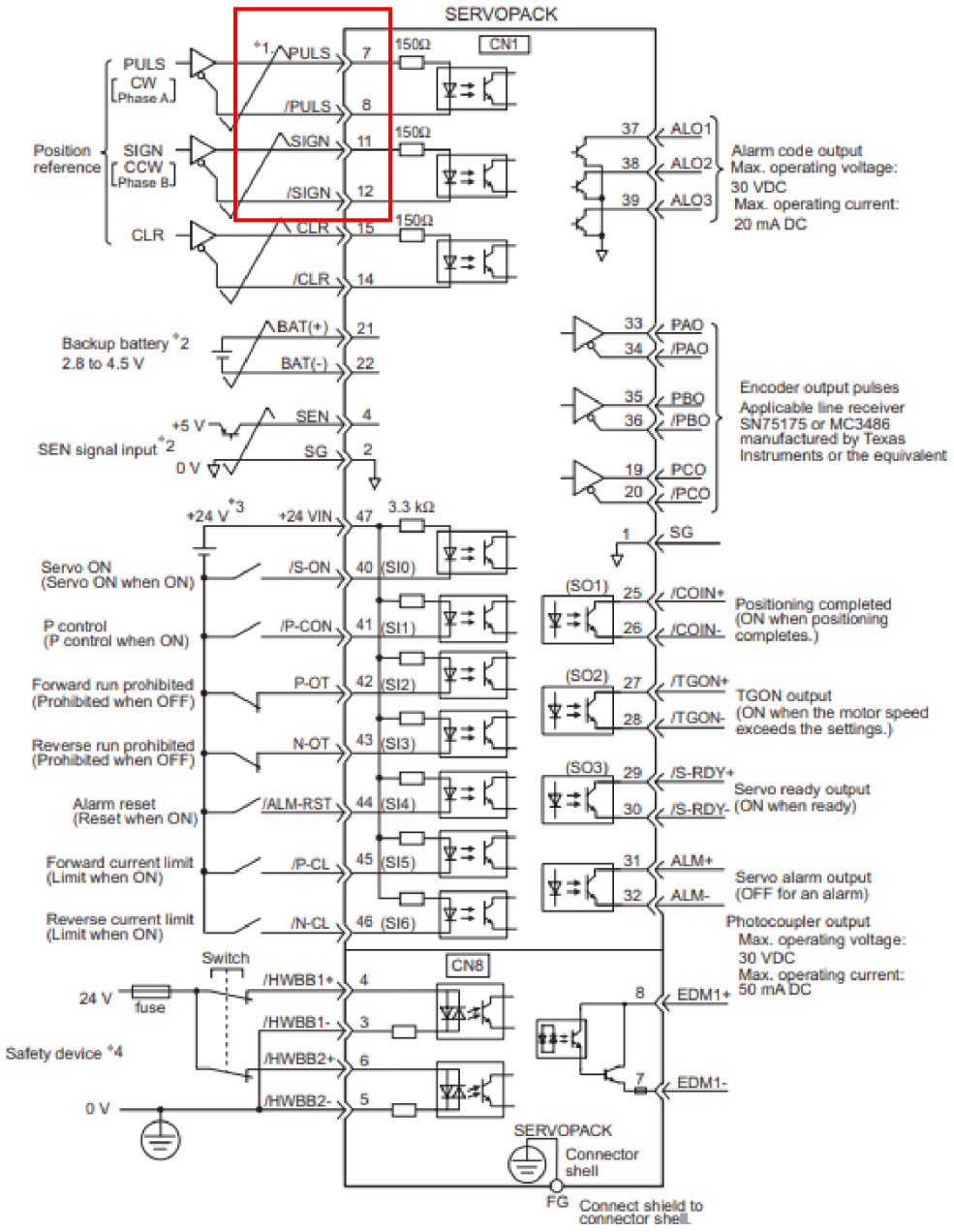


Figure 73. Pul/Dir wiring diagram on sevo drive SGDV Sigma V - Yaskawa

2.6.1 X-Y axis wiring diagram (Pul/Dir)

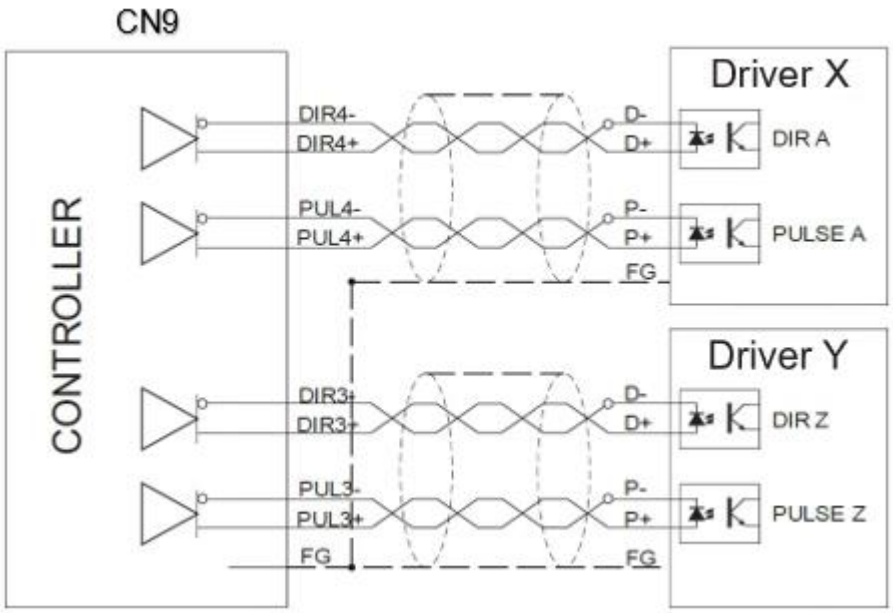


Figure 74. X,Y Pulse/Direction wiring diagram

2.6.2 Z-A axis wiring diagram (Pul/Dir)

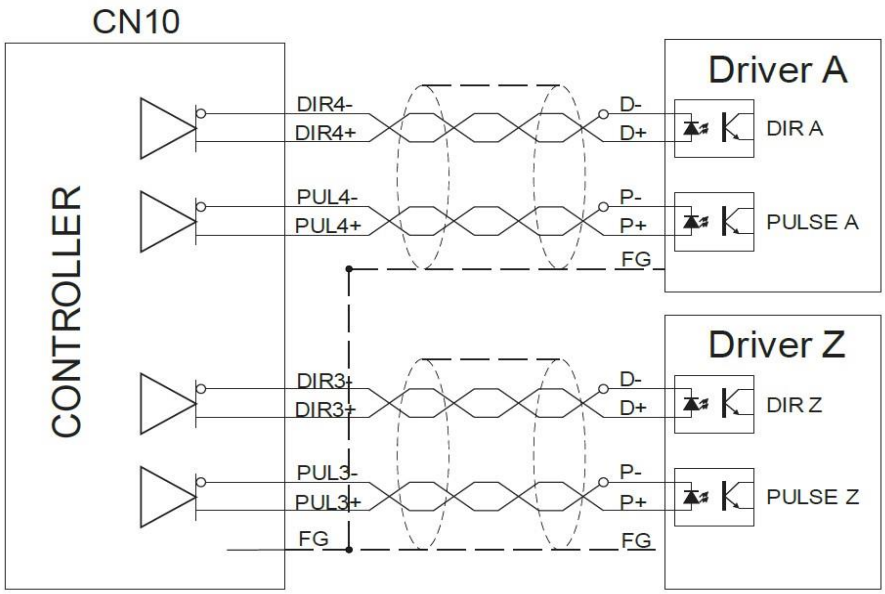


Figure 75. Z,A Pulse/Direction wiring diagram

2.6.3 Encoder feedback signal for phase Z from the servo drive of axes (X, Y, Z, A)

Note: Before connecting the encoder feedback signals for the motion axes, the installer must identify the servo type, model, and the corresponding product documentation.

Reference documentation: Encoder feedback signal wiring diagram for SGDV Sigma V servo drive – Yaskawa

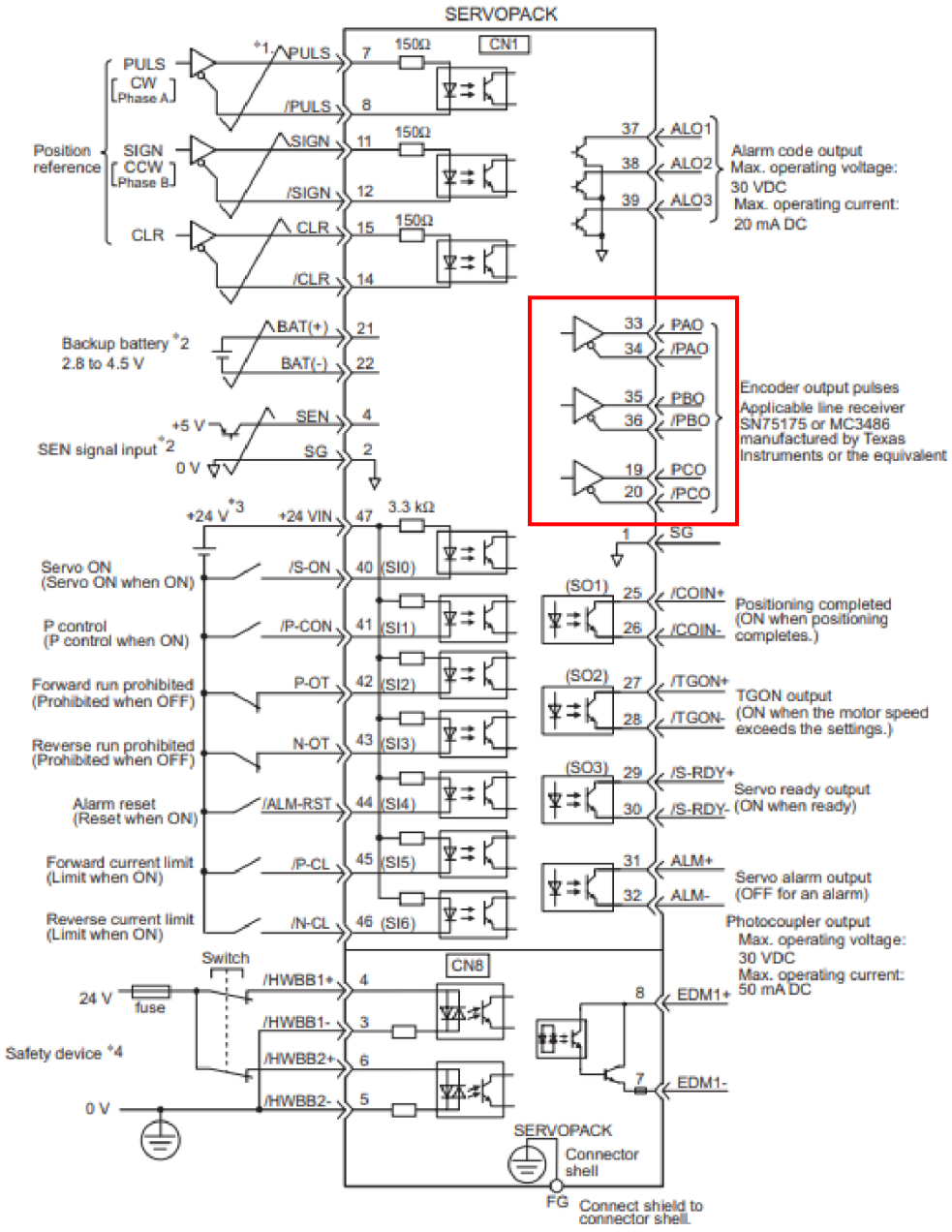


Figure 76. Encoder feedback wiring diagram on sevo drive SGDV Sigma V - Yaskawa

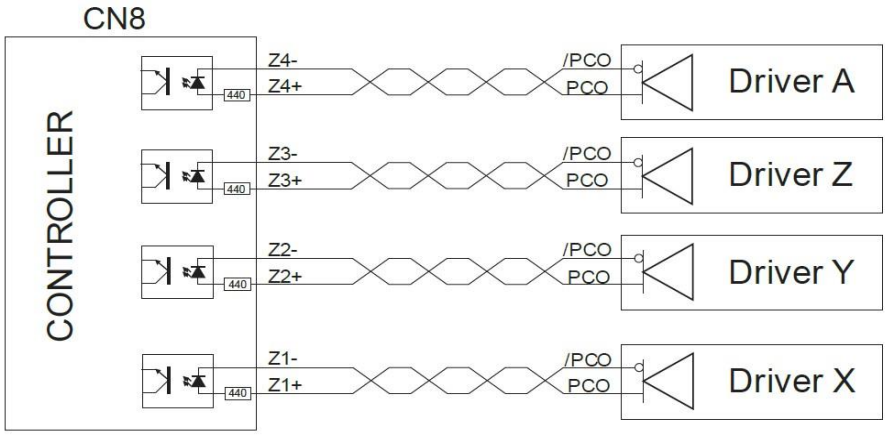


Figure 77. Z phase encoder feedback wiring diagram for axes (X,Y,Z,A)

2.6.4 Axis configuration

2.6.4.1 Enable/Disabe axis (X,Y,Z,A)

Note: After wiring the servo drives for axes X, Y, Z, and A, the installer must configure the following parameters to enable axis motion: AX1.ENA, AX2.ENA, AX3.ENA, AX4.ENA

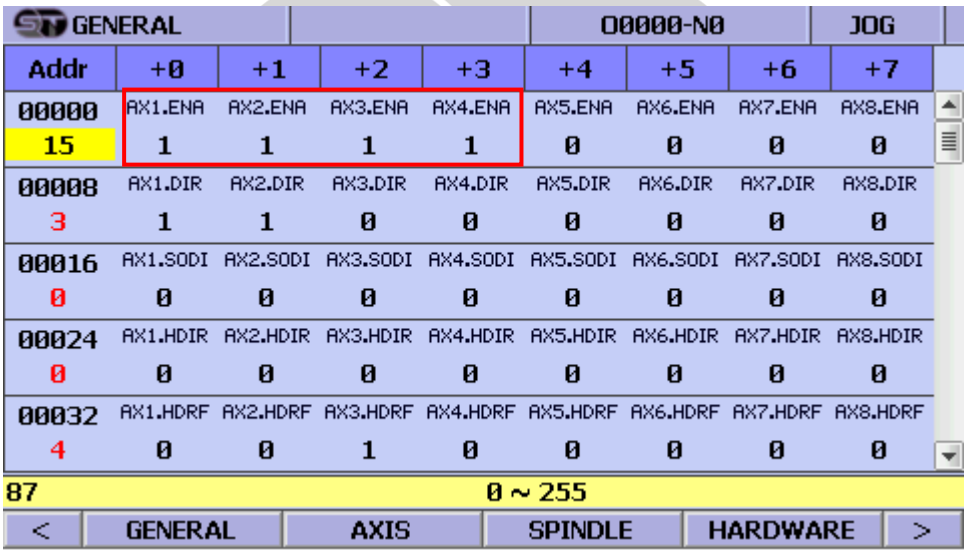


Figure 78. Axis enable/disable parameters

How to configure the parameter from 0 to 1:

- From the PARAMETER screen, use the arrows keys to select the parameter to configure.
- Then enter number 1 and press OK. A prompt will appear on the screen asking for the user password.
- Enter “1415” and press OK
- Next, enter number 1 again and press OK

- Press RESET button to save the configured parameters

Parameter status discription:

- **AX1.ENA:** X axis enable/disable
  - AX1.ENA = 0 : Axis disable
  - AX1.ENA = 1 : Axis enable
- **AX2.ENA:** Y axis enable/disable
  - AX2.ENA = 0 : Axis disable
  - AX2.ENA = 1 : Axis enable
- **AX3.ENA:** Z axis enable/disable
  - AX3.ENA = 0 : Axis disable
  - hAX3.ENA = 1 : Axis enable
- **AX4.ENA:** A axis enable/disable
  - AX4.ENA = 0 : A axis disable
  - AX4.ENA = 1 : A axis enable

2.6.4.2 Motor rotation direction for axes (X,Y,Z,A)

Note: To set the motor rotation direction for the axes (X, Y, Z, A), the installer needs to configure the following parameters: AX1.DIR, AX2.DIR, AX3.DIR, AX4.DIR

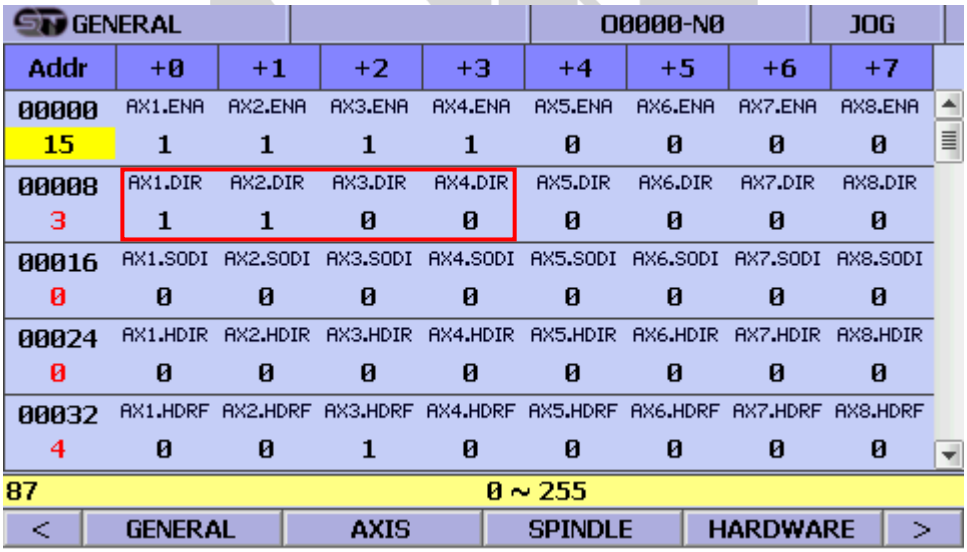


Figure 79. Thông số cài đặt chiều động cơ của trục (X,Y,Z,A)

How to configure the parameter from 0 to 1:

- From the PARAMETER screen, use the arrows keys to select the parameter to configure.
- Then enter number 1 and press OK. A prompt will appear on the screen asking for the user password.

- Enter “1415” and press OK
- Next, enter number 1 again and press OK
- Press RESET button to save the configured parameters

Parameter status description:

- **AX1.DIR:** Motor rotation direction of X axis
  - AX1.DIR= 0 : Motor rotates in the positive direction (X axis)
  - AX1.DIR = 1 : Motor rotates in the negative direction (X axis)
- **AX2.DIR:** Thông số cài đặt chiều động cơ trục Y
  - AX2.DIR= 0 : Motor rotates in the positive direction (Y axis)
  - AX2.DIR = 1 : Motor rotates in the negative direction (Y axis)
- **AX3.DIR:** Thông số cài đặt chiều động cơ trục Z
  - AX3.DIR = 0 : Motor rotates in the positive direction (Z axis)
  - AX3.DIR = 1 : Motor rotates in the negative direction (Z axis)
- **AX4.DIR:** Thông số cài đặt chiều động cơ trục A
  - AX4.DIR = 0 : Motor rotates in the positive direction (A axis)
  - AX4.DIR = 1 : Motor rotates in the negative direction (A axis)

2.6.4.3 Software limit for axes (X,Y,Z,A)

Note: To set the software limits for the axes (X, Y, Z, A), the installer needs to configure the following parameters: AXI1.SODI, AXI2.SODI, AXI3.SODI, AXI4.SODI)

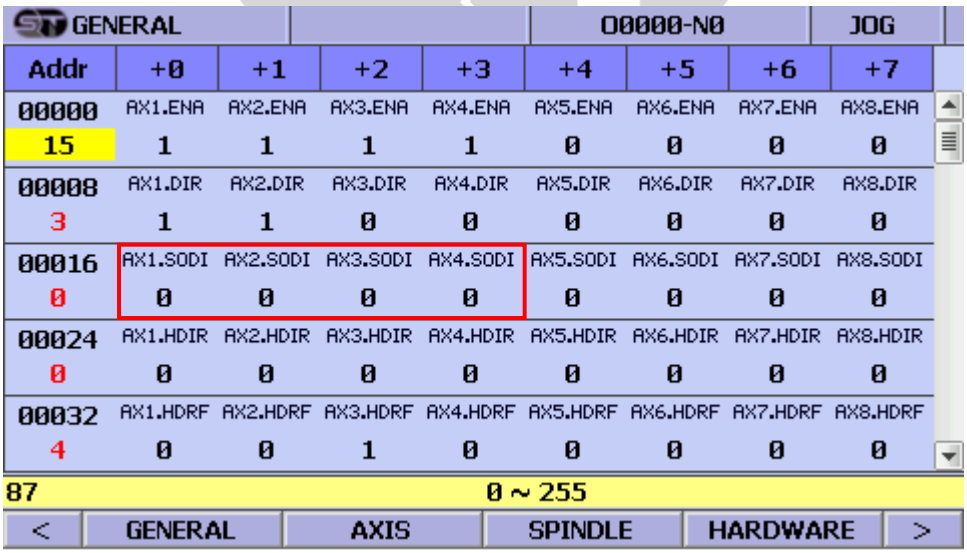


Figure 80.Software limit setting parameters for axes(X,Y,Z,A)

How to configure the parameter from 0 to 1:

- From the PARAMETER screen, use the arrows keys to select the parameter to configure.

- Then enter number 1 and press OK. A prompt will appear on the screen asking for the user password.
- Enter “1415” and press OK
- Next, enter number 1 again and press OK
- Press RESET button to save the configured parameters

Parameter status description:

- **AX1.SODI:** Software limit of X axis
  - AX1.SODI = 0 : Enable software limit of X axis
  - AX1.SODI = 1 : Disable software limit of X axis
- **AX2.SODI:** Software limit of Y axis
  - AX2.SODI = 0 : Enable software limit of Y axis
  - AX2.SODI = 1 : Disable software limit of Y axis
- **AX3.SODI:** Software limit of Z axis
  - AX3.SODI = 0 : Enable software limit of Z axis
  - AX3.SODI = 1 : Disable software limit of Z axis
- **AX4.SODI:** Software limit of A axis
  - AX4.SODI = 0 : Enable software limit of A axis
  - AX4.SODI = 1 : Disable software limit of A axis

2.6.4.4 Axes direction when moving to home position

Note: To set the axis direction for moving to the home position, the installer needs to configure the following parameters: AXI1.HDIR, AXI2.HDIR, AXI3.HDIR, AXI4.HDIR)

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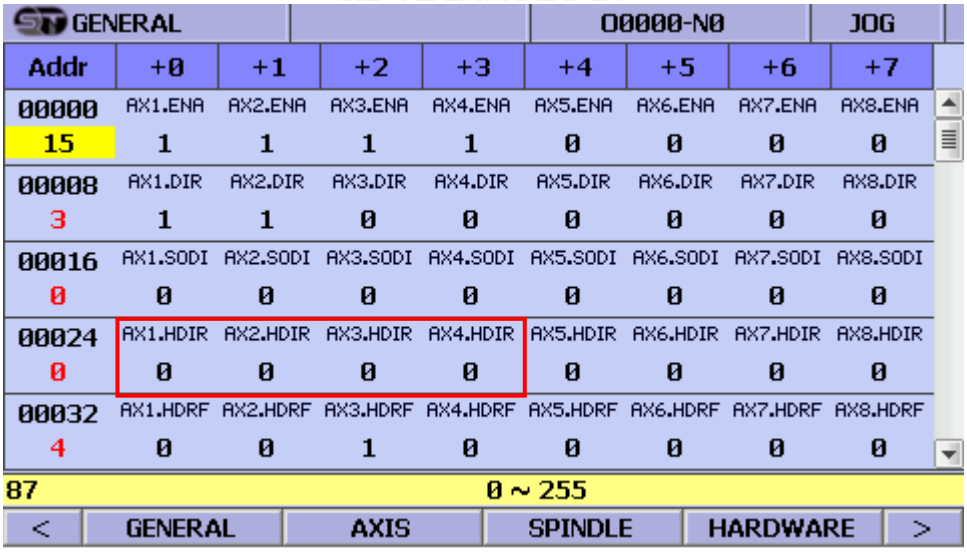


Figure 81. Axes direction when moving to home position parameters

How to configure the parameter from 0 to 1:

- From the PARAMETER screen, use the arrows keys to select the parameter to configure.
- Then enter number 1 and press OK. A prompt will appear on the screen asking for the user password.
- Enter “1415” and press OK
- Next, enter number 1 again and press OK
- Press RESET button to save the configured parameters

Parameter status description:

- **AX1.HDIR:** X axis direction when moving to home position
  - AX1.HDIR = 0 : X axis run to home.pos in positive direction
  - AX1.HDIR = 1 : X axis run to home.pos In negative direction
- **AX2.HDIR:** Y axis direction when moving to home position
  - AX2.HDIR = 0 : Y axis run to home.pos in positive direction
  - AX2.HDIR = 1 : Y axis run to home.pos In negative direction
- **AX3.HDIR:** Z axis direction when moving to home position
  - AX3.HDIR = 0 : Z axis run to home.pos in positive direction
  - AX3.HDIR = 1 : Z axis run to home.pos In negative direction
- **AX4.HDIR:** A axis direction when moving to home position
  - AX4.HDIR = 0 : A axis run to home.pos in positive direction
  - AX4.SODI = 1 : A axis run to home.pos In negative direction

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2.6.4.5 Axis movement type (X,Y,Z,A)

Note: To set the axis movement type, the installer needs to configure the following parameters: #4140, #4141, #4142, #4143)

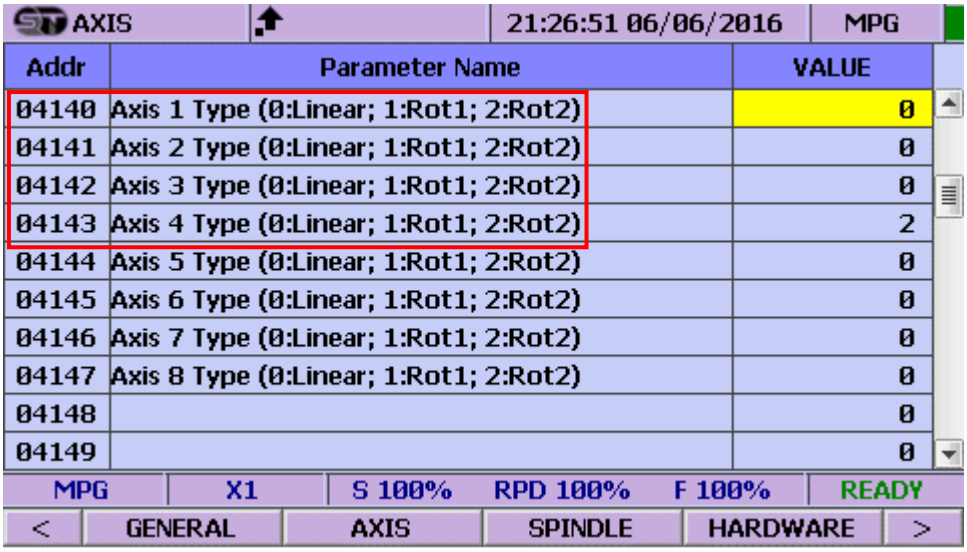


Figure 82. Thông số cài đặt kiểu trục

How to configure the parameter from 0 to 1:

- From the PARAMETER screen, use the arrows keys to select the parameter to configure.
- Then enter number 1 and press OK. A prompt will appear on the screen asking for the user password.
- Enter "1415" and press OK
- Next, enter number 1 again and press OK
- Press RESET button to save the configured parameters

Parameter status description:

- **#4140:** X axis movement type
  - #4140 = 0 : X axis moves in linear direction
  - #4140 = 1 : X axis moves in circular direction 0→360
  - #4140 = 2 : X axis moves in circular direction ± 360
- **#4141:** Y axis movement type
  - #4141= 0 : Y axis moves in linear direction
  - #4141 = 1 : Y axis moves in circular direction 0→360
  - #4141 = 2 : Y axis moves in circular direction ± 360
- **#4142:** Z axis movement type
  - #4142 = 0 : Z axis moves in linear direction

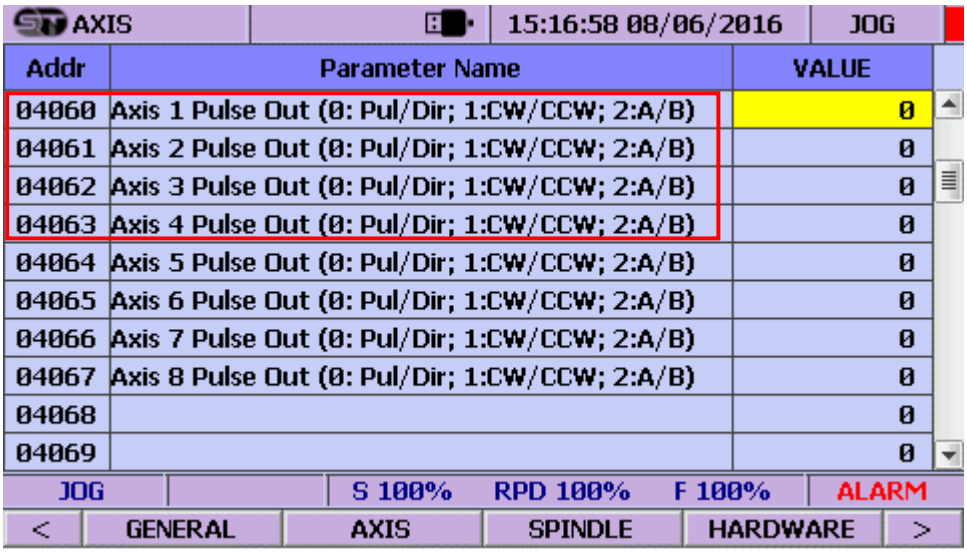
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- #4142 = 1 : Z axis moves in circular direction 0→360
- #4142 = 2 : Z axis moves in circular direction ± 360

- **#4143:** A axis movement type
  - #4143 = 0 : A axis moves in linear direction
  - #4143 = 1 : A axis moves in circular direction 0→360
  - #4143 = 2 : A axis moves in circular direction ± 360

2.6.4.6 Pulse generation type for axes (X,Y,Z,A)

Note: To set the pulse generation type for the axes (X, Y, Z, A), the installer needs to configure the following parameters: #4060, #4061, #4062, #4063)



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Figure 83. Pulse generation type for axes setting parameter

How to configure the parameter from 0 to 1:

- From the PARAMETER screen, use the arrows keys to select the parameter to configure.
- Then enter number 1 and press OK. A prompt will appear on the screen asking for the user password.
- Enter "1415" and press OK
- Next, enter number 1 again and press OK
- Press RESET button to save the configured parameters

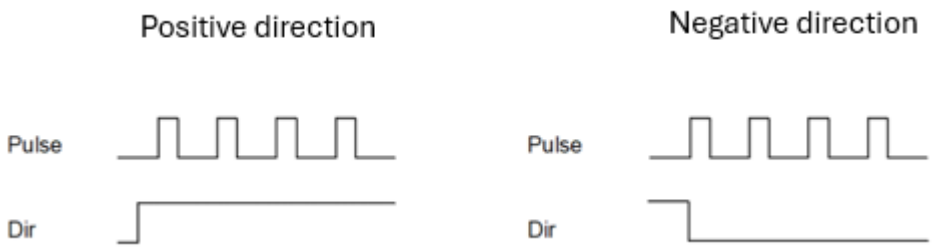
Parameter status description:

- **#4060:** X axis pulse generation type
  - #4060 = 0 : Pul/Dir type
  - #4060 = 1 : CW/CCW type
  - #4060 = 2 : A/B type

- **#4061:** Y axis pulse generation type
  - #4061 = 0 : Pul/Dir type
  - #4061 = 1 : CW/CCW type
  - #4061 = 2 : A/B type
- **#4062:** Z axis pulse generation type
  - #4062 = 0 : Pul/Dir type
  - #4062 = 1 : CW/CCW type
  - #4062 = 2 : A/B type
- **#4063:** A axis pulse generation tpye
  - #4063 = 0 : Pul/Dir type
  - #4063 = 1 : CW/CCW type
  - #4063 = 2 : A/B type

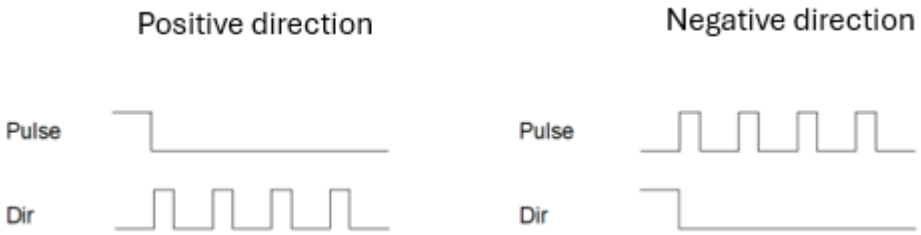
Note:

Pul/Dir pulse generation type:

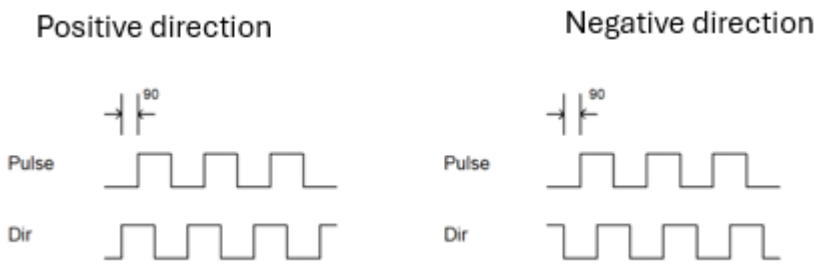


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CW/CCW pulse generation type:



Kiểu phát xung A/B



2.6.4.7 Electronic gear box setting

Note: To set the ratio for the electronic gearbox of the axes, the installer needs to configure and calculate the following parameters

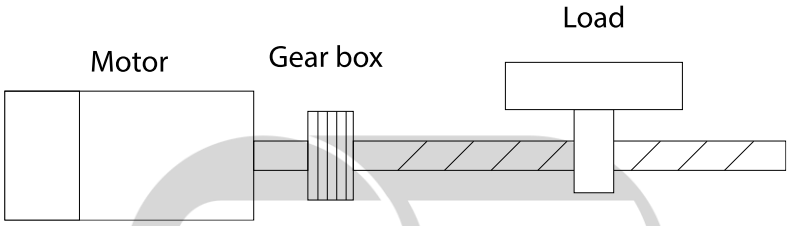
- Number of pulses per 1mm movement: (#12010, #12011, #12012, #12013)
- Electronic gear ratio (Axis Gear-Motor side): (#12340, #12341, #12342, #12343)
- Electronic gear ratio (Axis Gear-Screw side): (#12350, #12351, #12352, #12353)

How to calculate the electronic gear ratio

Formula:

Electronic gear ratio = (Pulse/mm*Axis Gear(Motor side)/Axis Gear (Screw side)

Case 1: Drive on lead screw



Example 1: The drive system uses a lead screw with a lead of 10 mm. The coupling is straight, with a transmission ratio of 1:1. The servo motor is set to 10,000 pulses per revolution.

From the above parameters, we can calculate the number of pulses required to move 1 mm

Number of pulses to move 1 mm = The number of pulses set on the servo motor / lead of the lead screw

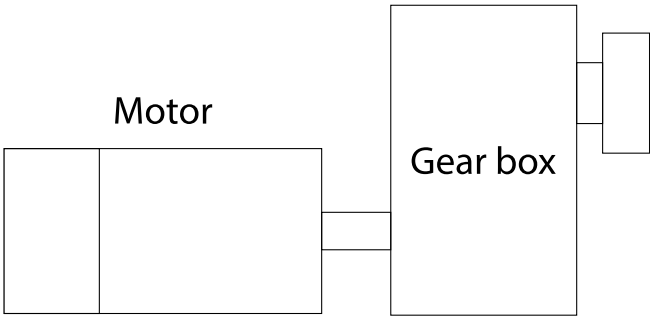
⇨ Number of pulse to move 1 mm = 10000/10 = 1000 pulse

When we calculated the number of pulses to move 1 mm

The installer will configure the following parameters

- #12010 = 1000 (Number of pulse for the machine work table to move 1mm)
- #12340 = 1 (Electronic gear ratio - Motor side)
- #12350 = 1 (Electronic gear ratio - Screw side)

Case 2: Drive on rotation axis (4th axis)



Example 2: The rotary drive system (axis 4), with a gearbox having a transmission ratio of 1:100, and the servo motor set to 360 pulses per revolution.

From the above parameters, we can calculate the number of pulses required for the load to complete one full rotation.

Number of pulses for the LOAD to make 1 rotation = Number of pulses set on the servo motor*gear ratio

⇨ Number of pulses for the LOAD to make 1 rotation = 360*100 = 36000 Pul/Rev

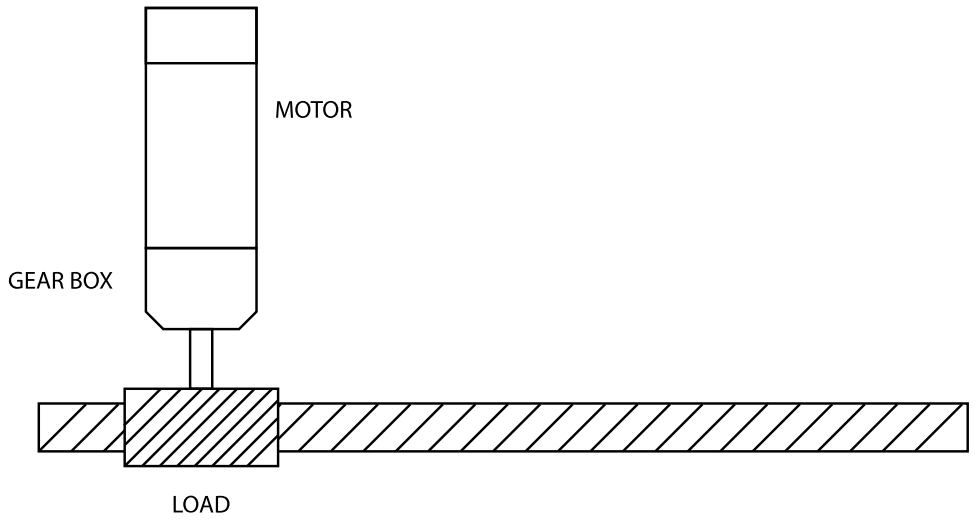
Then we can calculate the number of pulse for the machine work table to move 1 mm

⇨ Number of pulse for the machine work table to move 1mm = 36000/360 = 100 pulse

The installer will configure the following parameters:

- #12010 = 100 (Number of pulse for machine work table to move 1mm)
- #12340 = 1 (Electric gear ratio - Motor side)
- #12350 = 1 (Electric gear ratio - Screw side)

Case 3: Rack and pinion drive



Example 3: Rack and pinion drive system. The servo motor is configured with 1600 pulses per revolution. Gear ratio is 1:5

The rack has the following specifications:

- Pinion gear module: m=2
- Tooth pitch: $P = m*\pi = 2*\pi = 6{,}2831852$ mm.
- Number of teeth: Z=25

From that we can calculate the distance traveled in one full rotation of the gear

$= P*Z = 6{,}2831852 * 25 = 157{,}07963$ mm

- ⇨ To calculate one full rotation of the gear = 1600*5 = 8000 xung
- ⇨ It can de deduced that with one pulse, the machine table will move by
  $= 157{,}07963/8000 = 0{,}019643$mm
- ⇨ Similarly for 1 mm = 1/0,019643 =50,9424 pulse

Because the controller cannot be set to fractional values, we will round the value by multiplying the pulse/ mm by 10,000.

After obtaining the necessary values, the installer will configure the following parameters

- ⇨ #12010 = 1
- ⇨ #12340 = 10000
- ⇨ #12350 = 509424

2.6.4.8 Axis acceleration in JOG RAPID (G00)

Note: To set the acceleration parameters for the axes in JOG RAPID mode (G00), the installer needs to configure the following parameters (#12020, #12021, #12022, #12023).

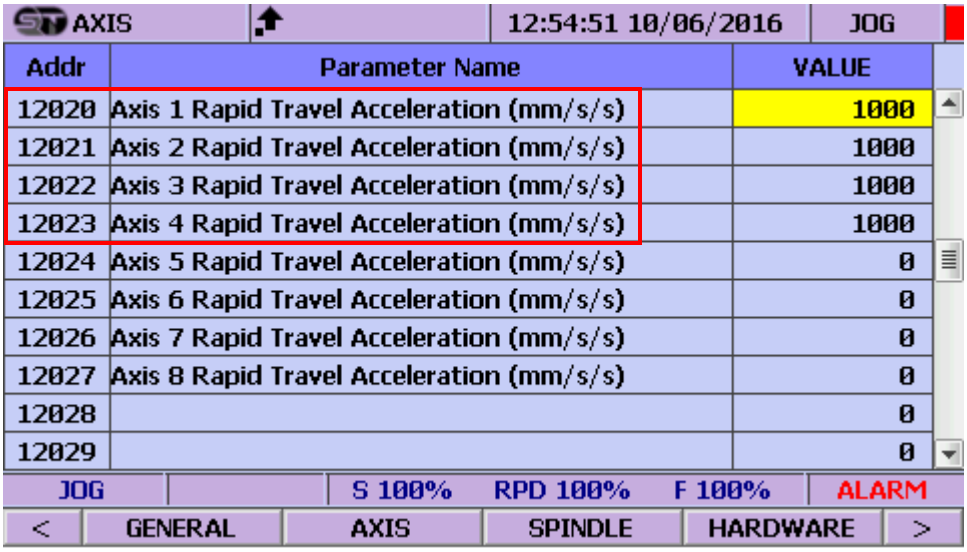


Figure 84. Axes acceleration setting parameters in JOG RAPID (G00)

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#12020:** X axis acceleration in JOG RAPID (G00)
- **#12021:** Y axis acceleration in JOG RAPID (G00)
- **#12022:** Z axis acceleration in JOG RAPID (G00)
- **#12023:** A axis acceleration in JOG RAPID (G00)

QS TECHNOLOGY

2.6.4.9 Axis acceleraion for machining (G01)

Note: To set the acceleration parameters for the axes (X, Y, Z, A) during machining, the installer needs to configure the following parameters (#12030, #12031, #12032, #12033).

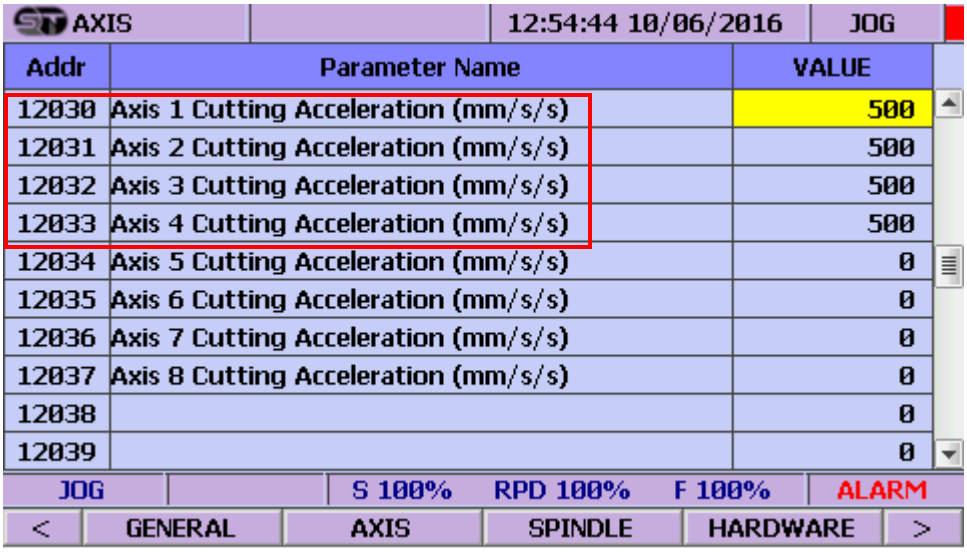


Figure 85. Axes acceleration for machining setting parameters.

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#12030:** X axis acceleration for machining (G01)
- **#12031:** Y axis acceleration for machining (G01)
- **#12032:** Z axis acceleration for machining (G01)
- **#12033:** A axis acceleration for machining (G01)

QS TECHNOLOGY

2.6.4.10 Axis acceleration in JOG mode

Note: To set the acceleration parameters for the axes (X, Y, Z, A) in JOG mode, the installer needs to configure the following parameters (#12330, #12331, #12332, #12333)..

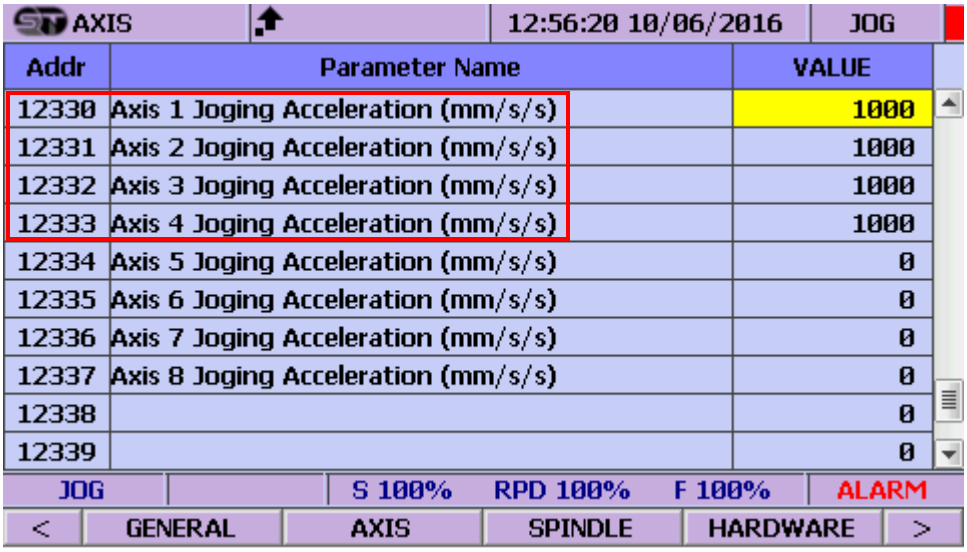


Figure 86. Axes acceleration in JOG mode setting parameters

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parametet status description:

- **#12330:** X axis acceleration in JOG mode
- **#12331:** Y axis acceleration in JOG mode
- **#12332:** Z axis acceleration in JOG mode
- **#12333:** A axis acceleration in JOG mode

2.6.4.11 Machine speed in JOG RAPID mode

Note: To set the machine speed parameters in JOG RAPID mode, the installer needs to configure the following parameters (#12040, #12041, #12042, #12043).

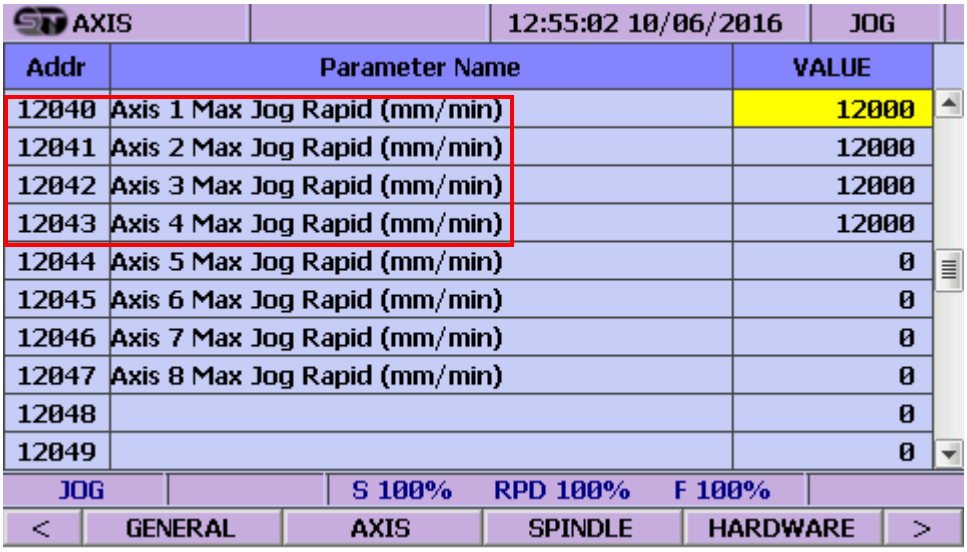


Figure 87. Machine speed in JOG RAPID mode parameter setting

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#12040:** X axis speed in JOG RAPID mode
- **#12041:** Y axis speed in JOG RAPID mode
- **#12042:** Z axis speed in JOG RAPID mode
- **#12043:** A axis speed in JOG RAPID mode

QS TECHNOLOGY

2.6.4.12 Axis max speed in JOG mode

Note: To set the machine speed parameters in JOG RAPID mode, the installer needs to configure the following parameters (#12050, #12051, #12052, #12053).

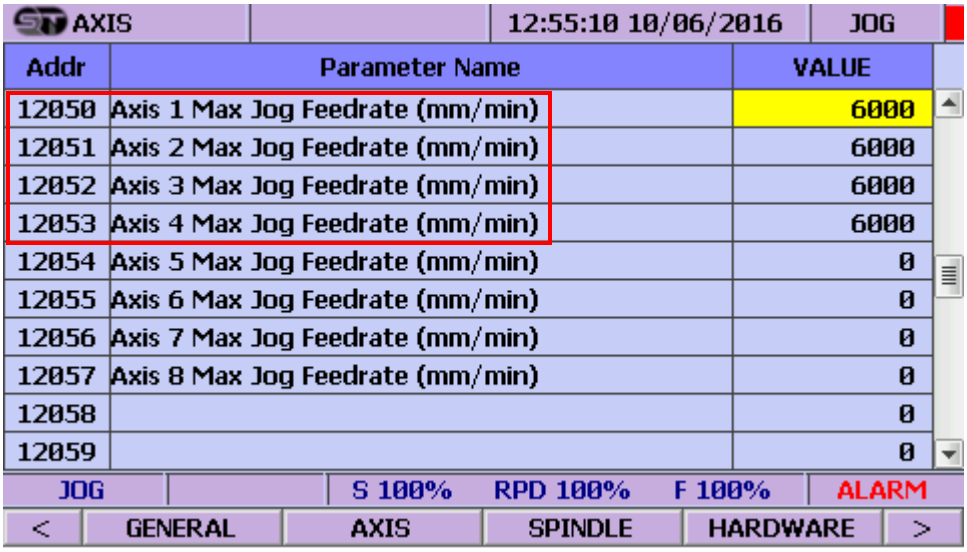


Figure 88. Axes max speed in JOG mode

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#12050:** X axis max speed in JOG mode
- **#12051:** Y axis max speed in JOG mode
- **#12052:** Z axis max speed in JOG mode
- **#12053:** A axis max speed in JOG mode

QS TECHNOLOGY

2.6.4.13 Axis max speed in G00 program

Note: To set the maximum speed of the axes during G00 program execution, the installer needs to configure the following parameters (#12060, #12061, #12062, #12063).

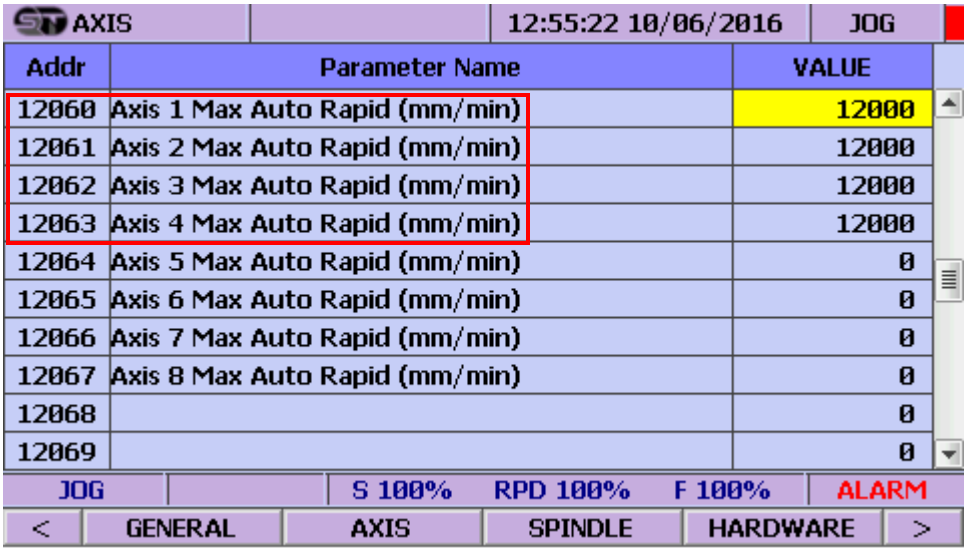


Figure 89. Axes max speed in G00 program setting parameters

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#12060:** X axis max speed in G00 program
- **#12061:** Y axis max speed in G00 program
- **#12062:** Z axis max speed in G00 program
- **#12063:** A axis max speed in G00 program

QS TECHNOLOGY

2.6.4.14 Axis max speed in G01 program

Note: To set the maximum speed of the axes during G00 program execution, the installer needs to configure the following parameters (#12070, #12071, #12072, #12073).

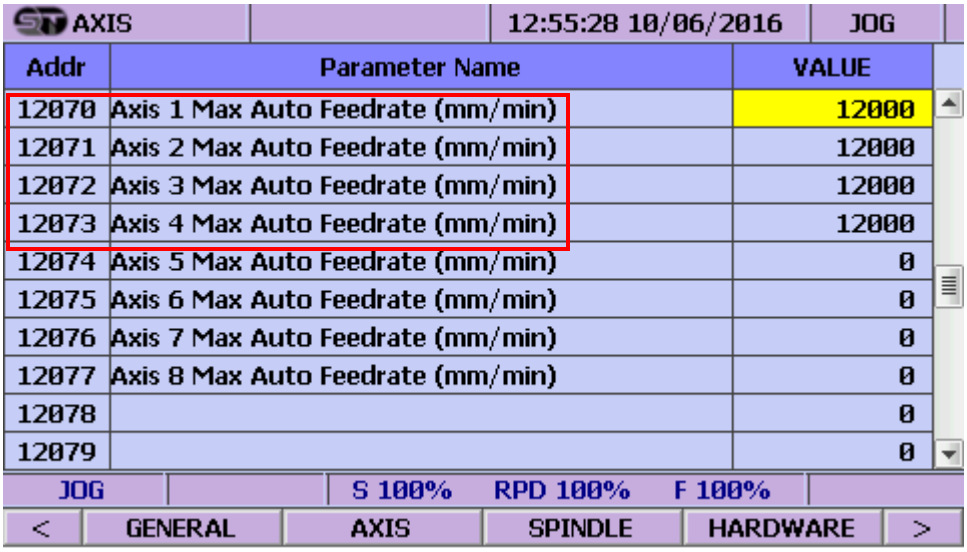


Figure 90. Axes max speed in G01 program setting parameters

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#12070:** X axis max speed in G01 program
- **#12071:** Y axis max speed in G01 program
- **#12072:** Z axis max speed in G01 program
- **#12073:** A axis max speed in G01 program

QS TECHNOLOGY

2.6.4.15 Axis homing type

Note: To set the homing type for the axes (X, Y, Z, A), the installer needs to configure the following parameters (#4020, #4021, #4022, #4023).

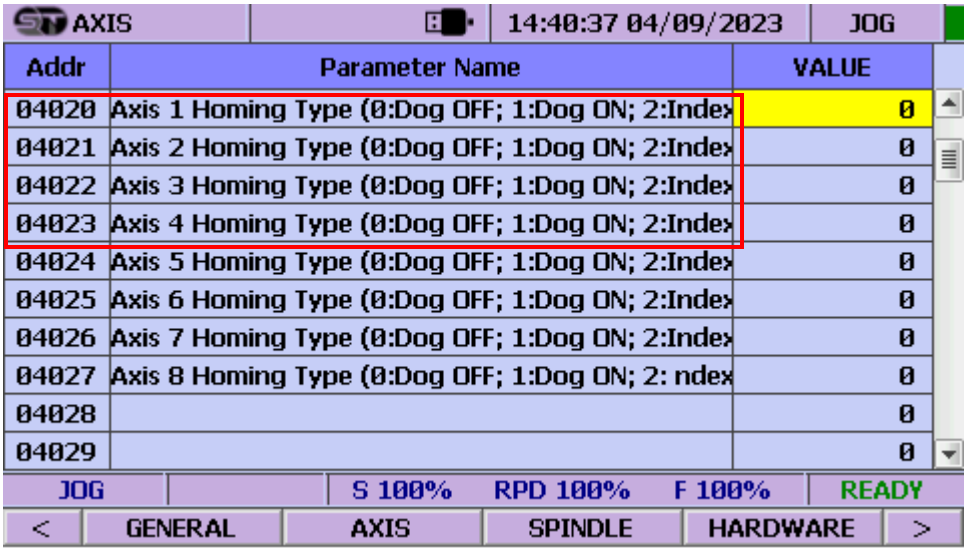


Figure 91. Axes homing type setting parameters

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#4020:** X axis homing type
- **#4021:** Y axis homing type
- **#4022:** Z axis homing type
- **#4023:** A axis homing type

QS TECHNOLOGY

DOG ON/DOG OFF homing type

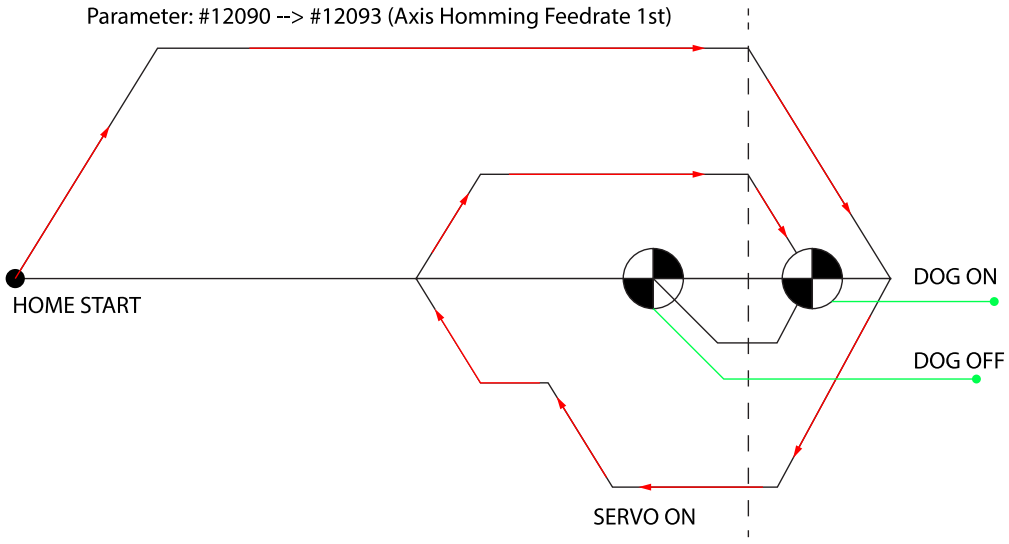


Figure 92. Diagram illustrating the axis movement when configured in DOG ON/DOG OFF mode.

INDEX homing type

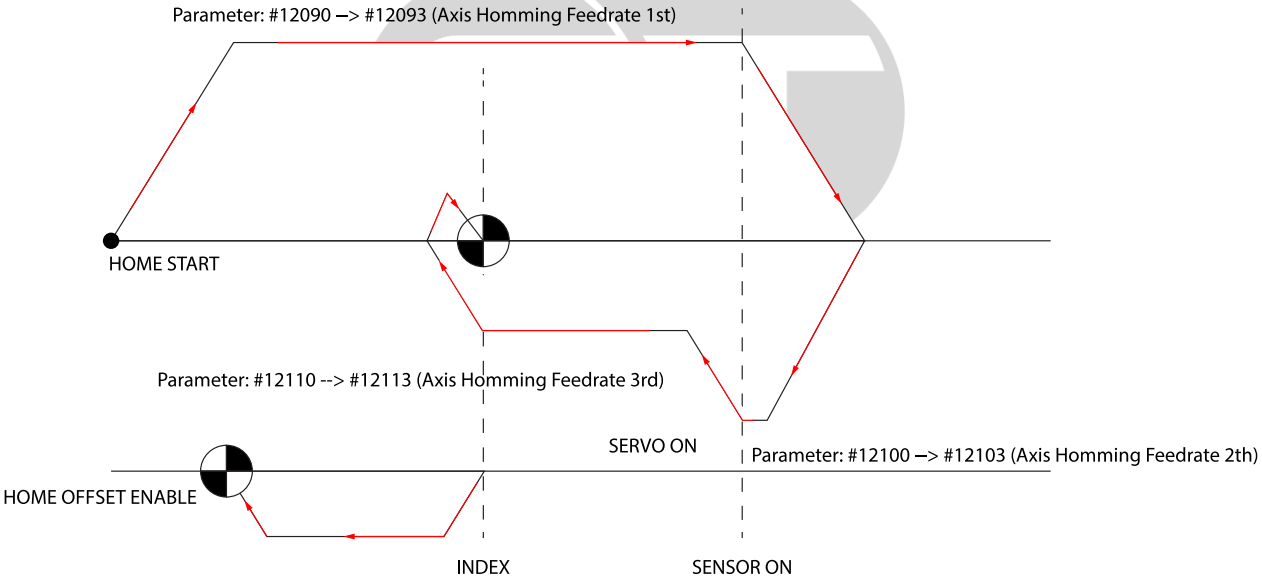


Figure 93. Diagram illustrating the axis movement when configured in INDEX mode.

2.6.4.16 Homing Feedrate 1st

Note: To set the homing speed for the axes during the first stage (Homing Feedrate 1st), the installer needs to configure the following parameters (#12090, #12091, #12092, #12093).

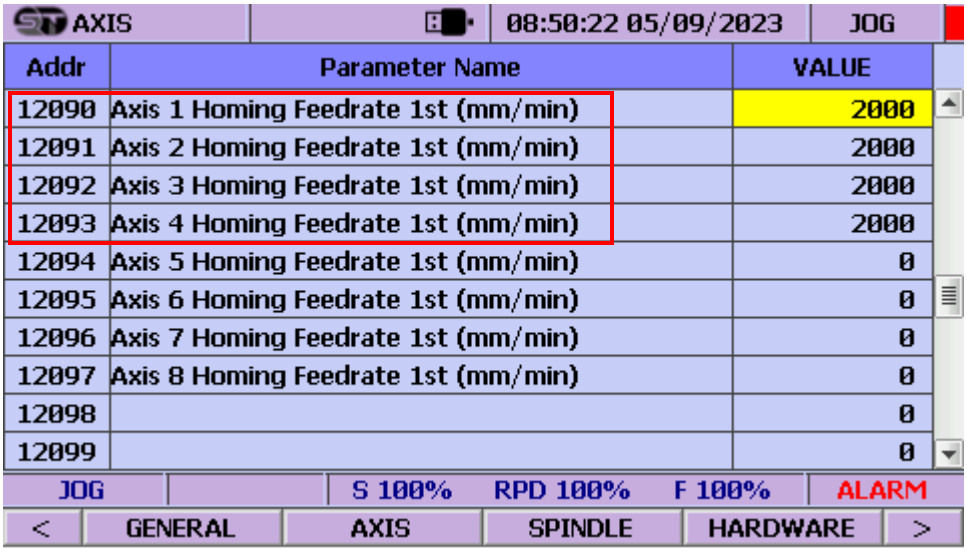


Figure 94. Axes homing feedrate 1st

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#12090:** X axis (Homing Feedrate 1st)
- **#12091:** Y axsi (Homing Feedrate 1st)
- **#12092:** Z axis (Homing Feedrate 1st)
- **#12093:** A axis (Homing Feedrate 1st)

QS TECHNOLOGY

2.6.4.17 Homing Feedrate 2nd

Note: To set the homing speed for the axes during the second stage (Homing Feedrate 2nd), the installer needs to configure the following parameters (#12100,#12101, #12102,#12103).

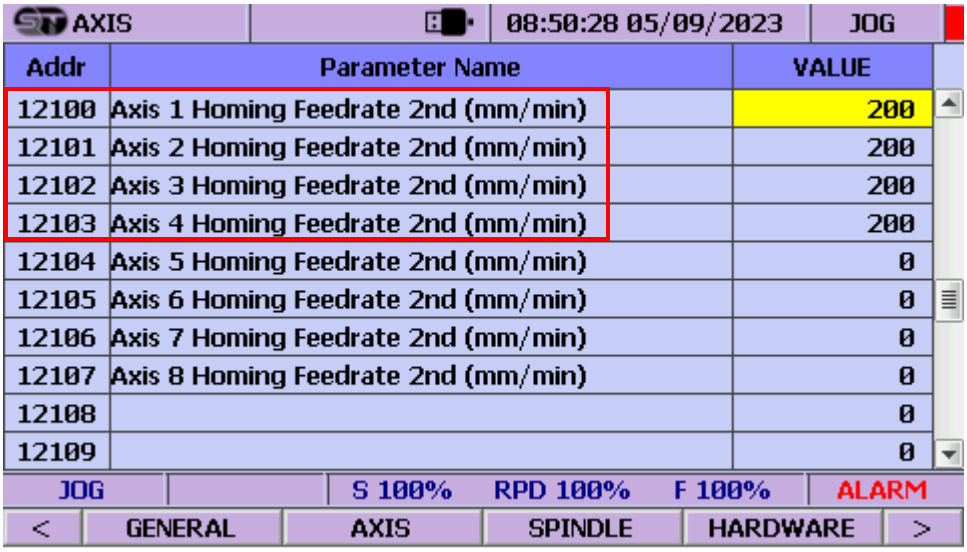


Figure 95. Axes homing feedrate 2nd

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#12100:** X axis (Homing Feedrate 2th)
- **#12101:** Y axis (Homing Feedrate 2th)
- **#12102:** Z axis (Homing Feedrate 2th)
- **#12103:** A axis (Homing Feedrate 2th)

QS TECHNOLOGY

2.6.4.18 Homing Feedrate 3rd

Note: To set the homing speed for the axes during the third stage (Homing Feedrate 3rd), the installer needs to configure the following parameters (#12110 #12111,#12112, #12113).

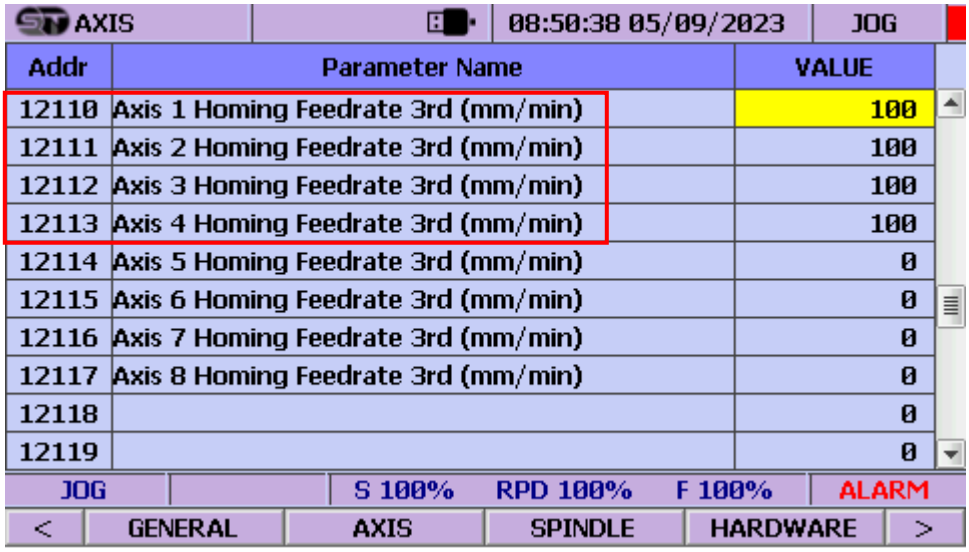


Figure 96. Axes homing feedrate 3rd

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#12110:** X axis (Homing Feedrate 3rd)
- **#12111:** Y axis (Homing Feedrate 3rd)
- **#12112:** Z axis (Homing Feedrate 3rd)
- **#12103:** A axis (Homing Feedrate 3rd)

QS TECHNOLOGY

2.6.4.19 Machine travel limit setting (Soft Limit)

Note: To set the machine travel limits, the operator needs to perform the following steps.

Example: Setting the travel limit for the X axis

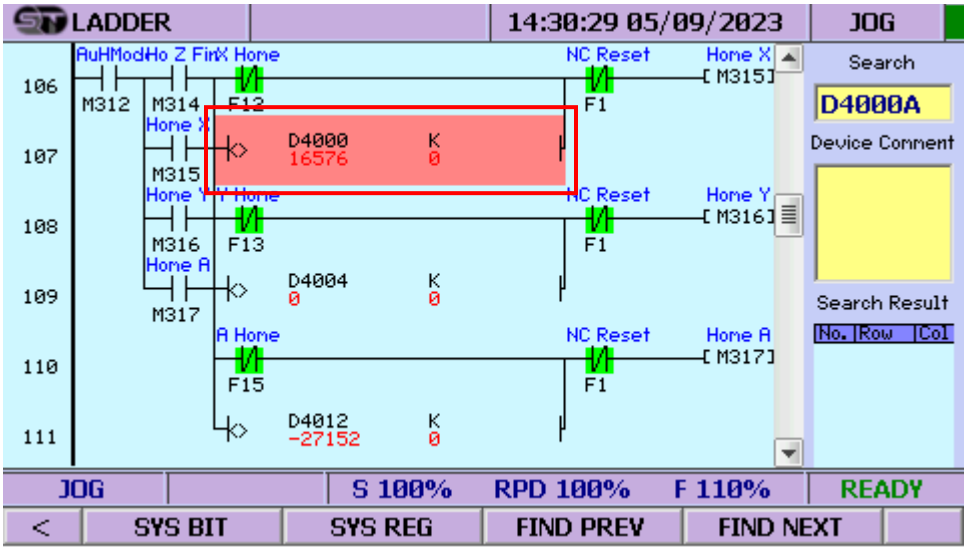
Note: Repeat the following steps to configure the remaining axes.

Step 1: Return the axes to home position

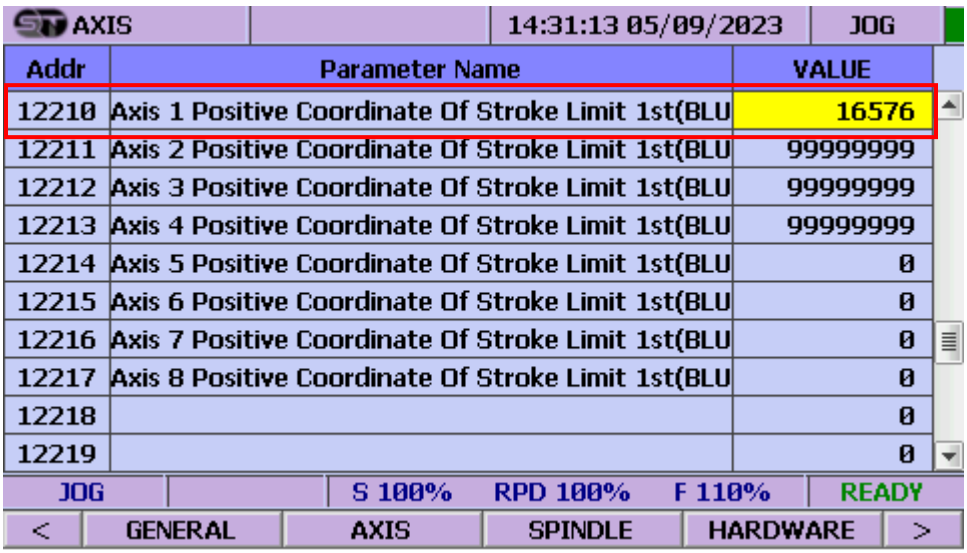
Step 2: Move the X axis to the end of the mechanical travel in the positive direction

Step 3: Access to PLC page

Step 4: Searching for D4000



Step 5: Enter the current value of D4000 into parameter #12210 in the PARAMETER page



Note:

- D4000: X axis coordinate – calculated based on the number of pulses
- D4004: Y axis coordinate – calculated based on the number of pulses
- D4008: Z axis coordinate – calculated based on the number of pulses
- D4012: A axis coordinate – calculated based on the number of pulses

2.6.4.19.1 Travel limit in negative direction

Note: To set the travel limits of the machine, the installer needs to configure the following parameters (#12200, #12201, #12202, #12203).

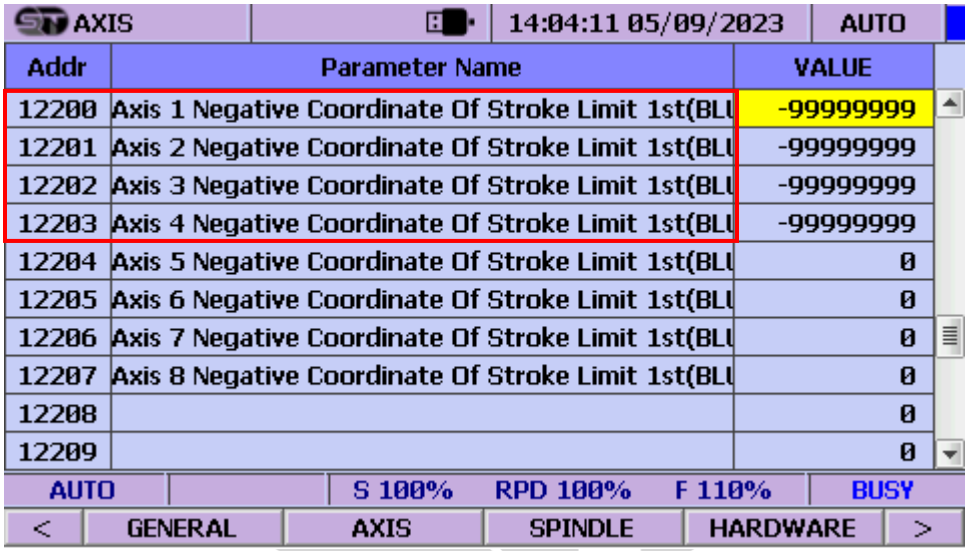


Figure 97. Axes travel limit in negative direction setting parameters

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#12200:** X axis travel limit in negative direction
- **#12201:** Y axis travel limit in negative direction
- **#12202:** Z axis travel limit in negative direction
- **#12203:** A axis travel limit in negative direction

2.6.4.19.2 Travel limit in positive direction

Note: To set the travel limits of the machine, the installer needs to configure the following parameters (#12210, #12211, #12212, #12213).

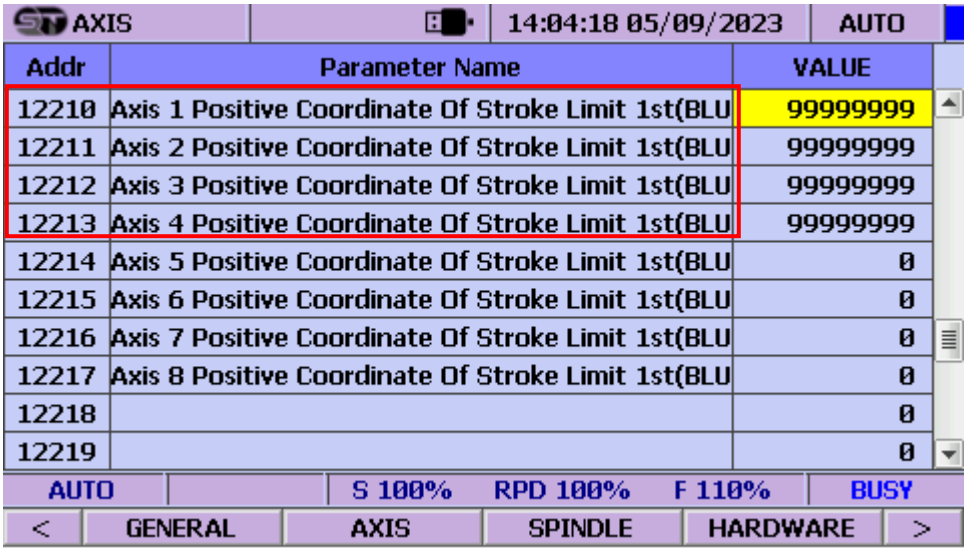


Figure 98. Axes travel limit in positive direction setting parameters

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#12210:** X axis travel limit in positive direction
- **#12211:** Y axis travel limit in positive direction
- **#12212:** Z axis travel limit in positive direction
- **#12213:** A axis travel limit in positive direction

QS TECHNOLOGY

2.6.5 Automatic tool setter

Note: To set the parameters for the automatic tool offset function, the installer needs to configure the following parameters (#14003 → #14012).

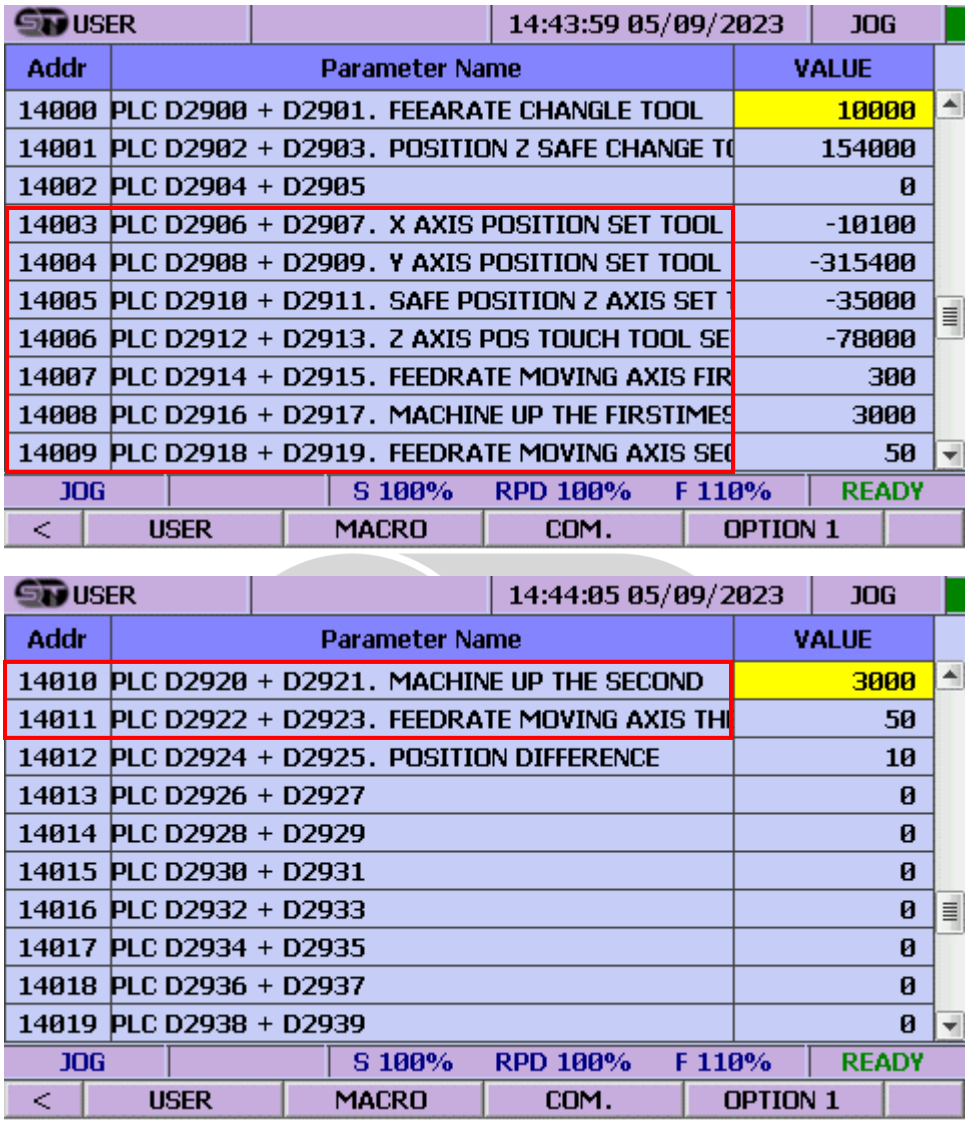


Figure 99. Automatic tool setter setting parameters

Parameter status description:

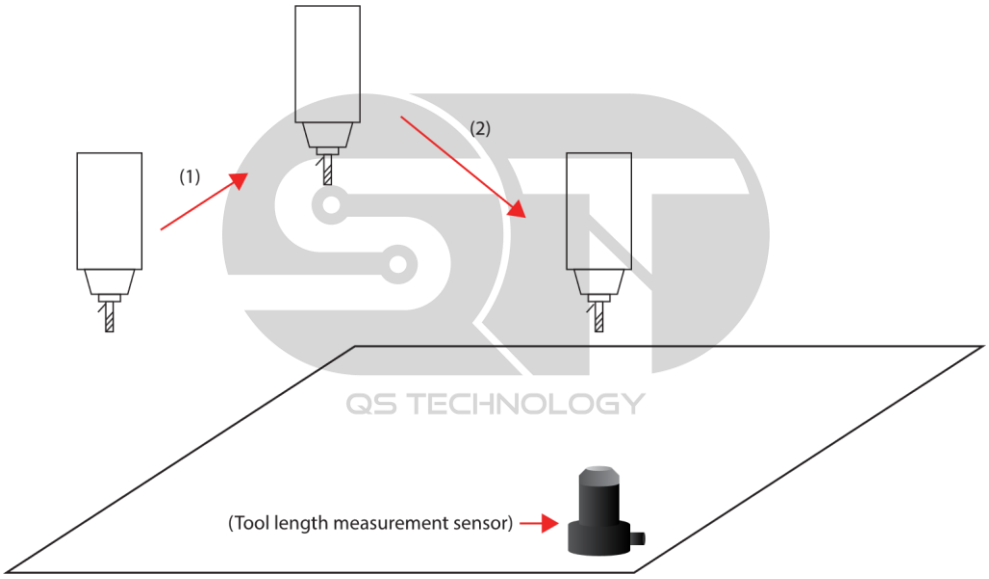
- #14003: X axis coordinates of tool setter device
- #14004: Y axis coordinates of tool setter device
- #14005: Safety Z axis coordinates of tool setter device
- #14006: Z axis coordinates when touching tool setter device
- #14007: Z axis speed when first touching to tool setter device (1st)
- #14008: Tool retract distance after the first touching tool setter device (1st)
- #14009: Z axis speed when second touching to tool setter device (2nd)

- #14010: Tool retract distance after the first touching tool setter device (2nd)
- #14011: Z axis speed when first touching to tool setter device (3rd)
- #14012: Allowable deviation after each contact with the automatic tool setter device

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Automatic tool setting process (M100)

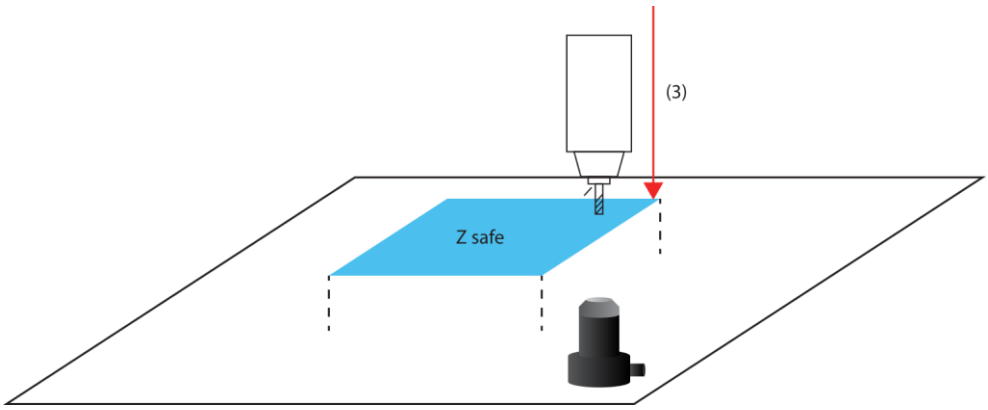


Stage 1:

When command M100 is called, the coolant system will be turned off. At this point, the Z axis, which is at any position on the machine table, will begin returning to the home position..

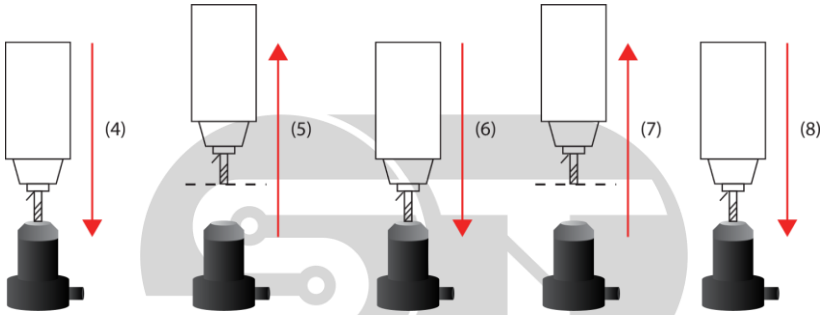
Stage 2:

At this point, the controller moves the X and Y axes, along with the tool head, to the tool setter position that has been set in parameters (#14003 and #14004).



Stage 3

Then, the Z axis will quickly move to the safe plane position that has been set in parameter (#14005).



Stage 4, 5:

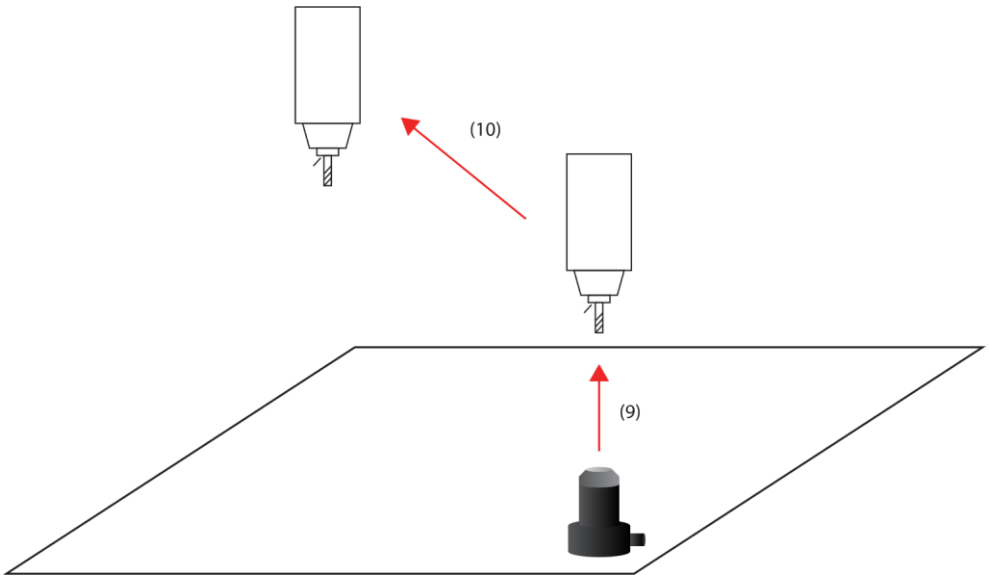
The Z axis will move and first touching the sensor at the speed set in parameter **#14007**. After touching the sensor, the Z axis will retract by a distance set in parameter **#14008.**

Stage 6, 7:

The Z axis will move and make a second touching with the sensor at the speed set in parameter **#14009**. After touching the sensor, the Z axis will retract by a distance set in parameter **#14010.**

Stage 8:

At this point, the Z axis will continue to move and make the third touching with the sensor at the speed set in parameter **#14011.**



Stage 9:

After completing the third touching with the sensor, the Z axis will retract and return to the initial home position, completing the tool setting process.

Stage 10:

After completing the tool setting process, the Macro program will take the measured values from the second and third contacts for comparison. If the error between the two measurements is greater than the value set in #14012, the controller will report an error and will not record the result in the corresponding (Offset H). If the error between the two contacts is within the allowed range, the controller will calculate the average value between the two contacts and record the result in the corresponding (Offset H).

Note:

At Stage (4,5): If the Z axis has moved to the position to contact the sensor, as set in parameter **#14006**, but has not yet made contact with the sensor, the controller will report an error.

- The cause is that the travel limit set in parameter **#14006** is incorrect. The operator needs to review and check the value set in parameter **#14006.**

If the Z axis has retracted the tool by a distance set in parameter **#14008** but the tool setter signal has not disappeared, the controller will report an error.

- The cause is that the tool retract distance of the Z axis is too small, or the tool setter sensor is malfunctioning. The operator needs to review and check the value set in parameter **#14008** and the tool setter sensor.

At Stage (5,6). If the Z axis has moved to the position to touch the sensor set in parameter #14006 but has not yet made contact with the sensor, the controller will report an error.

- The cause is that the travel limit set in parameter **#14006** is incorrect. The operator needs to review and check the value set in parameter **#14010.**

If the Z axis has retracted the tool by a distance set in parameter **#14008** but the tool setter signal has not disappeared, the controller will report an error.

- The cause is that the tool retract distance of the Z axis is too small, or the tool setter sensor is malfunctioning. The operator needs to review and check the value set in parameter **#14008** and the tool setter sensor.

At Stage (7). If the Z axis has moved to the position to contact the sensor set in parameter #14011 but has not yet made contact with the sensor, the controller will report an error.

- The cause is that the travel limit set in parameter **#14006** is incorrect. The operator needs to review and check the value set in parameter **#14006.**

At Stage (10)

- **Situation 1:** M100 executing
  - The average value will be set to the (Offset H) corresponding to the current tool in the spindle.
  - Ex: If the current tool in the spindle is T5, the average value will be set to (Offset H5)
- **Situation 2:** M100Hxxx command
  - The average value will be set to (Offset H) corresponding to the previously called H value..
  - Ex: If M100H3 is called, the average value will be set to (Offset H3)

QS TECHNOLOGY

2.6.6 Backlash compensation – Without load (G00)

Note: To configure the mechanical backlash compensation for the moving axes, the installer needs to set the following parameters (#8190, #8191, #8192, #8193).

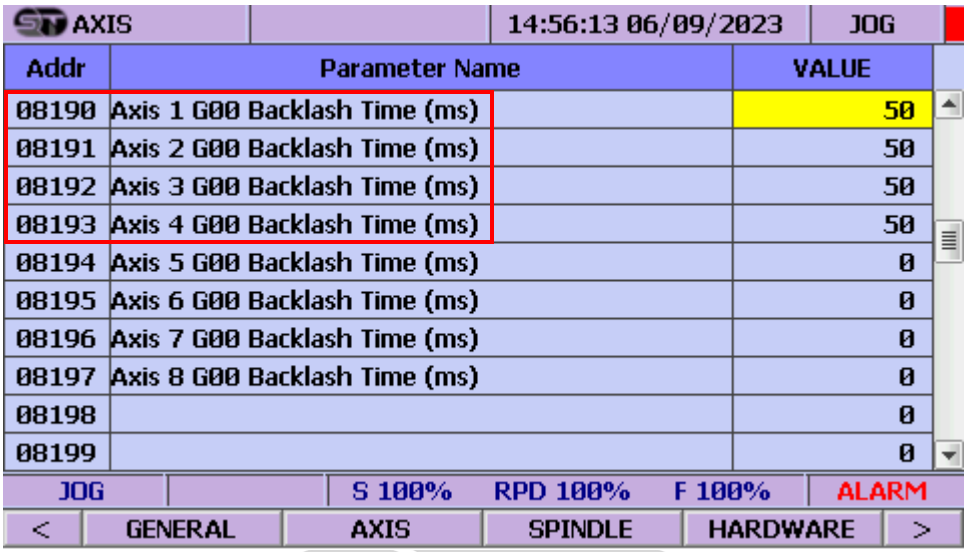


Figure 100. Backlash compensation setting values

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#8190:** X axis backlash compensation – Without load
- **#8191:** Y axis backlash compensation – Without load
- **#8192:** Z axis backlash compensation – Without load
- **#8193:** A axis backlash compensation – Without load

QS TECHNOLOGY

2.6.7 Backlash compensation – With load (G01)

Note: To configure the mechanical backlash compensation for the moving axes, the installer needs to set the following parameters (#8200, #8201, #8202, #8203).

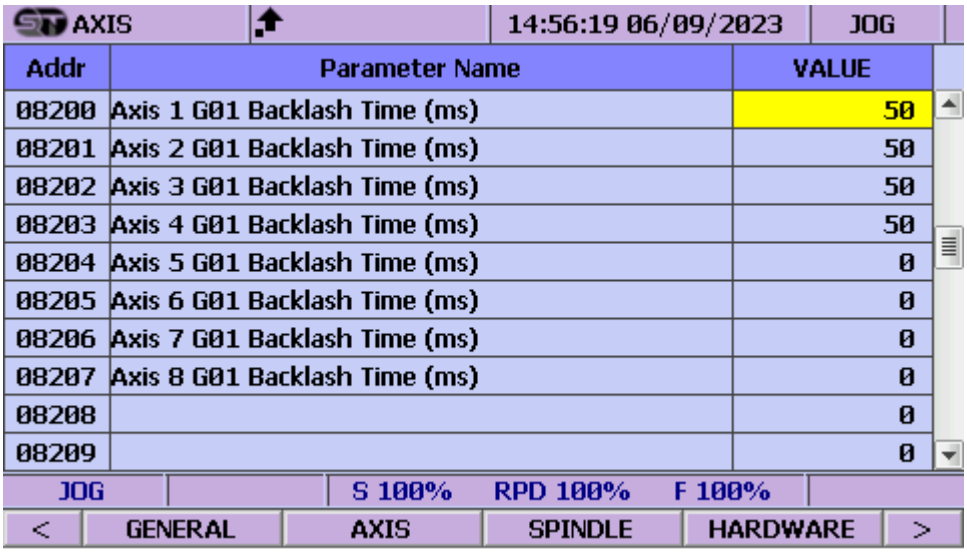


Figure 101. Backlash compensation – With load (G01)

Parameter setting:

- From the PARAMETER screen, use the arrow keys to select the parameter to configure.
- Then, enter number 1 and press OK. At this point, a dialog box will appear requesting the user password.
- Enter the user password is" 1415 "
- Next, enter the desire value into the corresponding parameter and press OK.
- Press RESET button to save the configrured parameter.

Parameter status description:

- **#8200:** X axis compensation – With load
- **#8201:** Y axis compensation – With load
- **#8202:** Z axis compensation – With load
- **#8203:** A axis compensation – With load

QS TECHNOLOGY

PART 3 SPECIAL FUNCTIONS

3.1 Auto home

Note: This function allows the operator to automatically return the axes to the home position. To use this function, the operator needs to configure parameter M195..

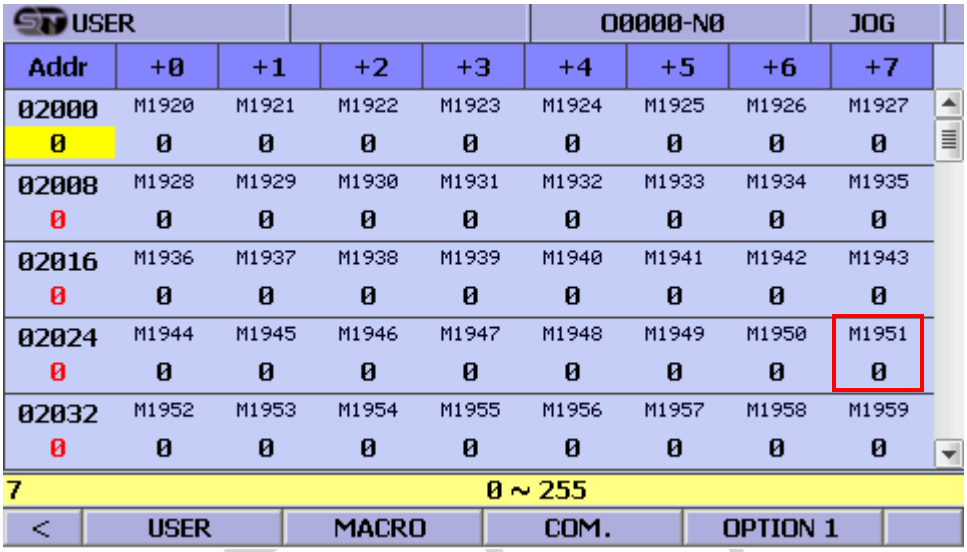


Figure 102. Auto home setting parameter

How to configure the parameter from 0 to 1:

- From the PARAMETER screen, use the arrows keys to select the parameter to configure.
- Then enter number 1 and press OK. A prompt will appear on the screen asking for the user password.
- Enter “1415” and press OK
- Next, enter number 1 again and press OK.

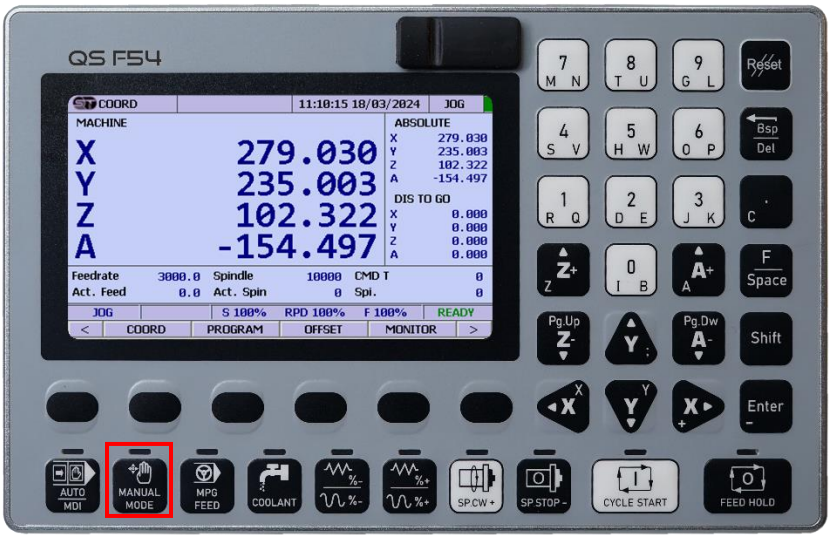
Parameter status description:

- **M1951:** Enable/Disable Auto Home feature for axes
  - M1951 = 0 : Disable
  - M1951 = 1 : Enable

QS TECHNOLOGY

Operating instructions:

Step 1: On the main screen, select MANUAL MODE button.



Step 2: Press MANUAL MODE 3 times to switch to HOME mode.



Step 3: Check the axes on the machine. Then press the CYCLE START button on the controller to return the axes to the home position.



After pressing the CYCLE START button, the machine will first move the Z axis to the home position. Once this cycle is completed, the machine will return the remaining axes to their home positions simultaneously.

3.2 Workpiece homing

Note: This function requires the operator to set parameter M1950.

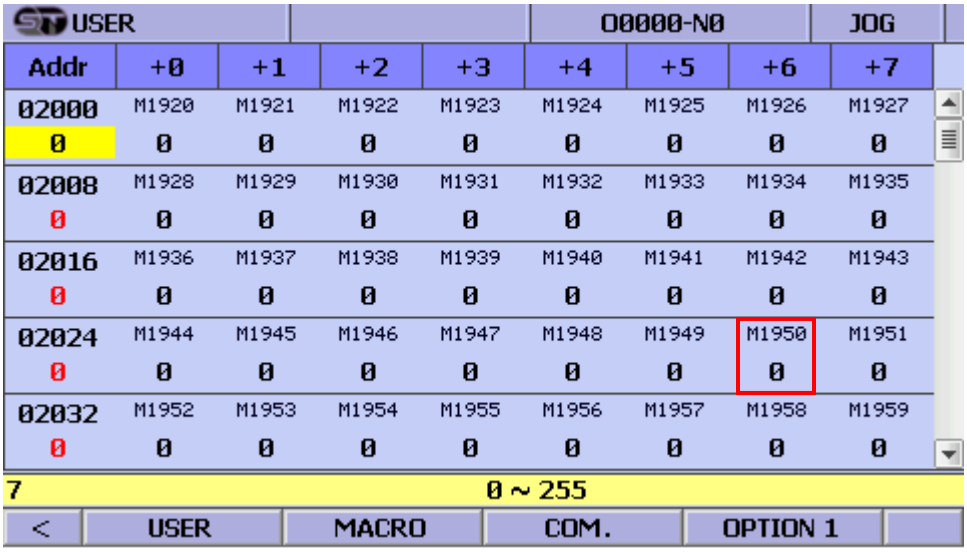


Figure 103. Thông số cài đặt chức năng về góc phôi

How to configure the parameter from 0 to 1:

- From the PARAMETER screen, use the arrows keys to select the parameter to configure.
- Then enter number 1 and press OK. A prompt will appear on the screen asking for the user password.
- Enter “1415” and press OK
- Next, enter number 1 again and press OK.

Parameter status description:

- **M1951:** Enable/Disable workpiece homing feature
  - M1950 = 0 : Disable
  - M1950 = 1 : Enable

QS TECHNOLOGY

Operation instructions:

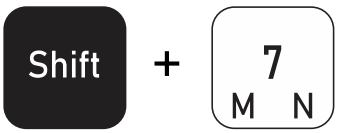
Step 1: From manin screen press MANUAL MODE button



Step 2: Press MANUAL MODE button to switch to JOG mode



Step 3: Press combine SHIFT và key number 7

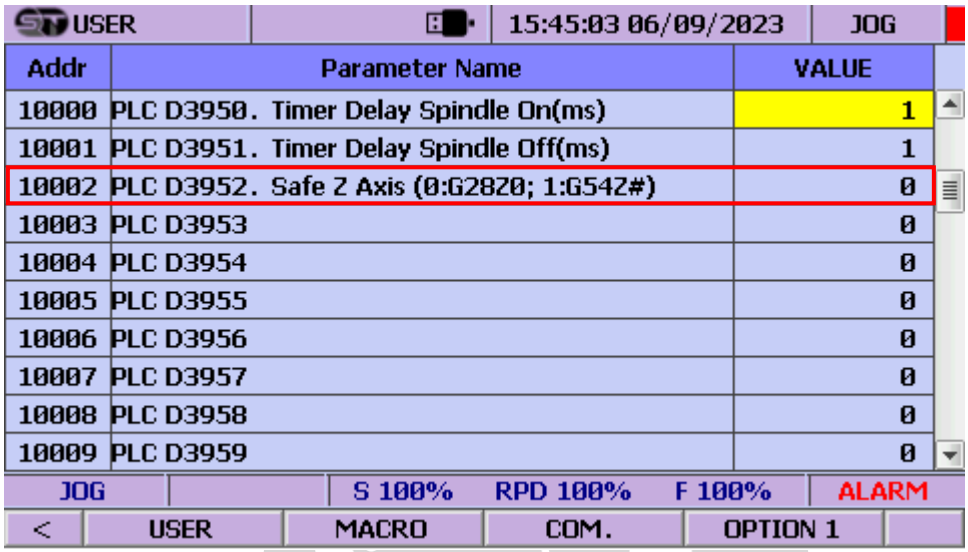


Explanation:

Stage 1: When executing M100 command, spindle and coolant motor system will be turned off.

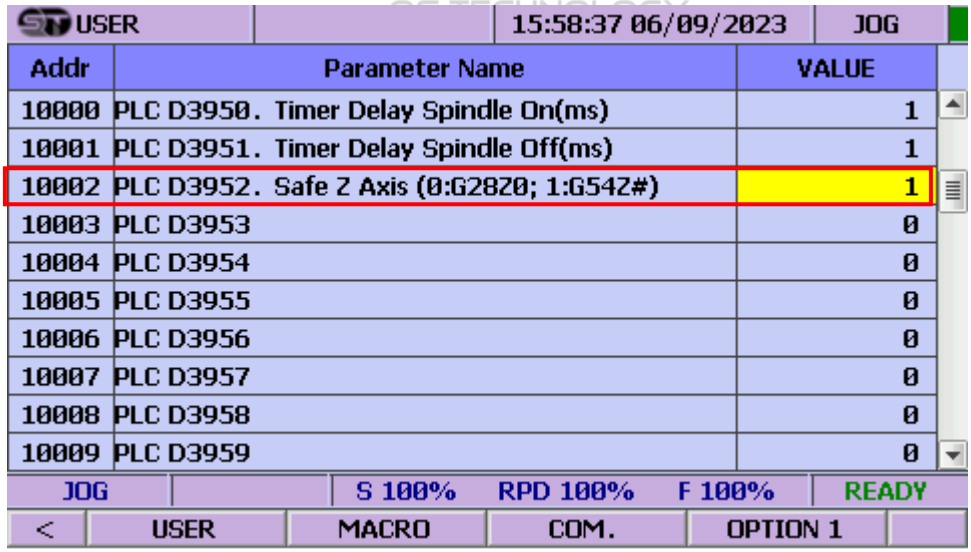
Stage 2: There are 2 cases when changing the parameter #10002

- **Case 1:** When #10002 = 0



"At this point, the Z-axis will move to the machine zero position (Machine pos = 0). At the same time, the X and Y axes will move to bring the tool to the workpiece origin at X0Y0. The Z-axis will default to the commands G91 G28 Z0.

- **Case 2:** When #10002 = 1

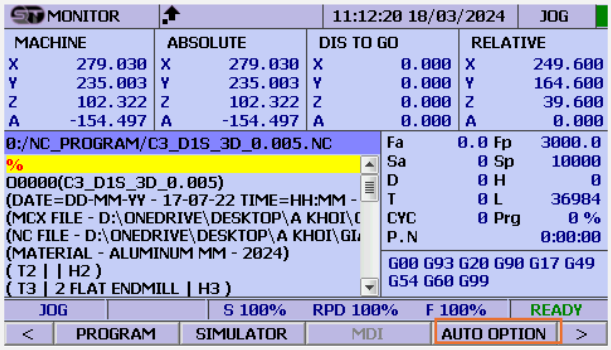


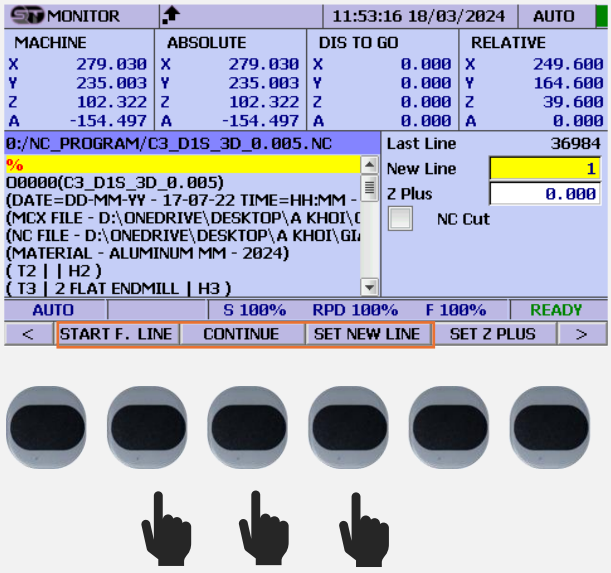
At this point, the Z-axis will move to the machine zero position (Machine pos = 0). Simultaneously, the X and Y axes will move to bring the tool to the workpiece origin at X0Y0. After that, the Z-axis will continue moving to a position slightly above the workpiece using the G54 Z command.

3.3 Resume the program when sudden power failure occurs

These automatic features assist users in handling unexpected incidents during the machining process.

Description images
Model: F54





Instructions
Model: F54

Step 1:

From MONITOR screen, select AUTO OPTION button

Step 2:

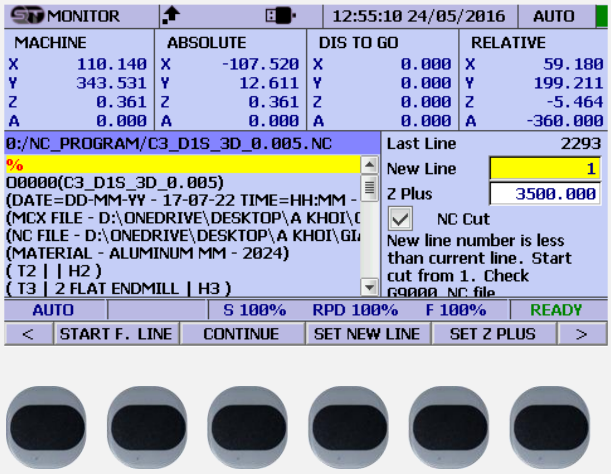
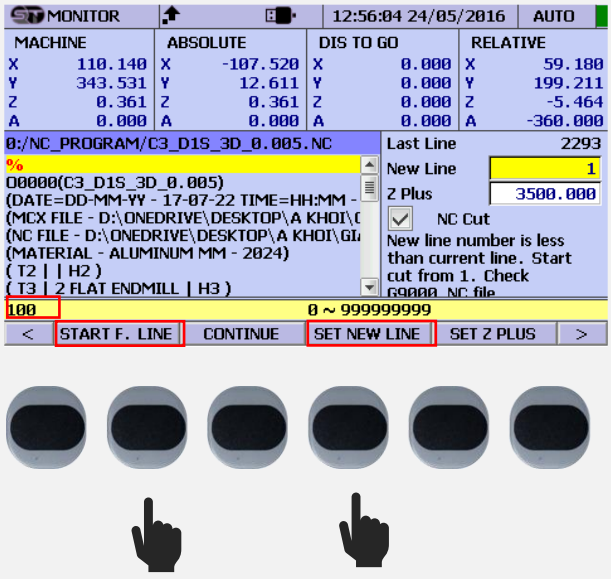
START FROM LINE: Resume the program from any command line.

CONTINUE: Resume the machining program from the line where it was previously stopped.

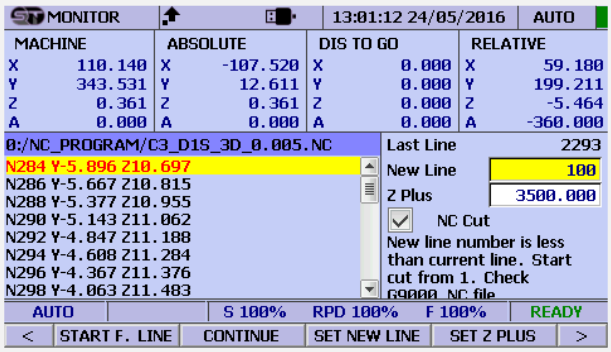
SET NEW LINE: Enter the line number where you want to start the program, then click the SET NEW LINE button to confirm.

NC CUT: Automatically generate a sub-program to activate the necessary G-code commands, such as starting the spindle, turning on the coolant, etc., when restarting the program after a machine fault, before moving to the line where it was previously stopped.

3.4 Start program from any line

Description images Model: F54	Instructions Model: F54
 	**Step 1:** Ex: We want to run program from line 100.
 	**Step 2:** Enter number 100 and press SET NEW LINE box. Then press START FROM LINE button.

Description images
Model: F54



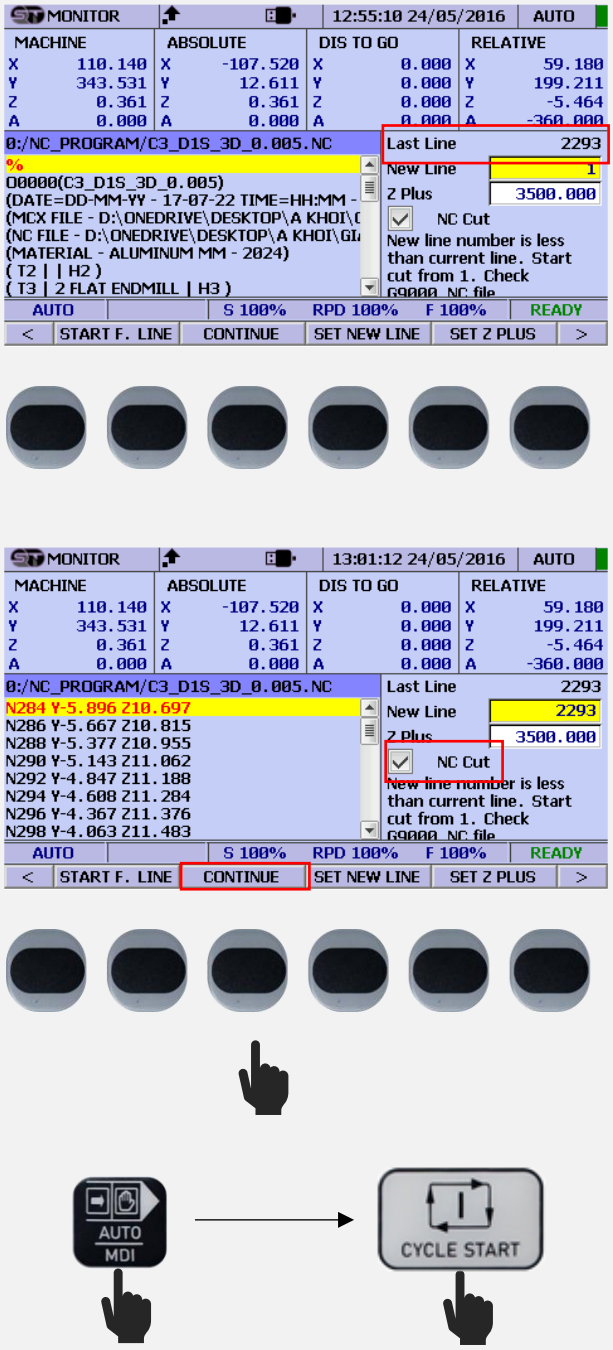
Instructions
Model: F54

Step 3:

The controller will move to line 100, but the machine will not run.

3.5 Resume the program from the line where it was previously stopped

Description images
Model: F54



Instructions
Model: F54

Step 1:

Check the line where it was previously stopped in the LAST LINE button.

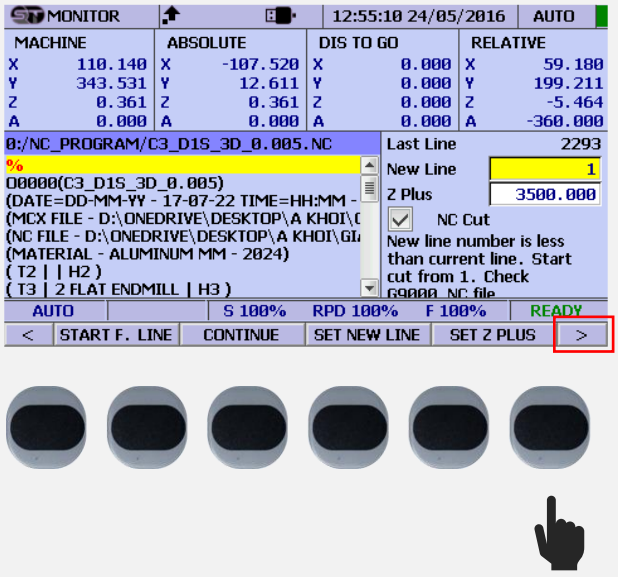
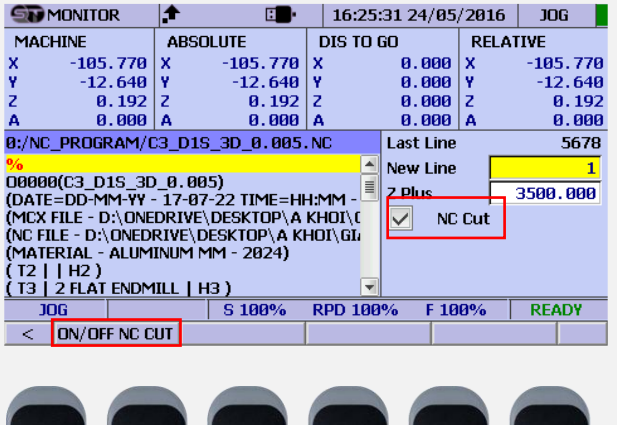
Step 2:

Click the CONTINUE button. Then switch the controller to AUTO mode by pressing the AUTO button on the control panel. Finally, check the workpiece and parameters, and press the CYCLE START button to begin the machining process.

Note:

NC CUT-Enable: When restarting the program, the controller will automatically generate a sub-program to activate the necessary G-code commands before moving to the line where it was previously stopped.

NC CUT-Disable: When restarting the program, the controller will not generate a sub-program to activate the necessary G-code commands and will move and run directly from the line where it was previously stopped.

Description images Model: F54	Instructions Model: F54
 	**Step 1:** From the monitor screen, click the right arrow button to go to the next page.
 	**Step 2:** Press ON/OFF NC CUT button to enable or disable the feature.

PART 4 OPERATION INTERFACE

4.1 Main screen

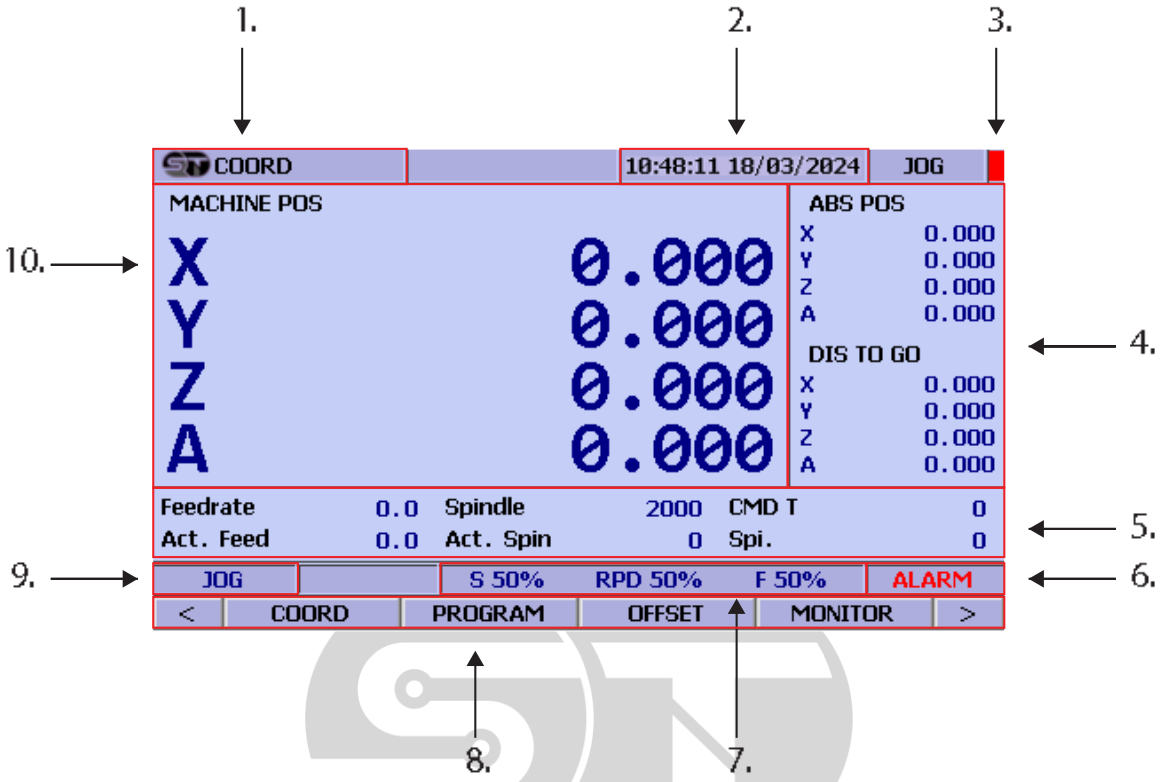


Figure 104. Main screen of F54 controller

MODEL	F54
1	The name of the coordinate display screen
2	Date-Time
3	Status light
4	Absolute cooridnates (ABS) và Distance to go coordinates
5	Feedrate and spindle speed in programming and in practice. Also, the tool currently in the spindle.
6	Operation status (Alarm/Busy/Warning)
7	Spindple speed percentage, Rapid, Feedrate
8	Function bar
9	Operation mode
10	Machine coordinates system

Note

- **Distance to go:** The remaining distance the machine needs to move to complete a command or machining program.
- **Tọa độ Absolute:** Là hệ tọa độ để xác định vị trí của công cụ cắt dựa trên một điểm gốc cố định

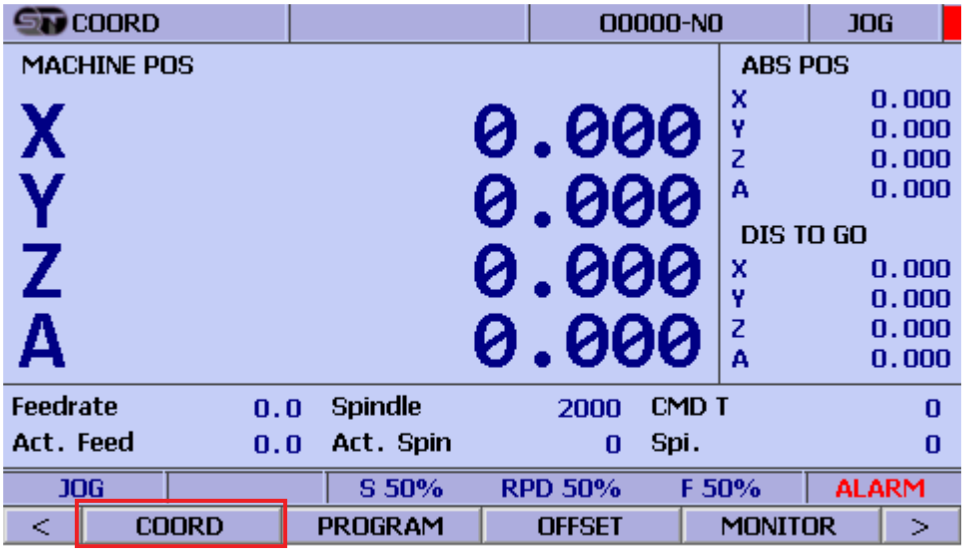
4.2 Coordinate sreen

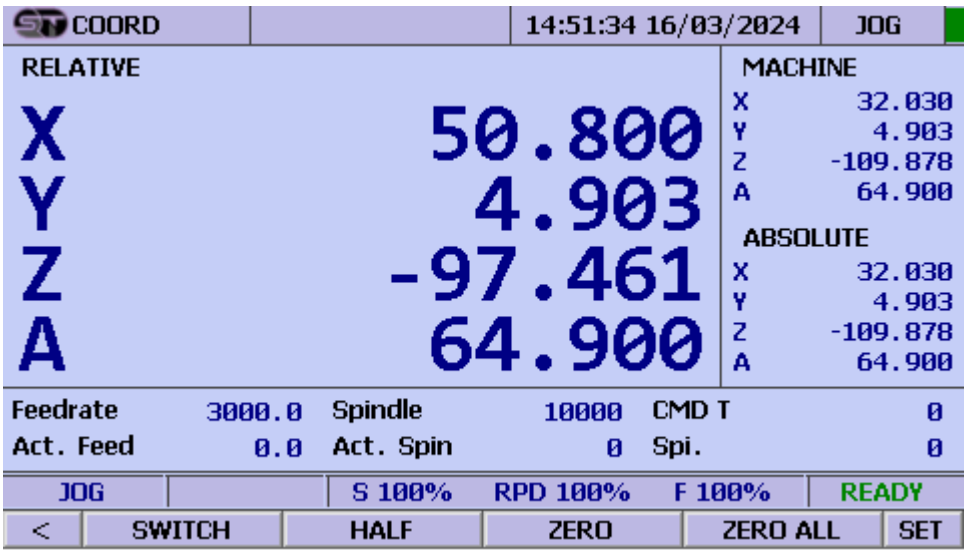
On the coordinate screen, the user can view and monitor the machine's operational parameters and the machine's coordinate system, including: Absolute coordinate system (ABS), Relative coordinate system (REL), and Distance to go.

Functions in the coordinate screen:

- **Switch:** Switching display screen (MACH/ABS/REL/DIS)
- **Half:** Divide the specified axis coordinates in the relative coordinates (REL) by 2
- **Zero:** Set the specified axis coordinates in the relative coordinates (REL) to position 0
- **Zero all:** Set the coordinates of axes in the relative coordinates (REL) to position 0.

Model: F54 controller



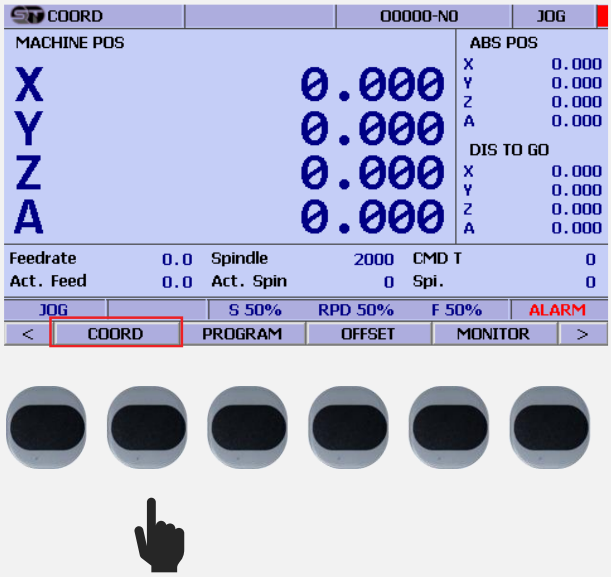
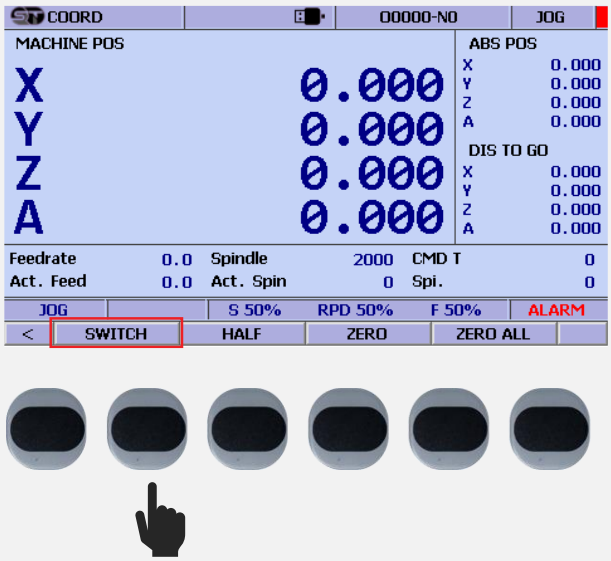


Instructions:

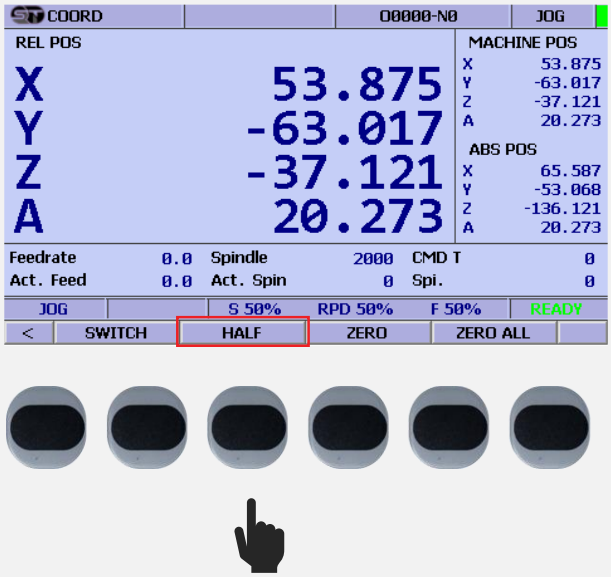
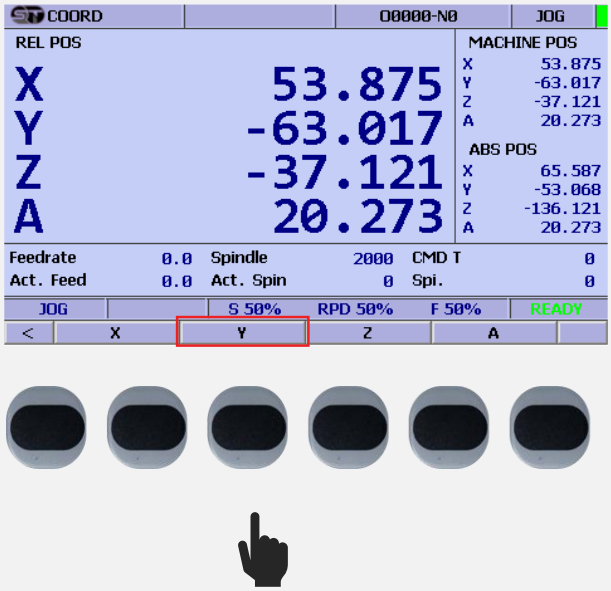
- Switching coordinate system screen **(4.2.1)**
- Divide the specified axis coordinates in the relative coordinates (REL) by 2 **(4.2.2)**
- Set the specified axis coordinates in the relative coordinates (REL) to position 0. **(4.2.3)**

QS TECHNOLOGY

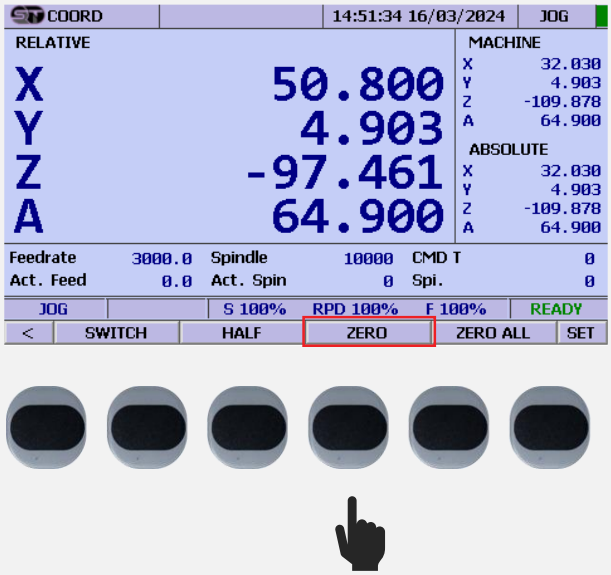
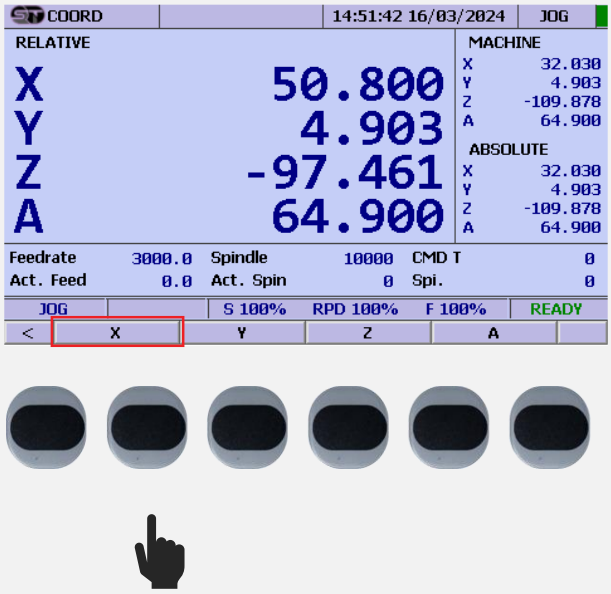
4.2.1 Switching coordinate system screen

Description images Model: F54	Instructions Model: F54
	**Step 1:** From main screen, select COORD button
	**Step 2:** Press the SWITCH button to toggle the coordinate display. **Note:** Order of coordinate display: Machine→Absolute→DIS→Relative

4.2.2 Divide the specified axis coordinates in the relative coordinates (REL) by 2

Description images Model: F54	Instructions Model: F54
 	**Step 1:** From relative coordinates screen, select HALF.
 	**Step 2:** Select the axis you want to divide the coordinates by 2. **Ex:** For Y axis

4.2.3 Set the specified axis coordinates in the relative coordinates (REL) to position 0

Description images Model: F54	Instructions Model: F54
 	**Step 1:** From relative coordinates screen (REL), select ZERO button.
 	**Step 2:** Select the axis you want to set to position 0. **Ex:** For X axis
	Note: Follow the same steps as above when you want to set the coordinates of all axes in the relative coordinate system (REL) to position 0.

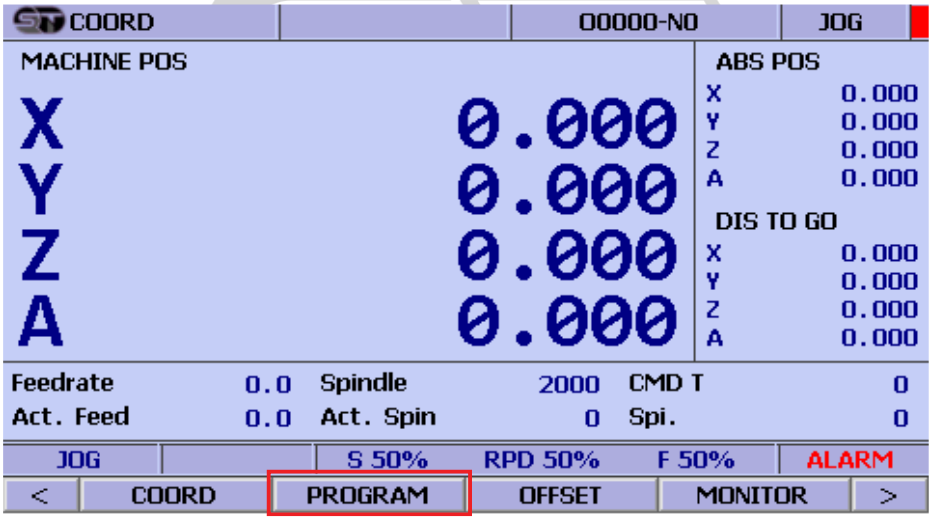
4.3 Program screen

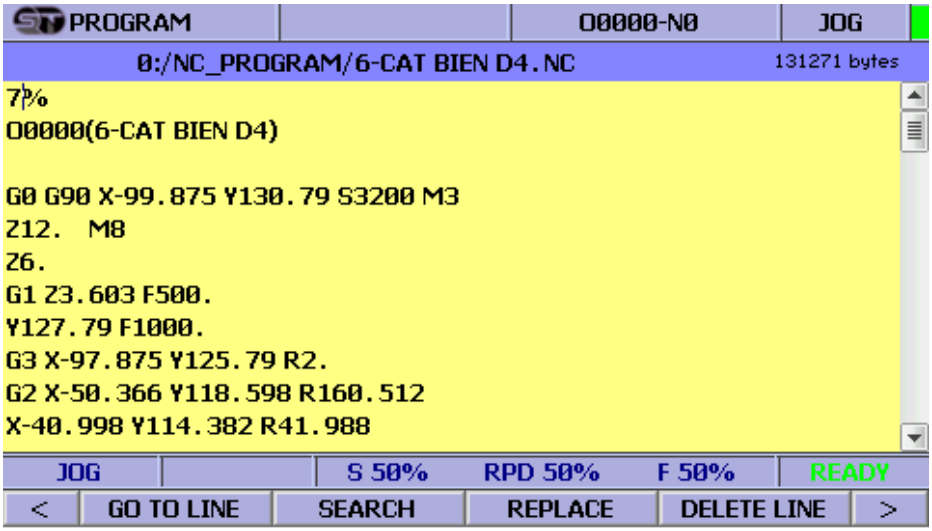
On this page, the user can store, edit, or load the program to execute.

Functions in program screen:

- **Execute**: Load the program for execution.
- **New file**: Create a new file.
- **Open file**: Access the device's memory or open a file/folder.
- **Go to line**: Move to the specified line.
- **Search**: Search for an object within the program.
- **Replace**: Replace an existing object in the program with another object.
- **Delete line**: Delete a line of code.
- **Copy from**: Copy an object from the current position.
- **Copy to**: Specify where to paste the object.
- **Paste**: Confirm and paste the object to the specified location

Model: F54 controller



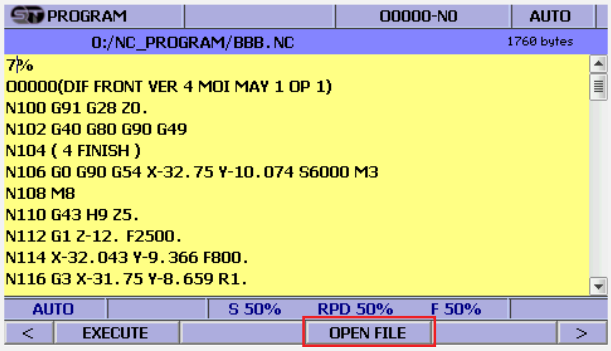
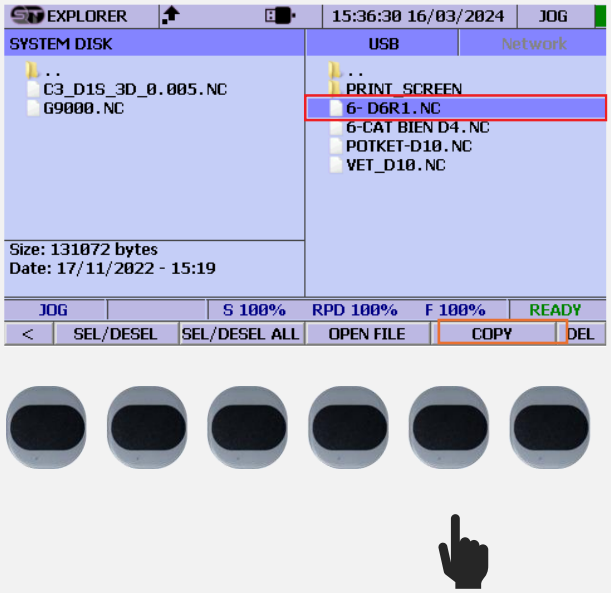


Instructions:

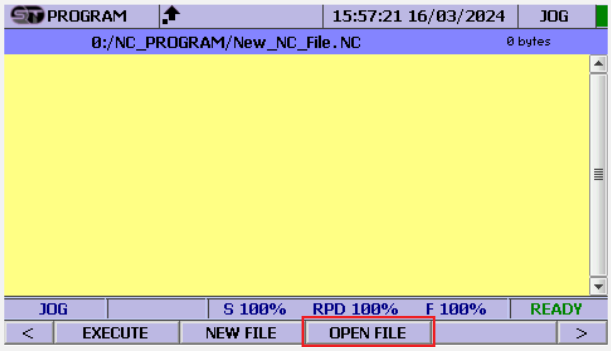
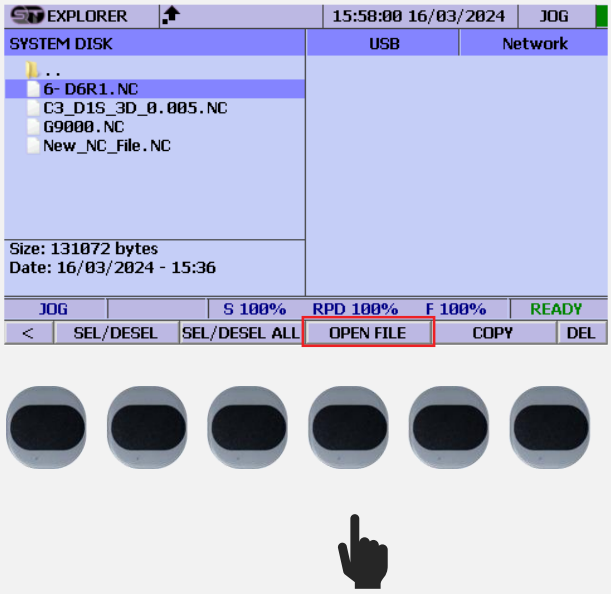
- Copy data from USB to the controller **(4.3.1)**
- Execute program **(4.3.2)**
- Go to any line in the program **(4.3.3)**
- Search for and replace objects in the program **(4.3.4)**
- Delete a program line **(4.3.5)**
- Copy and paste **(4.3.6).**

QS TECHNOLOGY

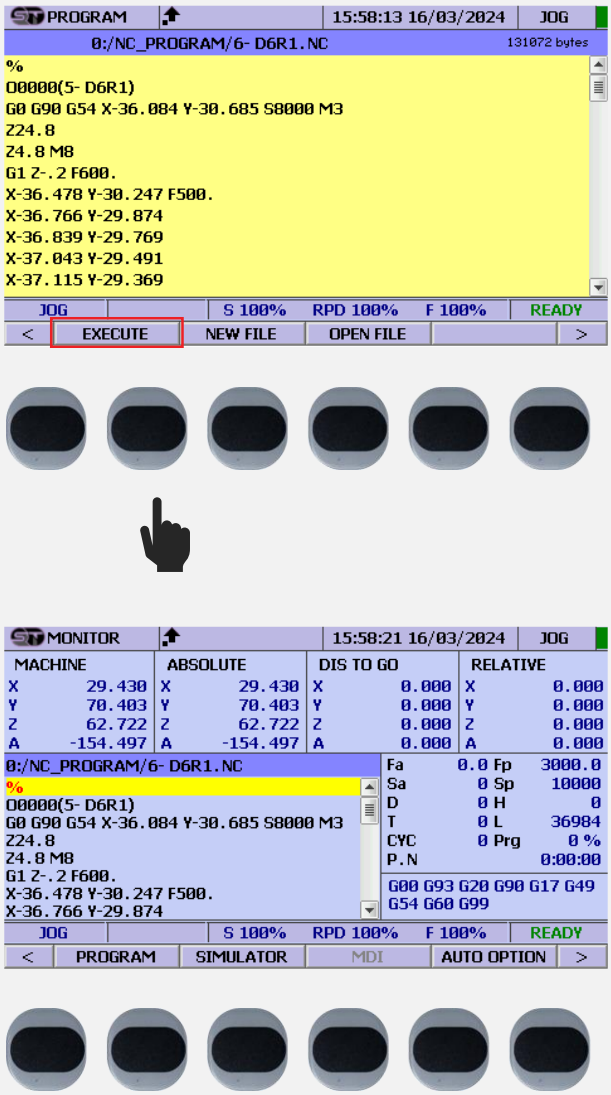
4.3.1 Copy data from USB to the controller

Description images Model: F54	Instructions Model: F54
	**Step 1:** From the editor screen, select the OPEN FILE button to access the controller's memory.
	**Step 2:** Insert the USB device into the controller, then select the program/data and press the COPY button.

4.3.2 Execute program

Description images Model: F54	Instructions Model: F54
	**Step 1:** From the editing screen, select the OPEN FILE button to access the controller's memory.
	**Step 2:** Select the program and then press the OPEN FILE button to continue.

Description images
Model: F54



Instructions
Model: F54

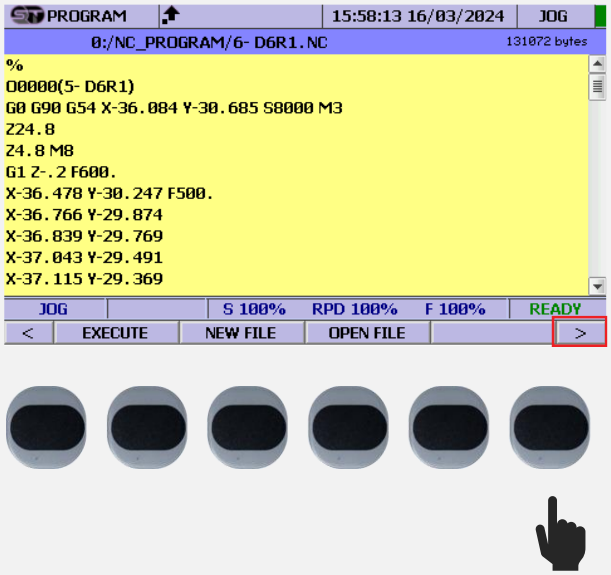
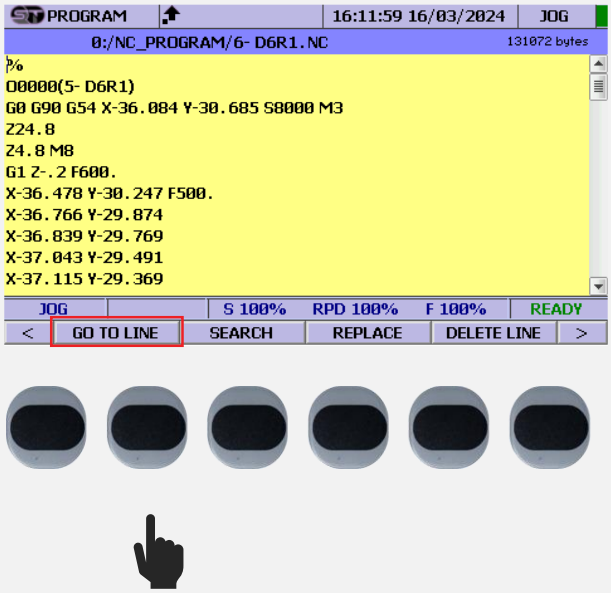
Step 3:

Next, press the EXECUTE button to execute the program.

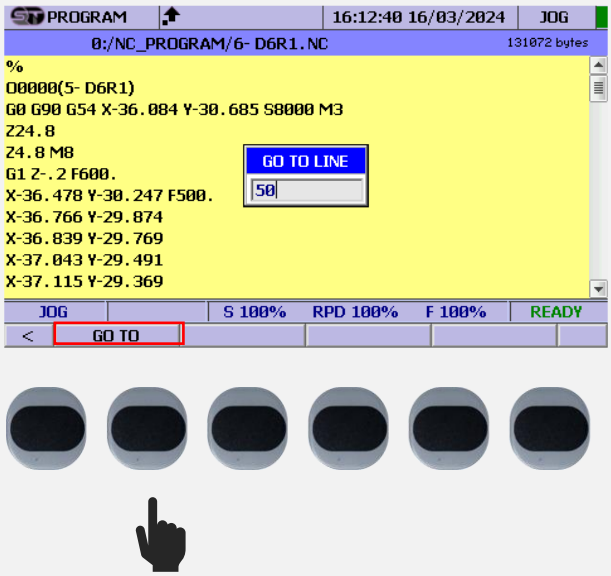
Step 4:

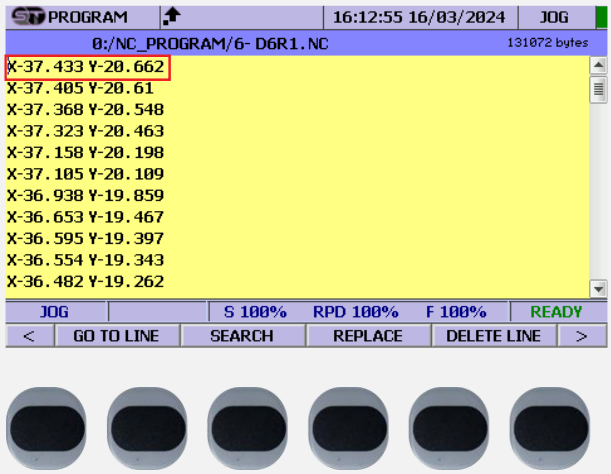
The controller will switch to the monitoring screen interface. The operator must check the program and the workpiece before starting the machining process.

4.3.3 Go to any command line

Description images Model: F54	Instructions Model: F54
 	**Step 1:** From the editing screen, press the right arrow button to move to the next page.
 	**Step 2:** Press GO TO LINE button

Description images
Model: F54





Instructions
Model: F54

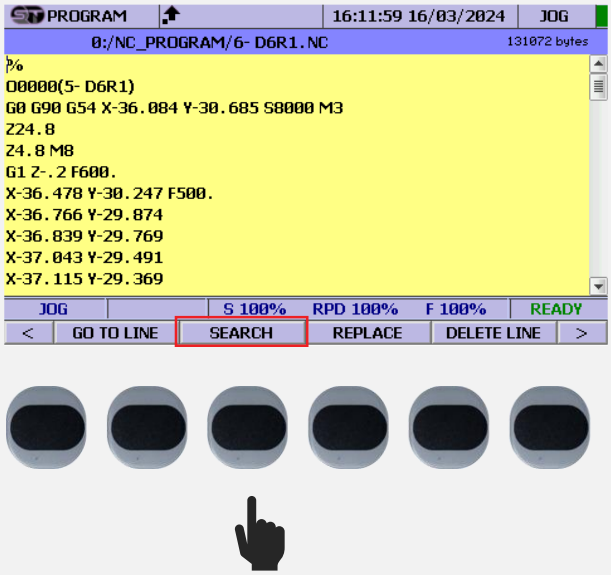
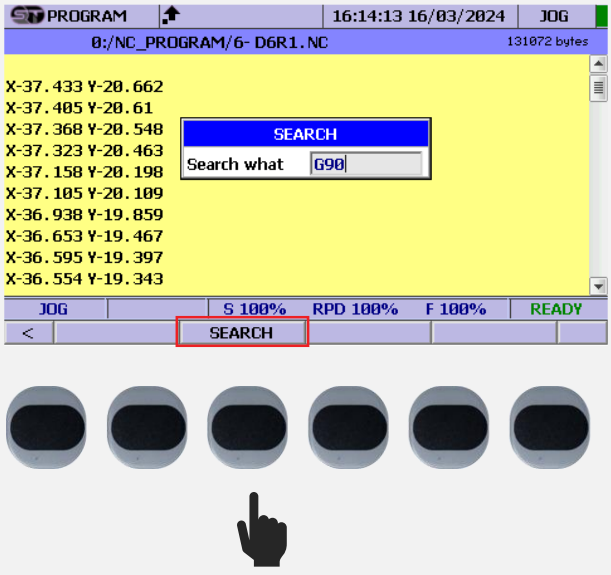
Step 3:

Enter the line number you want to jump to, then press the GO TO button.

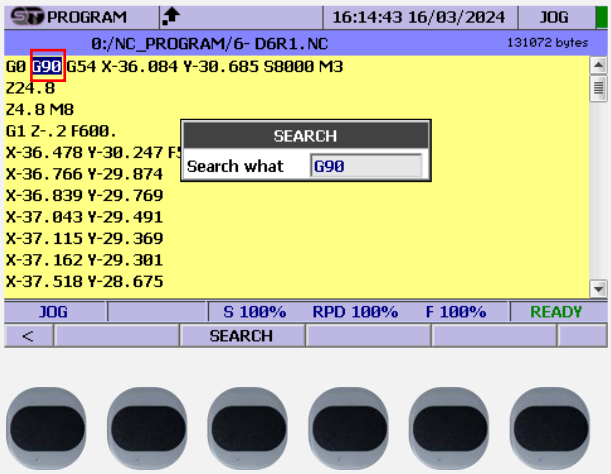
Step 4:

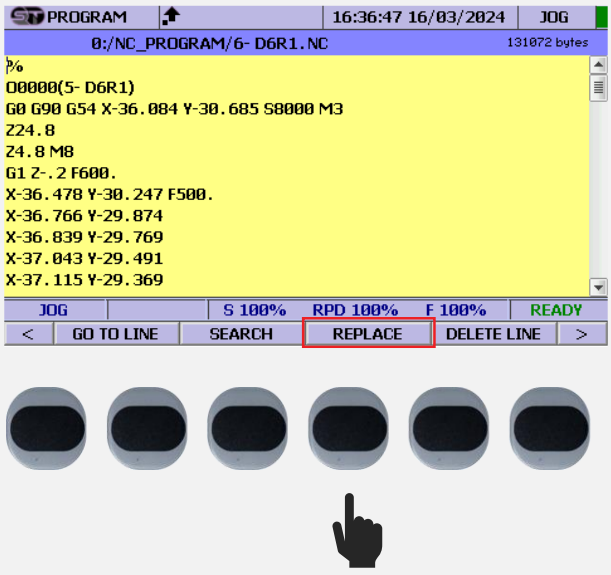
After pressing the GO TO button, the controller will automatically locate and jump to the specified line.

3.3.4 Search and replace object in the program

Description images Model: F54	Instructions Model: F54
	**Step 1:** On the second editing page, press the SEARCH button.
	**Step 2:** Enter the name of the object you want to search for and press the SEARCH button.

Description images
Model: F54





Instructions
Model: F54

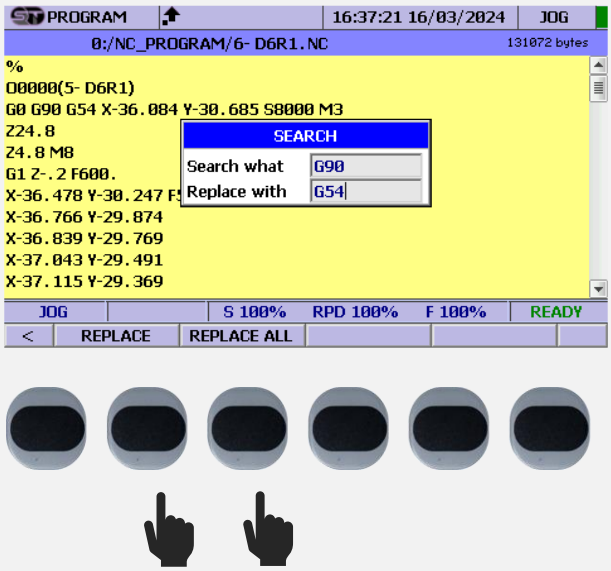
Step 3:

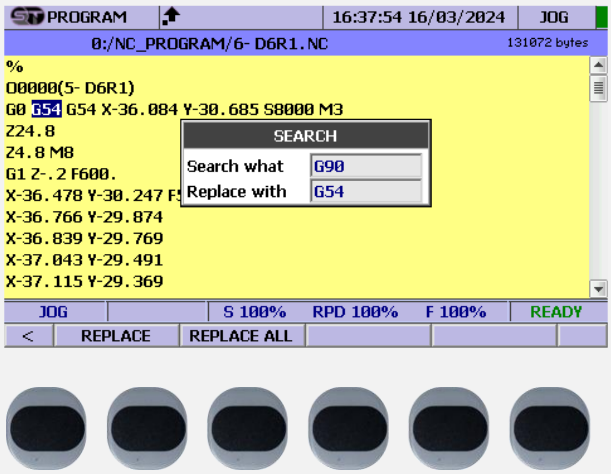
If the object exists in the program, the controller will find it and provide the result.

Step 4 (Find & Replace):

To replace the existing object in the program with another object, select the REPLACE button.

Description images
Model: F54





Instructions
Model: F54

Step 5:

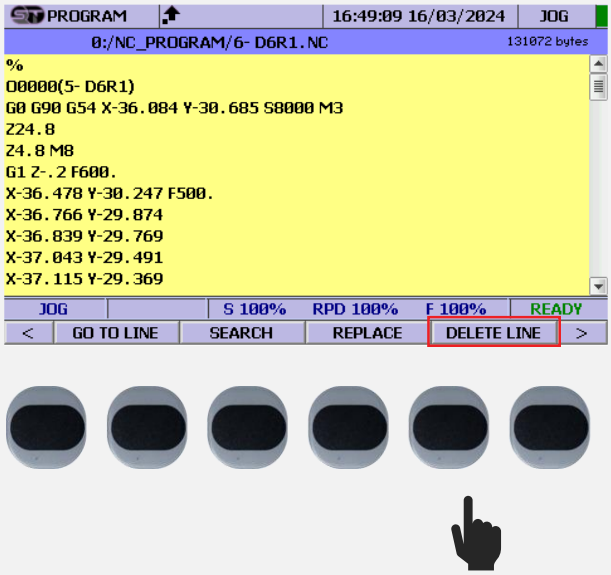
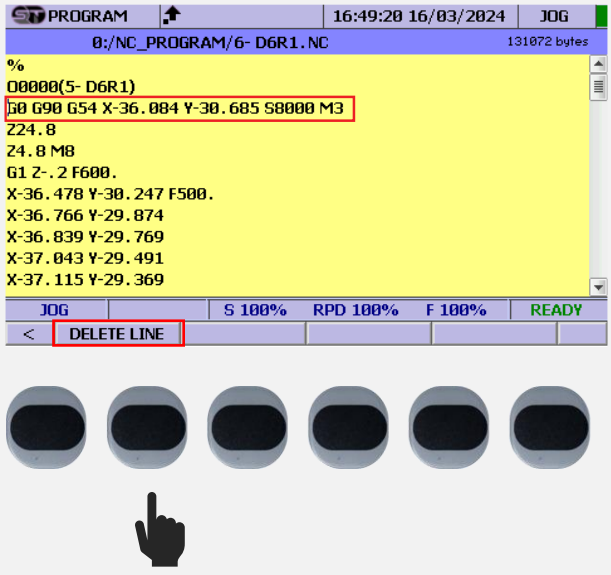
Enter the name of the object to search for in the SEARCH WHAT field and enter the name of the object to replace it with in the REPLACE WITH field.

Note:

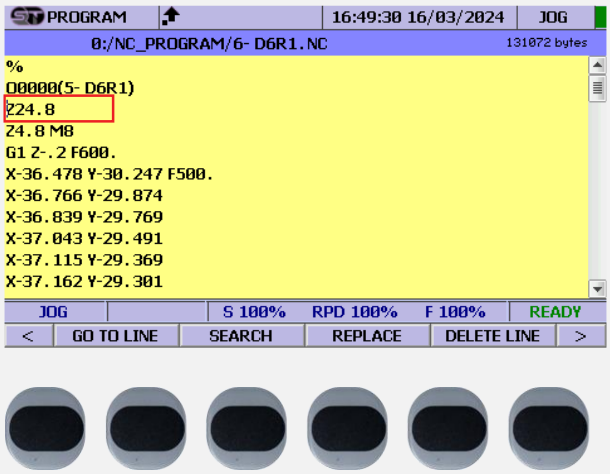
REPLACE: Only replace for 1 object

REALCE ALL: Replace for all object in the program

3.3.5 Delete command line

Description images Model: F54	Instructions Model: F54
	**Step 1:** On the second editing page, select the DELETE LINE button.
	**Step 2:** Then, select the line you want to delete and move the cursor to the beginning of the line using the arrow keys on the device's control panel. After that, press the DELETE LINE button.

Description images
Model: F54

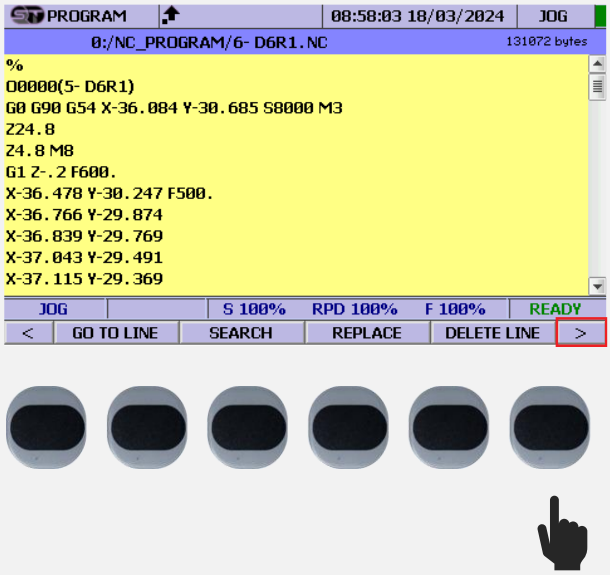
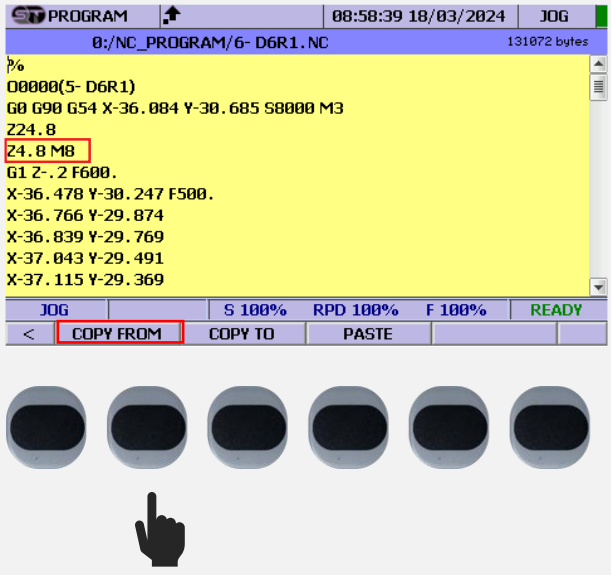


Instructions
Model: F54

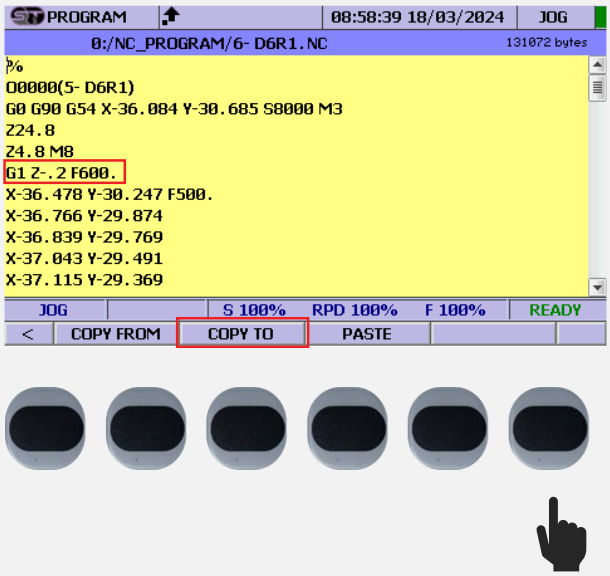
Step 3:

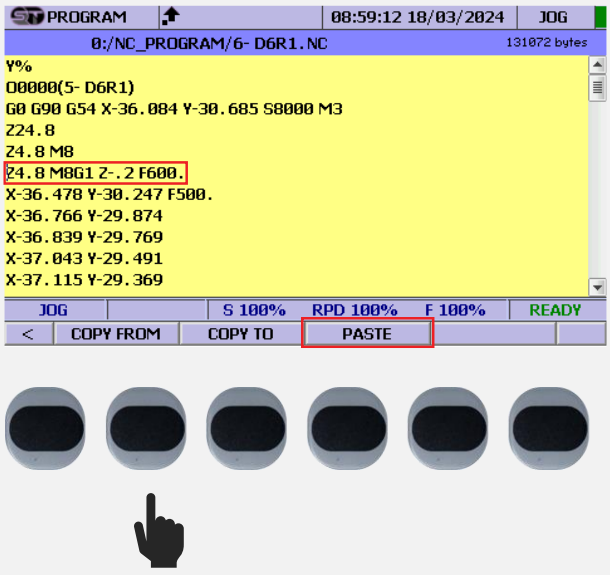
Check the line that was just deleted.

4.3.6 Copy and Paste

Description images Model: F54	Instructions Model: F54
	**Step 1:** From the second editing page, press the right arrow button to move to the third editing page.
	**Step 2:** Select and move the cursor to the beginning of the command, then press the COPY FROM button.

Description images
Model: F54





Instructions
Model: F54

Step 3:

Select the location where you want to paste the command and press the COPY TO button to confirm the paste position.

Step 4:

Press the PASTE button to paste the command at the specified location.

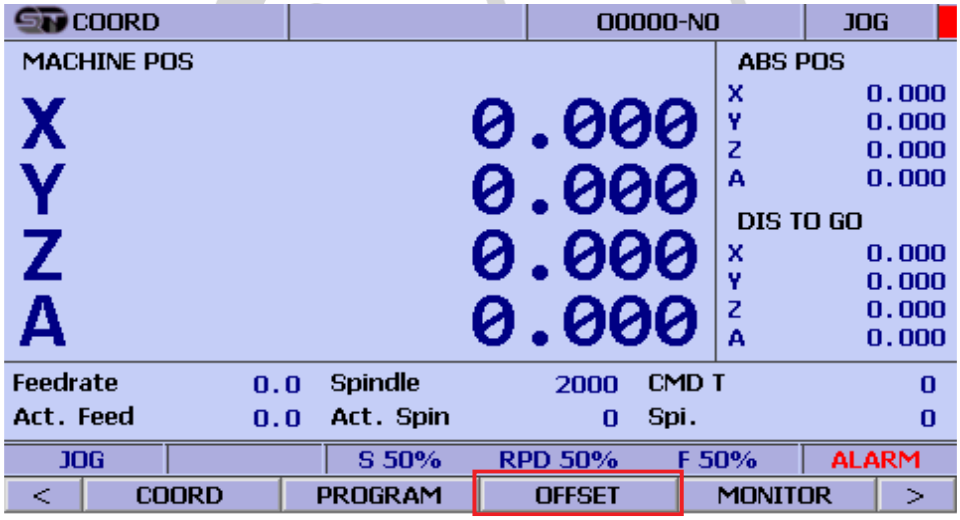
4.4 Offset screen

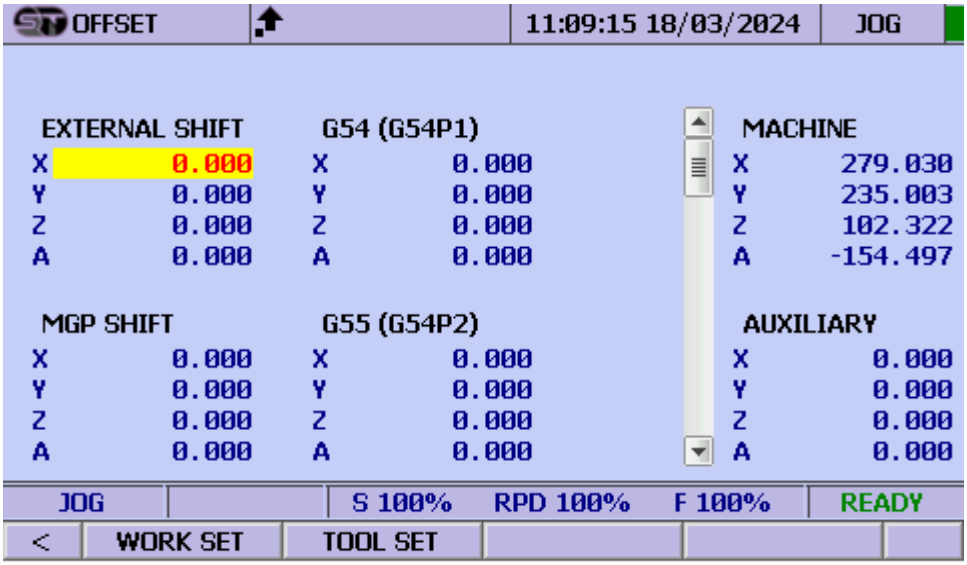
This page allows the user to set up the parameters of the workpiece as well as the height and diameter of the cutting tool.
The F54 controller provides the user with 6 coordinate systems, including: G54, G55, G57, G58, and 18 extended coordinate systems from G59.1 (G54P7) to G59.12 (G54P18).

Functions in offset screen:

- **Work set**: Set the parameters of the workpiece.
- **Tool set**: Set the height and diameter of the cutting tool.
- **Tool NO**: Display of the tools available in the tool holder.
- **Apply AUX**: Assign the coordinates calculated from the rapid center function to the specified coordinate system.
- **Apply MACH**: Assign the machine's coordinates to the specified coordinate system.
- **Middle function**: Calculate the center coordinates based on 4 points for rectangular workpieces and 3 points for circular workpieces

Model: F54 controller



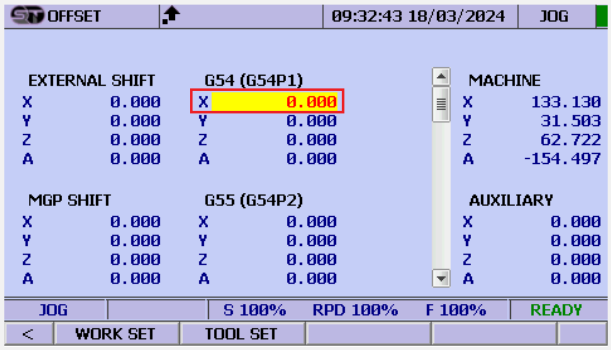
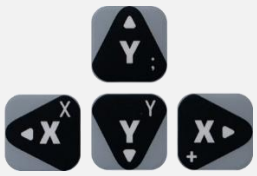
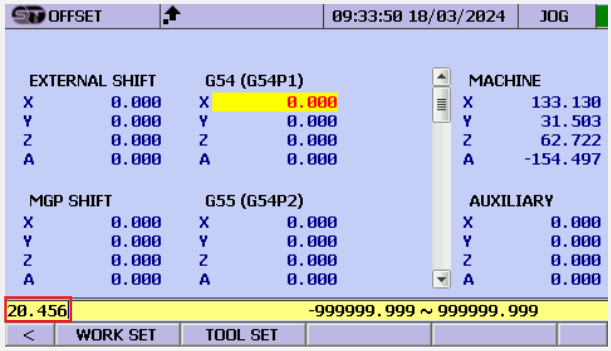


Instructions:

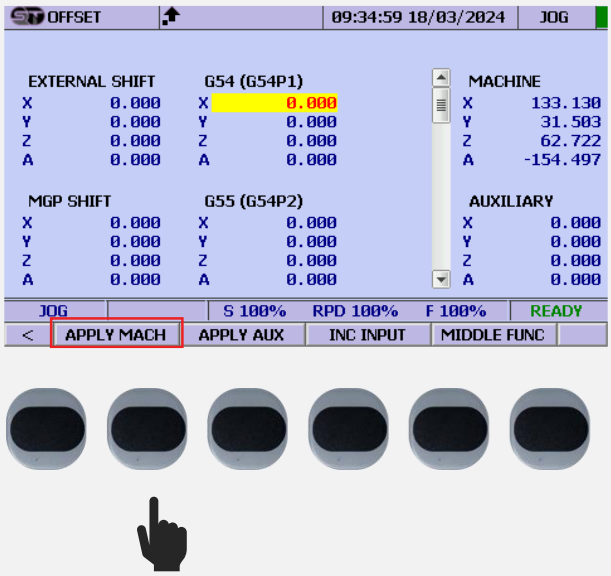
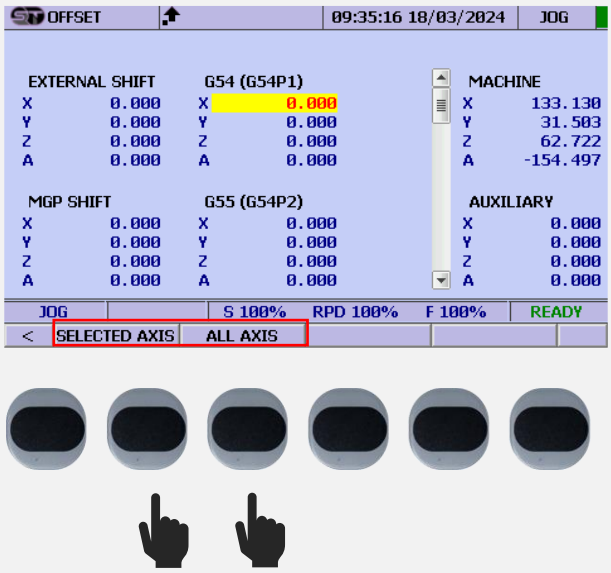
- Manual data input **(4.4.1)**
- Assign machine coordinates and auxiliary coordinates to the specified coordinate system **(4.4.2)**
- Quickly determine the center coordinates based on 4 points and 3 points of the workpiece. **(4.4.3)**
- Cutting tool parameter setting **(4.4.4).**

QS TECHNOLOGY

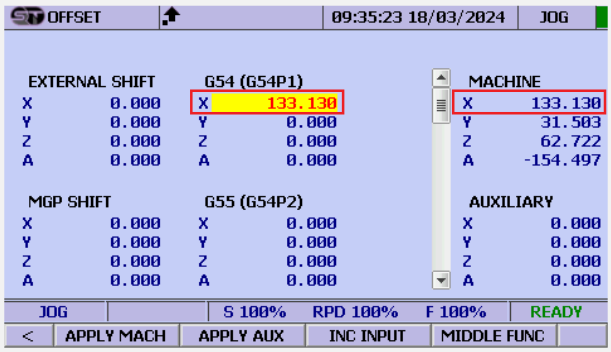
4.4.1 Manual data input

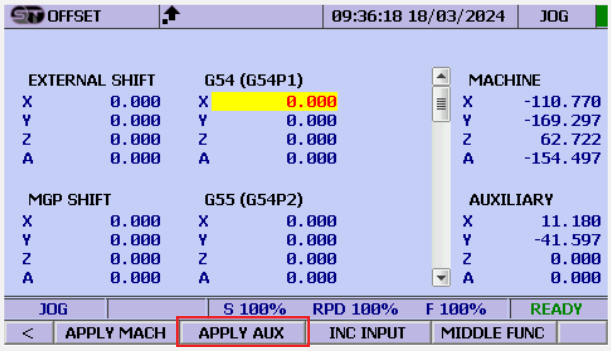
Description images Model: F54	Instructions Model: F54
     	**Step 1:** Use the arrow keys on the control panel to select the coordinate system. **Example:** here we select the coordinate system (G54) and the X-axis coordinate. **Step 2:** Enter the coordinate values using the numeric and character keys on the control panel, then press the Enter button.

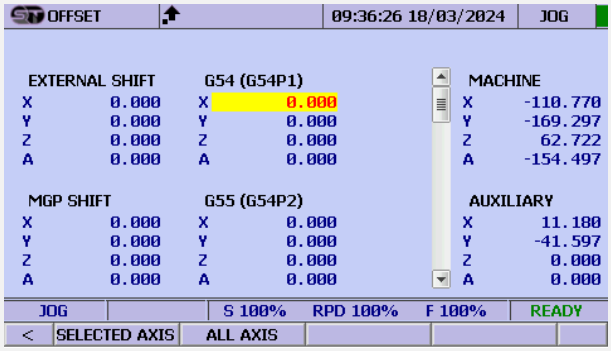
4.4.2 Assign the coordinates and auxiliary coordinates to the specified coordinate system

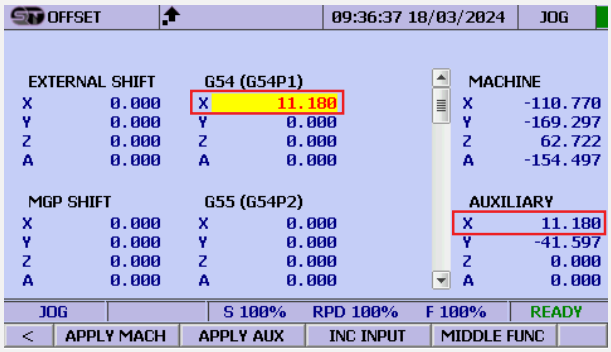
Description images Model: F54	Instructions Model: F54
	**Step 1:** Select the coordinate system and axis where you want to assign the machine coordinates. Then press the APPLY MACH button. **Example:** We assign the machine coordinates to the coordinate system (G54), specifically for the Xaxis.
	**Step 2:** Press the SELECTED AXIS button to assign the machine's axis coordinates to the corresponding axis of the specified coordinate system. Select ALL AXIS to assign all of the machine's axis coordinates to the corresponding axes of the specified coordinate system.

Description images
Model: F54









Instructions
Model: F54

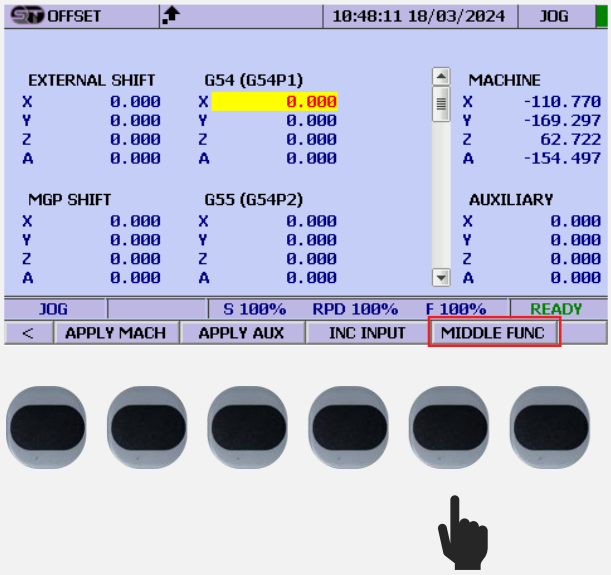
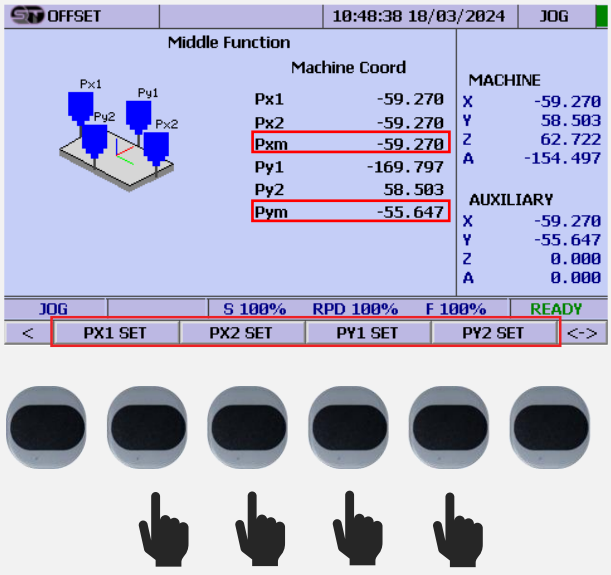
Step 3:

Check the coordinates that were just assigned from the machine's axis coordinates.

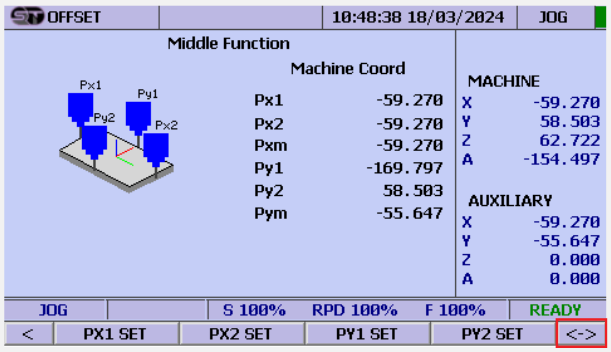
Note:

Follow the same steps as above when assigning auxiliary coordinates to the coordinate system. Instead of pressing APPLY MACH, you will press APPLY AUX. The AUX coordinates will be calculated using the MIDDLE FUNCTION, see section **(4.4.3)** below for more details.

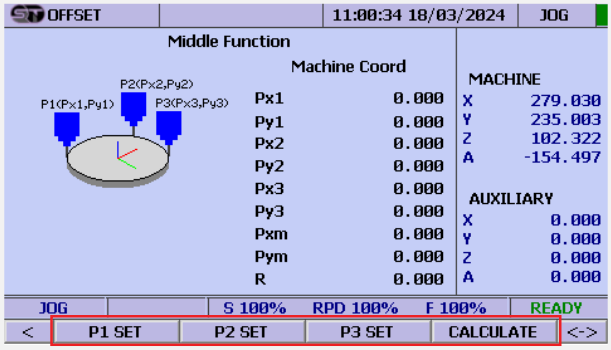
4.4.3 Quickly determine the center coordinates for a 3-point or 4-point workpiece

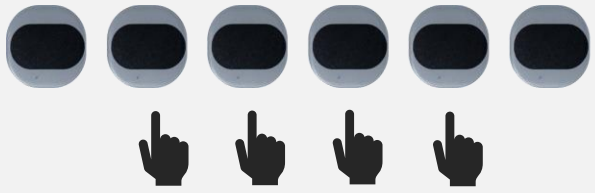
Description images Model: F54	Instructions Model: F54
	**Determine the center coordinates using 4 points of the workpiece.** **Step 1:** Select MIDDLE FUNCTION button
	**Step 2:** Move the axes and touch the 4 edges of the workpiece (PX1, PX2, PY1, PY2) as shown in the diagram. Once the process is complete, the center coordinates of the workpiece will be automatically calculated and displayed at (PXM, PYM) and the auxiliary coordinate system (AUX).

Description images
Model: F54









Instructions
Model: F54

Determine the center coordinates using 3 points of the workpiece.

Step 1:

From the center calculation page using 4 points, select the right arrow button to move to the next page.

Step 2:

Move the machine axes and touch the 3 points of the circular workpiece (P1, P2, P3) as shown in the diagram. Once the process is complete, the center coordinates of the workpiece will be automatically calculated and displayed at (PXM, PYM, R) and the auxiliary coordinate system (AUX), where R is the radius of the workpiece.

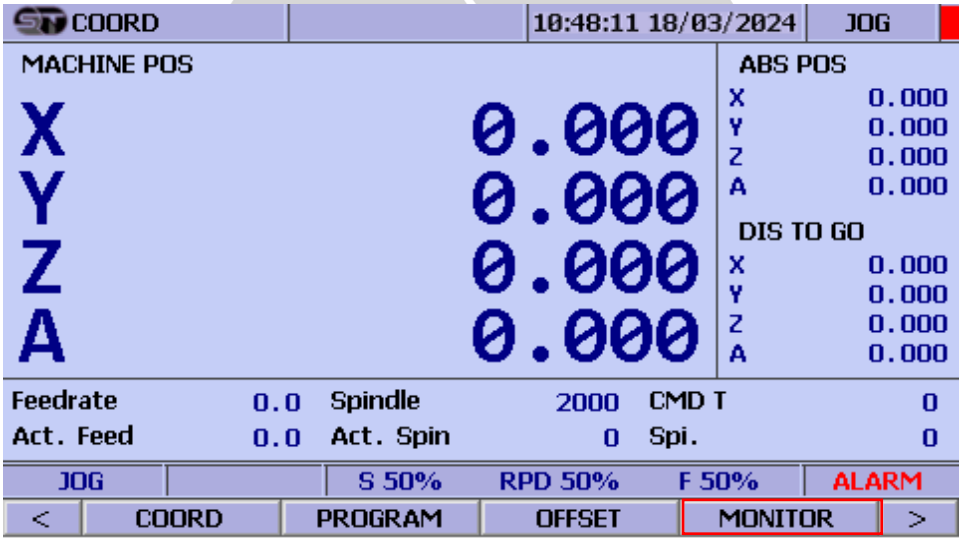
4.5 Monitor screen

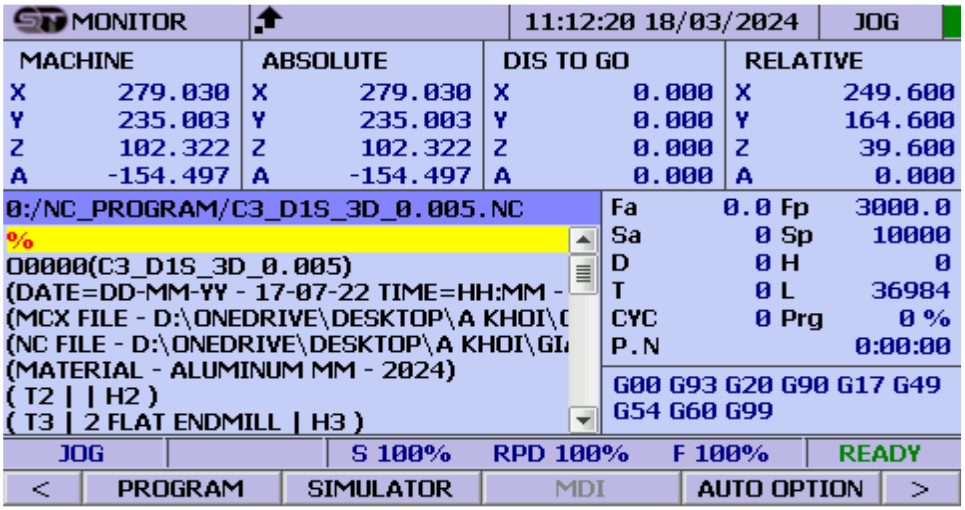
On the monitor screen, the operator can easily see information such as the machine's coordinate system, as well as: program name, operating mode, cutting speed settings, spindle,...

Functions in monitor screen

- **Program:** Open device's memory
- **Simulator:** Simulate the machining program and tool path
- **MDI:** Manual data input
- **Auto Option:** Automatic features
    - Allows resuming the program from any command line.
    - Continue the program from the last stopped command line after an unexpected issue, power failure, ...
    - Automatically generate a subprogram to activate all necessary G-code groups before resuming from the previously stopped command line.

Model: F54 controller



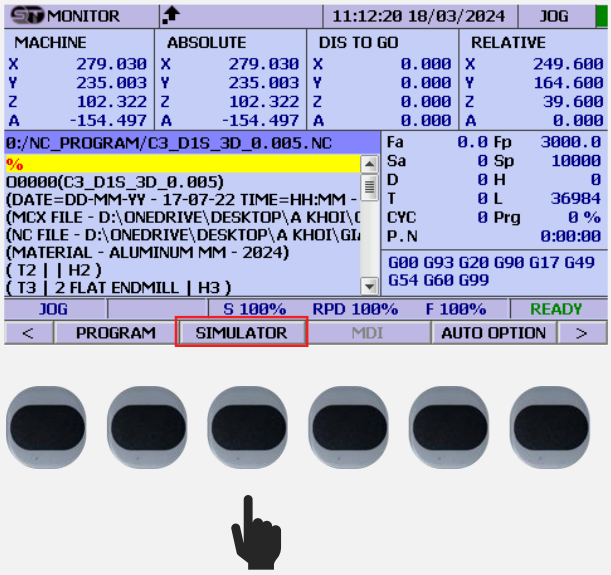


Instruction:

- Toolpath simulation **(4.5.1)**
- MDI mode **(4.5.2)**
- Working history **(4.5.3)**

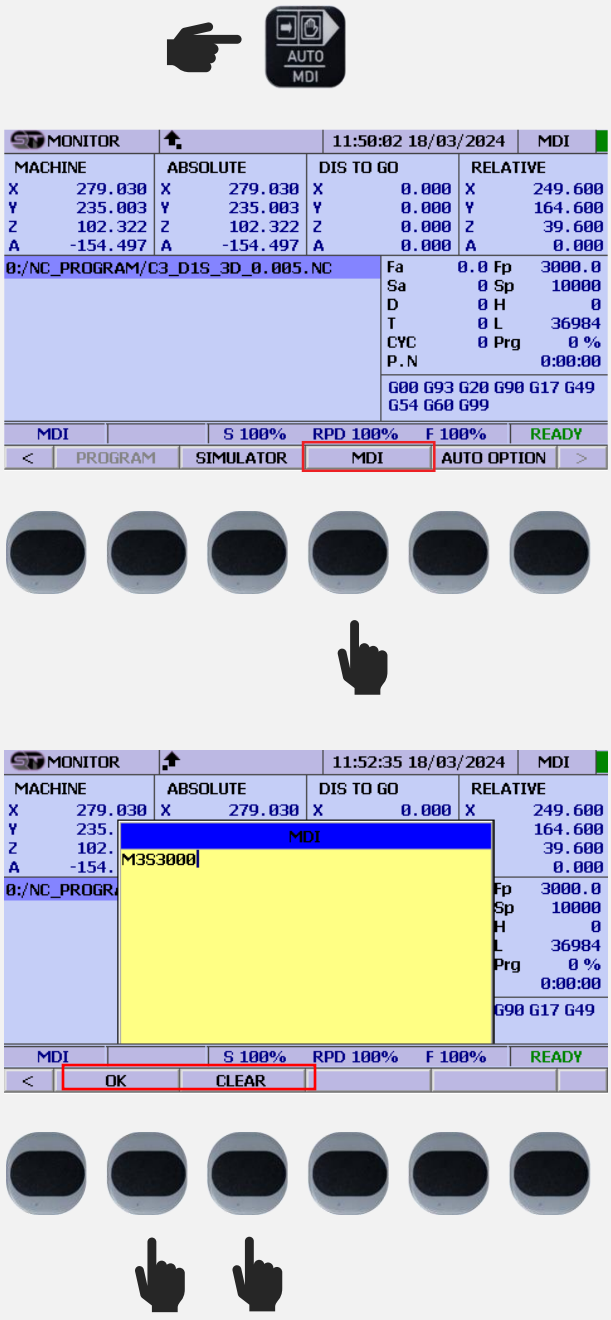
QS TECHNOLOGY

4.5.1 Toolpath simulation

Description images Model: F54	Instructions Model: F54
	**Step 1:** After calling the program for execution, on the monitor screen, press the SIMULATOR button. **Step 2:** ait for the simulation process to complete. Then press the AUTO SCALE button to fit the image to the screen. TOOL MODE: Simulation display modes — 2D Milling, 2D Engraving, 3D Milling, and 3D Engraving. INFORMATION: Displays the maximum and minimum machining travel values of the workpiece.

4.5.2 MDI mode

Description images
Model: F54



Instructions
Model: F54

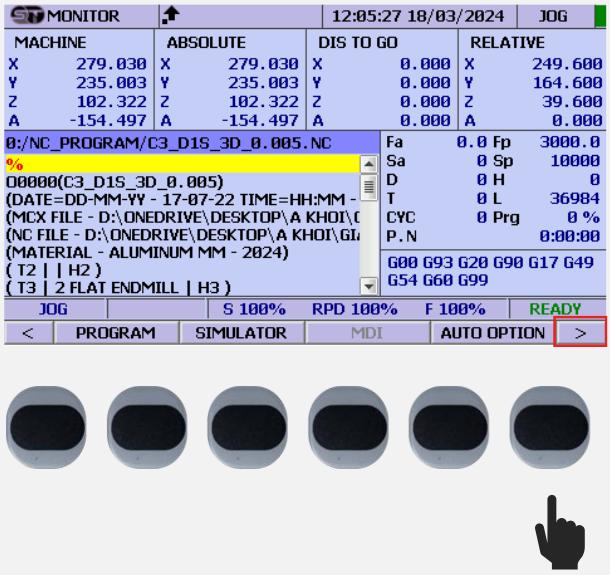
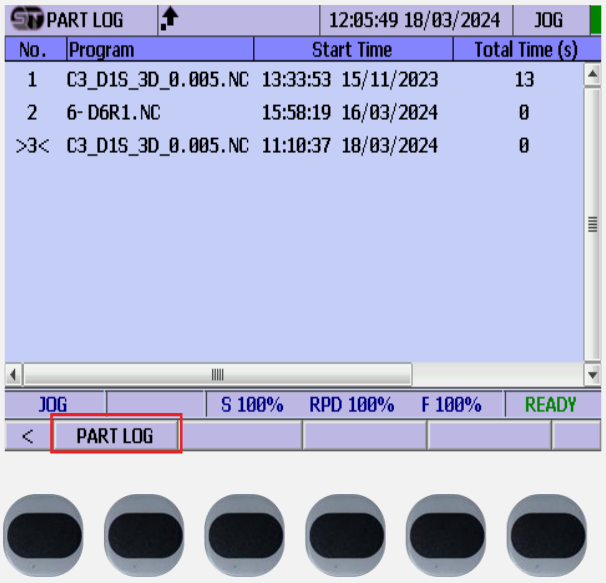
Step 1:

First, press the AUTO button on the device's control panel to switch from AUTO mode to MDI mode. At this point, the MDI function button on the monitor screen will light up, then press the MDI button.

Step 2:

Enter any command for the controller and press OK to confirm, or select CLEAR to re-enter.
For example: here we enter the command M8 to turn on the coolant.

4.5.3 Working history

Description images Model: F54	Instructions Model: F54
  	**Step 1:** From the main screen of the monitor page, press the right arrow button to move to the next page. **Step 2:** Press the PART LOG button to open the machine operation history table. In this table, the operator can view the operation history, program name, and the machining cycle completion time.

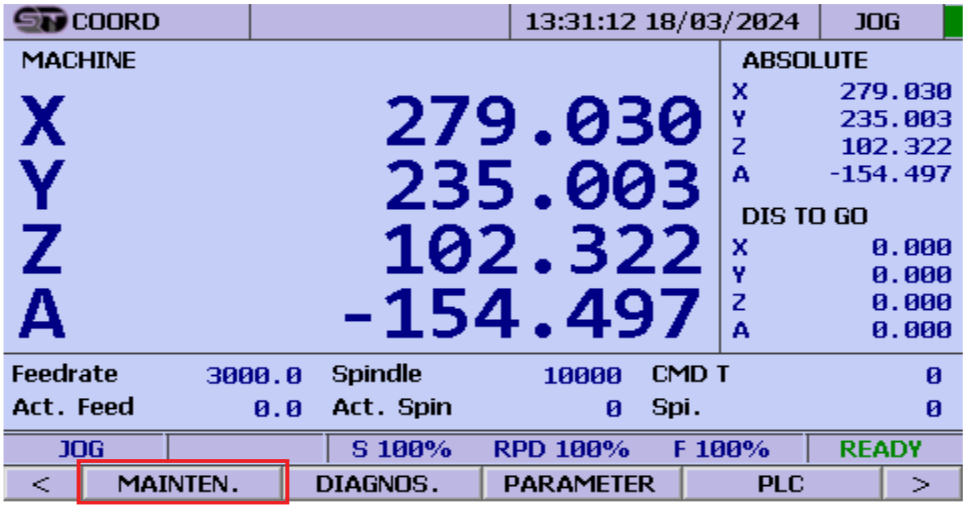
4.6 Maintenance screen

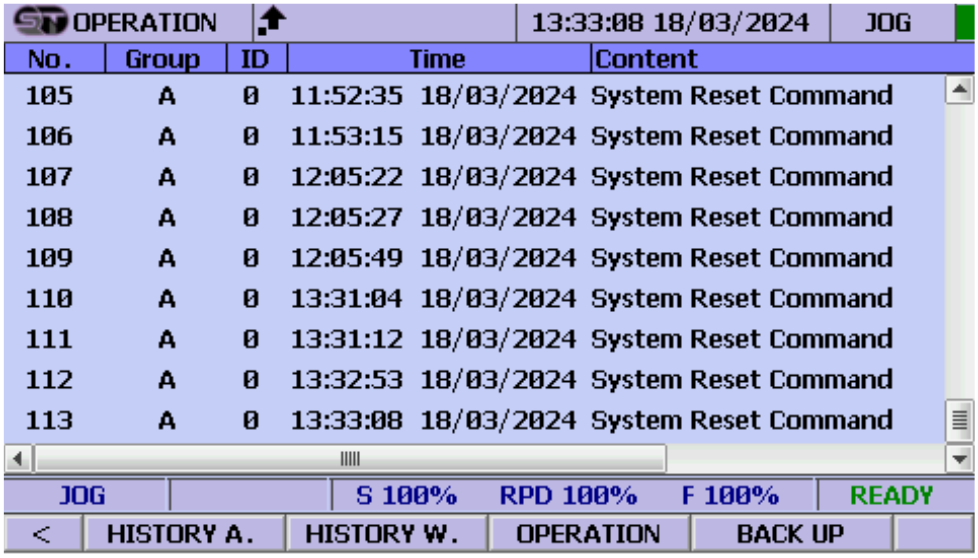
"On this screen, the operator can view the error and warning history of the controller. To access the maintenance page, press the MAINTEN button from the main screen. Additionally, the operator can save the error history for future reference.

Functions in maintenance screen:

- **Alarm:** Alarm history
- **Warning:** Warning history
- **Operation:** Openting history
- **Backup:** Alarm/History/Operation data backup.

Model: F54 controller





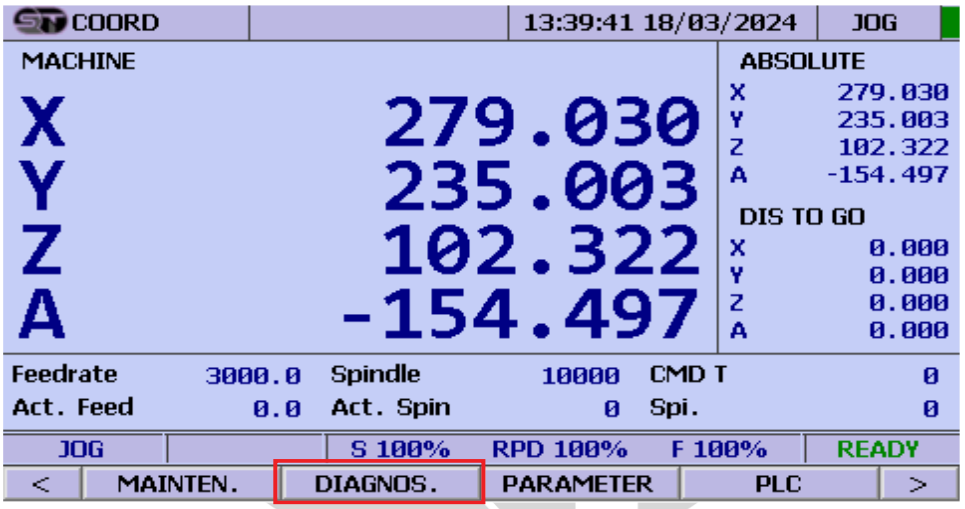
4.7 Diagnostic screen

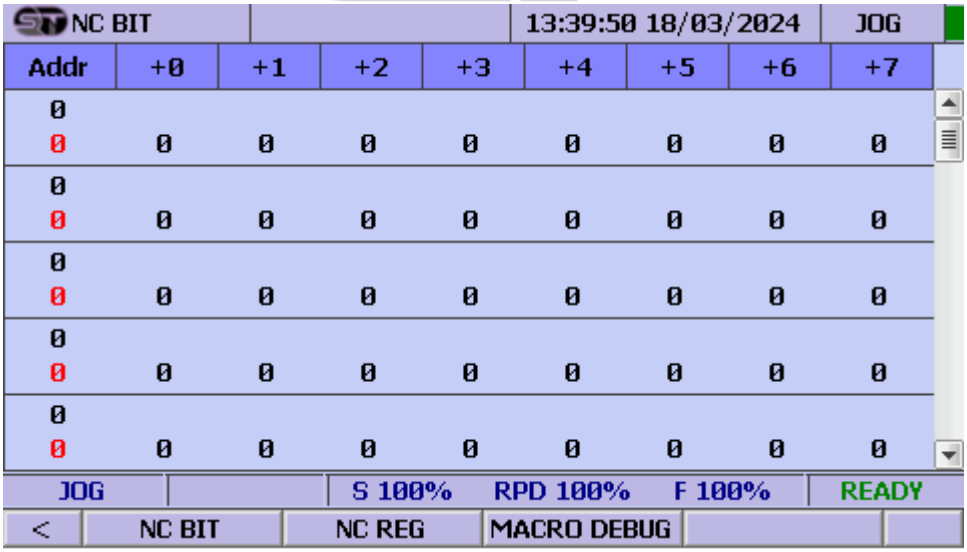
The controller of QS Technology allows users to deeply monitor system bits, registers, and macro variables through the diagnostic table.

Functions of diagnostic screen:

- **NC bit:** Open NC bit page
- **NC register:** Open NC register page
- **Macro debug:** Open Macro table (Local var & Global var)

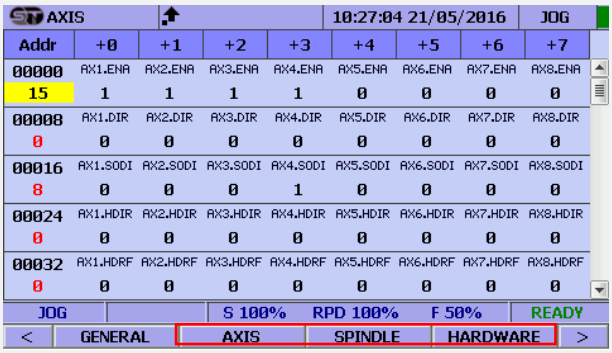
Model: F54 controller



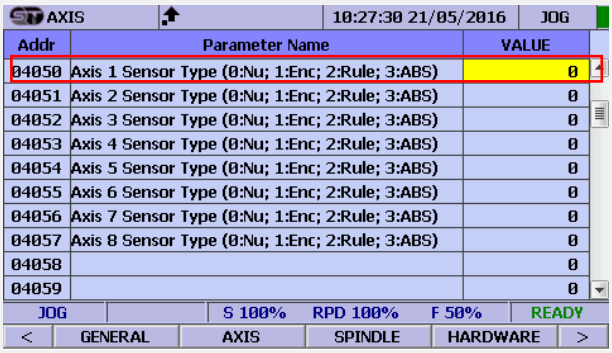


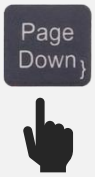
4.8 Paramters setting

Description images
Model: F54. F86, Astro 6A, Astro 10i









Instructions
Model: F54, F86, Astro 6A, Astro 10i

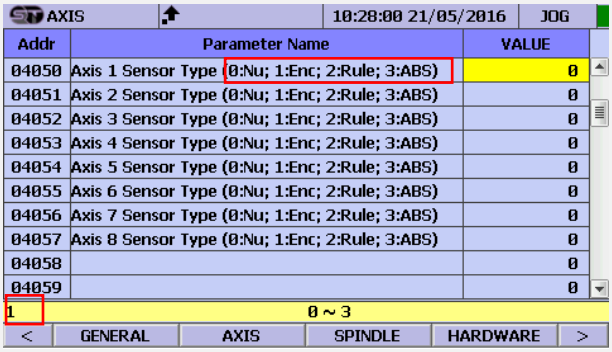
Step 1:

Select the parameter to be configured, such as axis parameters, spindle, or hardware.

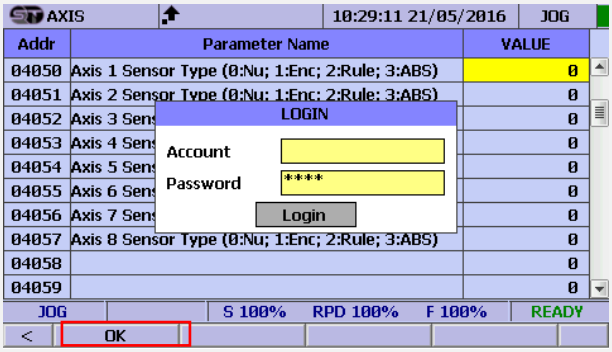
Step 2:

Press the PAGE DOWN button on the control panel to scroll down and find the required parameter.

Description images
Model: F54. F86, Astro 6A, Astro 10i











Instructions
Model: F54, F86, Astro 6A, Astro 10i

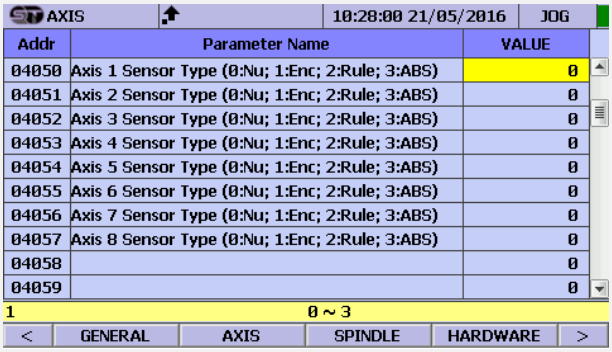
Step 3:

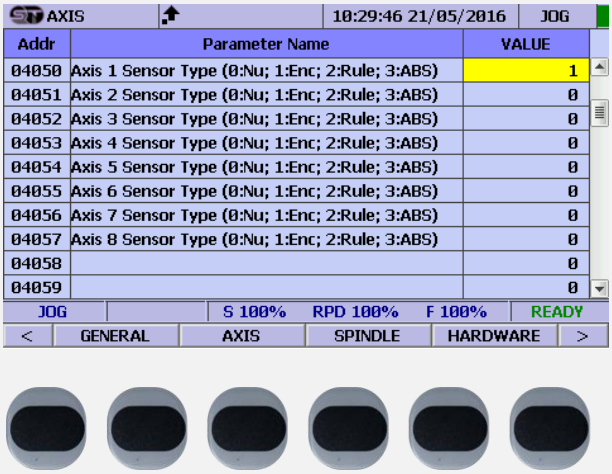
Check the annotation on the right side of the parameter before setting it.
For example: set the value to 1 by entering the number 1 using the numeric keypad on the control pane.

Step 4:

A window will appear on the screen requesting a password. At this point, enter the user password '1415' and press the OK button.

Description images
Model: F54. F86, Astro 6A, Astro 10i





Instructions
Model: F54, F86, Astro 6A, Astro 10i

Step 5:

Then, enter the value once again and press the RESET button on the control panel to save the parameter.

Step 6:

Finally, restart the controller.

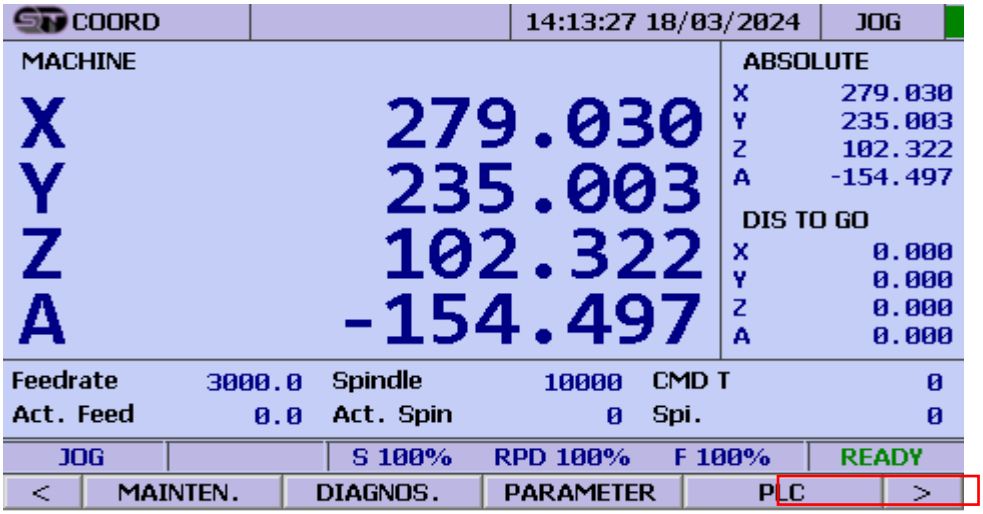
4.9 PLC screen

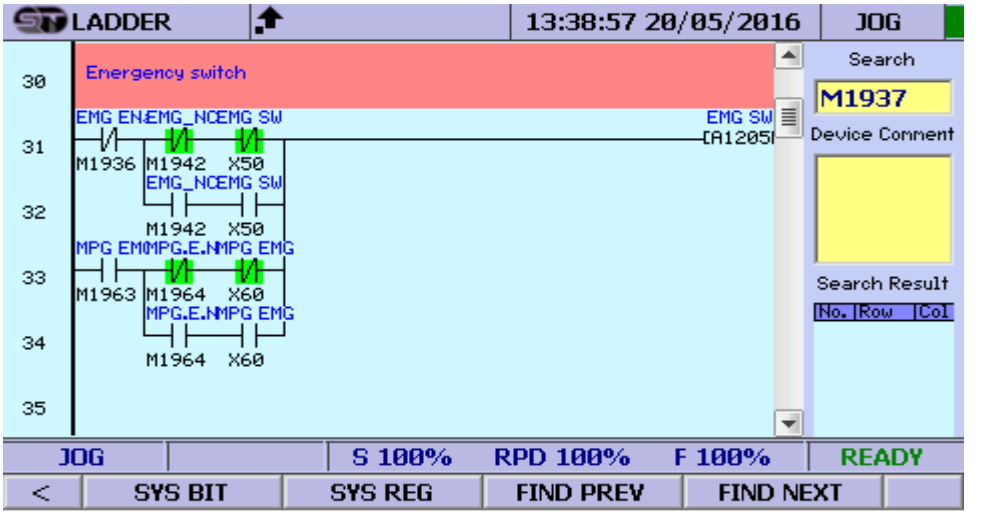
All controller models from QS Technology are integrated with a PLC processing module, allowing users to monitor the machine's operating status through the PLC ladder. Users can prepare the PLC ladder program and load it into the controller to ensure the machine operates according to the specified requirements.

Functions in PLC screen:

- **SYS BIT:** Open system bit page
- **SYS REG:** Open register page
- **FIND REV:** Find the previous object in the ladder program
- **FIND NEXT:** Find the next object in the ladder program.

Model: F54 controller



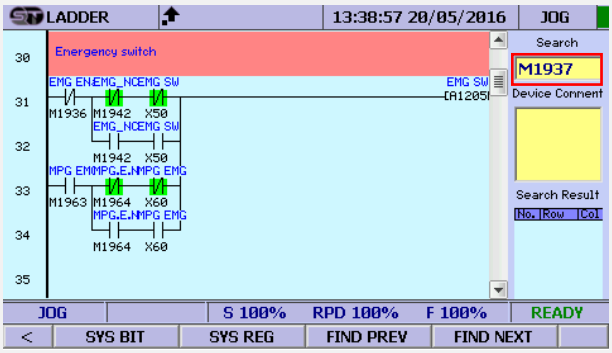
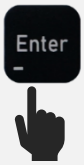
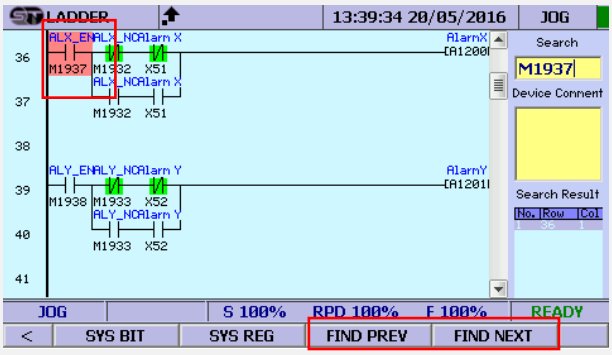
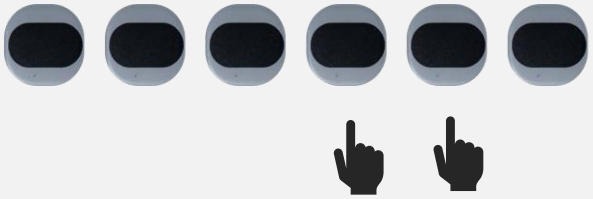


Instructions:

- Searching for an object in the program **(4.9.1)**
- PLC data loading **(4.9.2)**

QS TECHNOLOGY

3.9.1 Searching for an object in the program

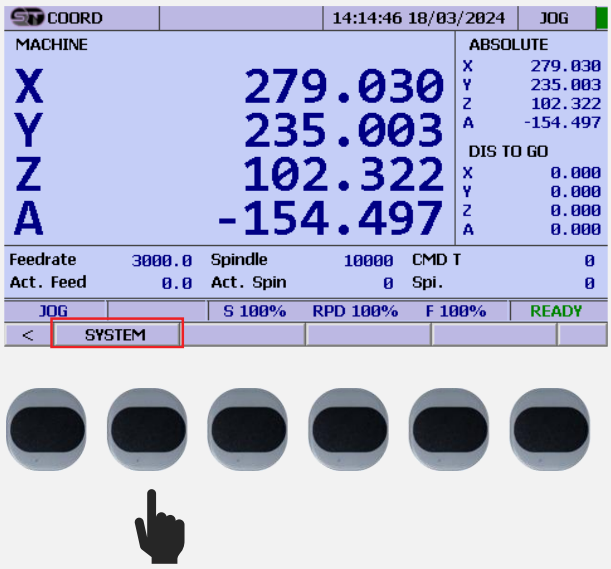
Description images Model: F54. F86, Astro 6A, Astro 10i	Instructions Model: F54, F86, Astro 6A, Astro 10i
   	**Step 1:** Enter the name of the object to search for in the SEARCH field, then press the ENTER button on the control panel. **Step 2:** The controller will search and return the result if the object exists in the program. If the operator wants to check if there are any other values with the same name in the program, press the FIND NEXT and FIND REV buttons to check.

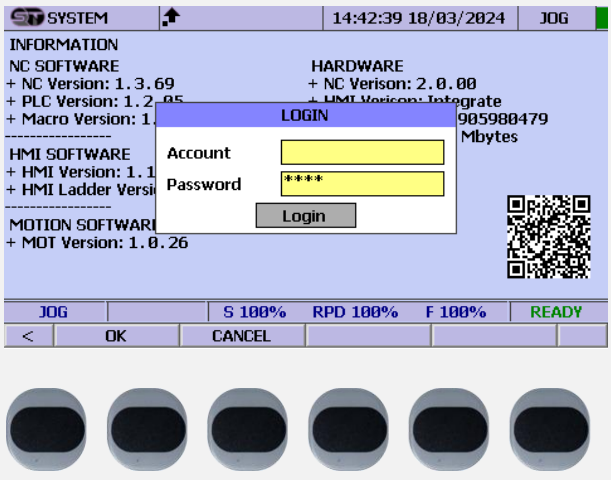
3.9.2 PLC data loading

To successfully load data into the controller, the operator must be aware of the file format and name. If the name and format are incorrect, the data cannot be loaded into the controller.

NO	Data type	File Name	Format	Final Format
1	PLC	ALM_PLC_USER PLC	.HPL	ALM_PLC_USER.HPL PLC.HPL

Description images
Model: F54. F86, Astro 6A, Astro 10i





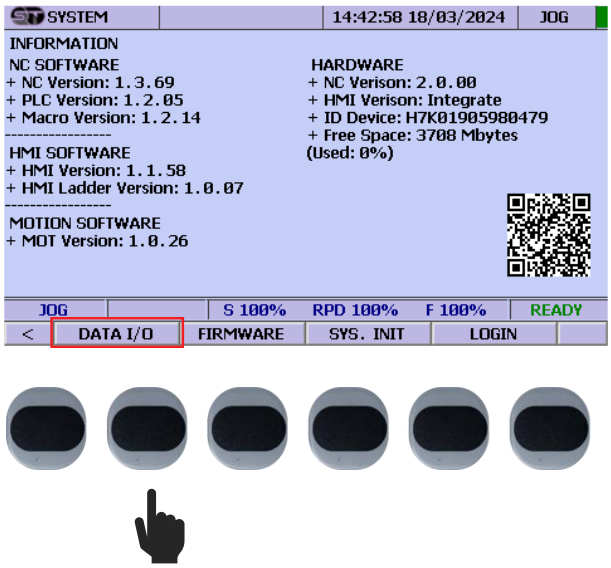
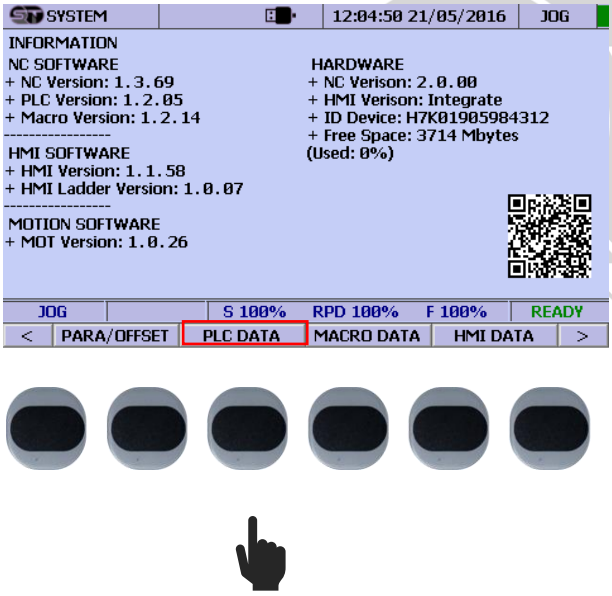
Instructions
Model: F54, F86, Astro 6A, Astro 10i

Step 1:

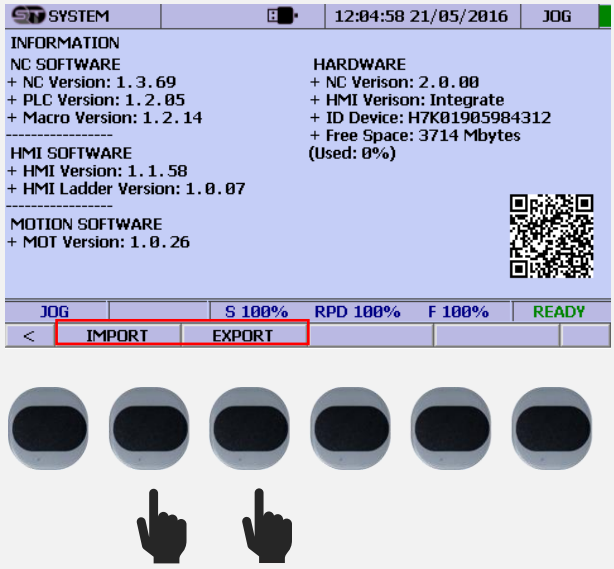
On main screen select SYSTEM

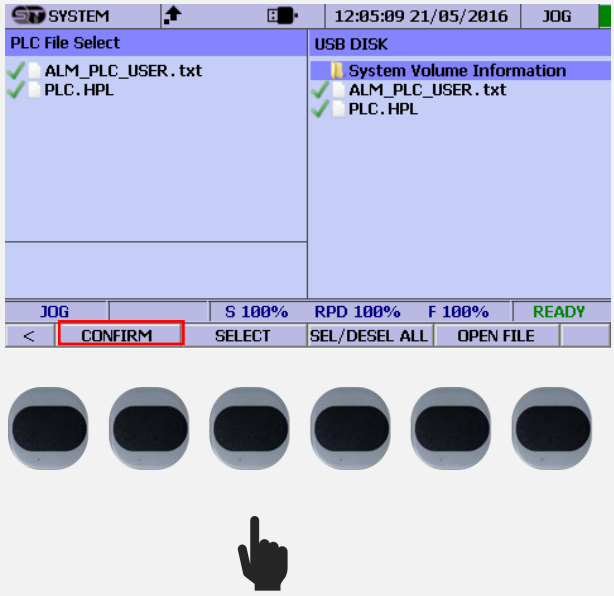
Step 2:

Enter the user password '1415' and press the OK button.

Description images Model: F54		Instructions Model: F54
		**Step 3:** Select the DATA I/O section. In this section, the user can load and export data for. (Macro/PLC/Parameter/Offset)
		**Step 4:** Press PLC DATA.

Description images
Model: F54





Instructions
Model: F54

Step 5:

Select IMPORT or EXPORT.

Step 6:

Insert the USB drive into the controller and select the folder containing the PLC data. At this point, the controller will automatically recognize the PLC data if the file is correctly named and in the proper format. Then, press the CONFIRM button to begin the data loading process into the controller.

4.10 System data

The operator accesses the system page to configure system parameters as well as to load or export data on the controller. It is important that the files are correctly named and formatted.

NO	Data Type	File Name	Format	Final Format
1	Macro alarm	ALM_MACRO_USER	.txt	ALM_MACRO_USER.txt
2	Parameter	PARAM_NC PARAM_USER	.txt	PARAM_NC.txt PARAM_USER.txt
3	Offset	OFFSET	.txt	OFFSET.txt
4	PLC	ALM_PLC_USER PLC	.HPL	ALM_PLC_USER.HPL PLC.HPL

Note:

Macro programs are typically created using a text editor such as Notepad and saved with a .txt file extension. Before loading the macro program into the controller, the programmer must remove the .txt file extension.

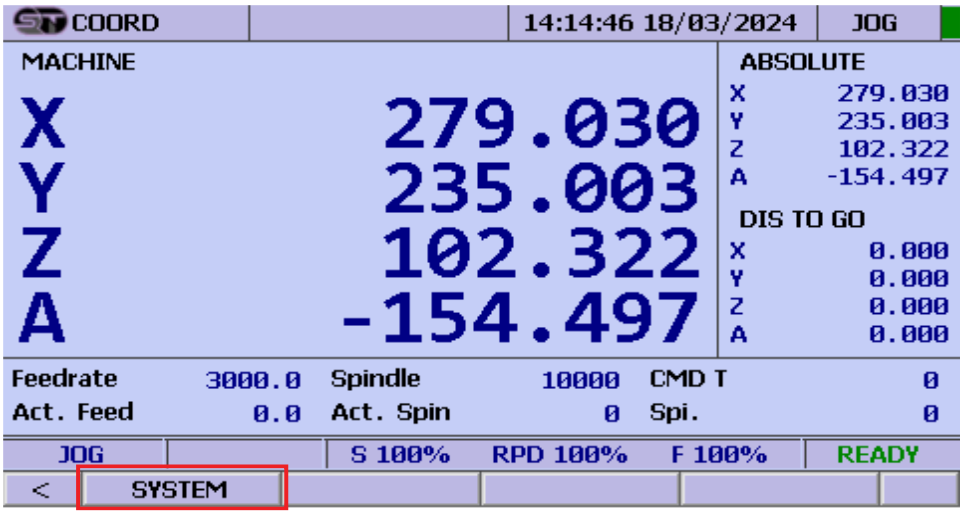
Example:

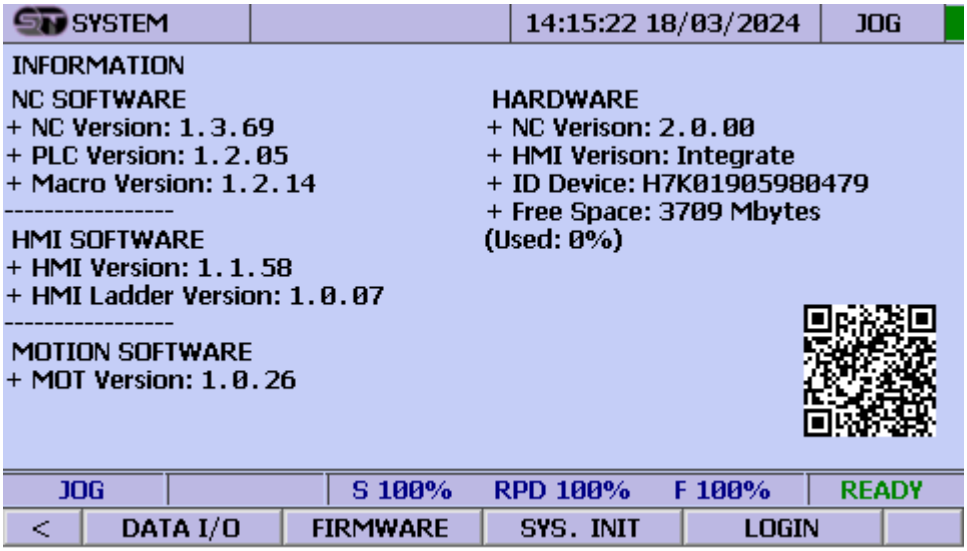
The macro program created using Notepad is named “M0100.txt”. The user must remove the .txt extension before loading the data into the controller.

M0100.txt ⟶ M0100

QS TECHNOLOGY

Model: F54 controller

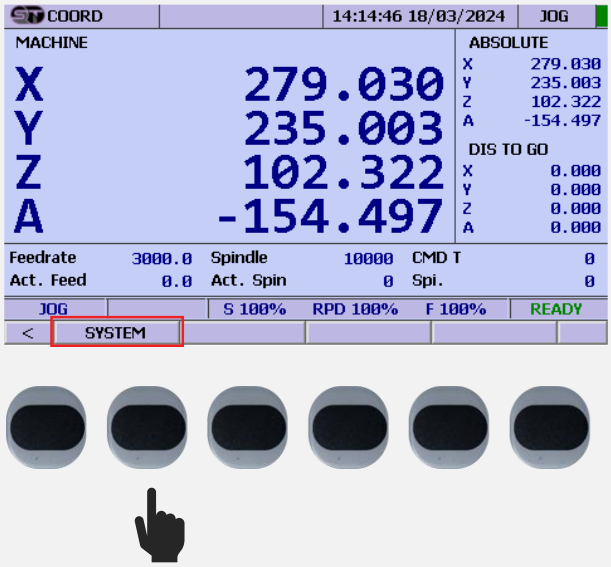
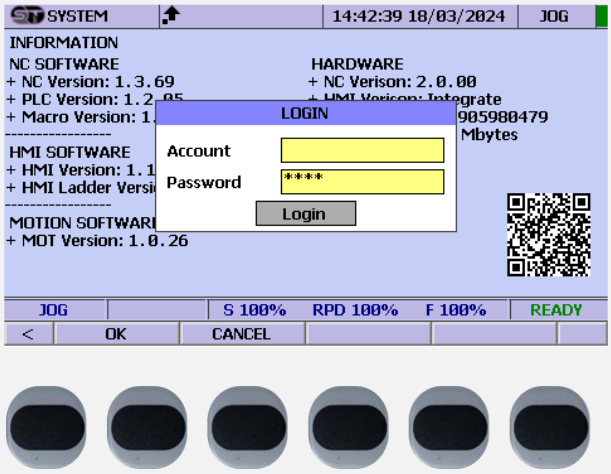




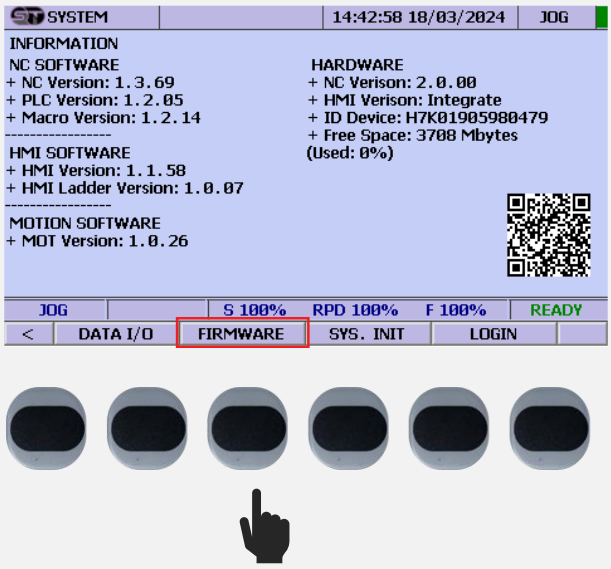
Instructions:

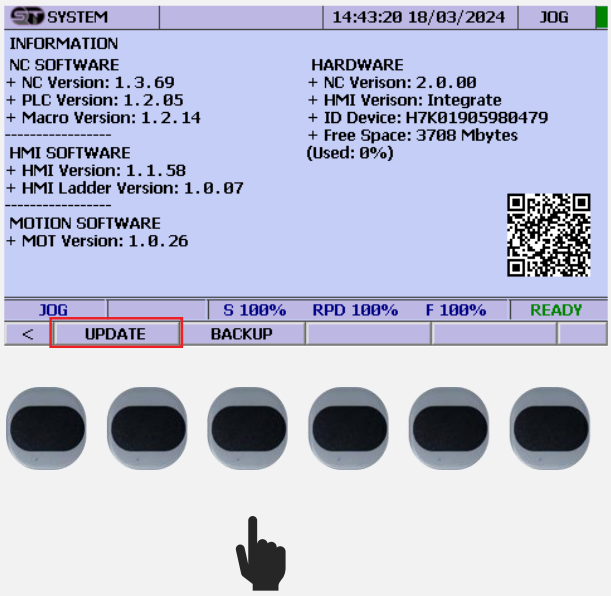
- Firmware update **(4.10.1)**
- Date & Time setting **(4.10.2)**

4.10.1 Firmware update

Description images Model: F54	Instructions Model: F54
	**Step 1:** Press SYSTEM button.
	**Step 2:** Enter user password “1415”

Description images
Model: F54





Instructions
Model: F54

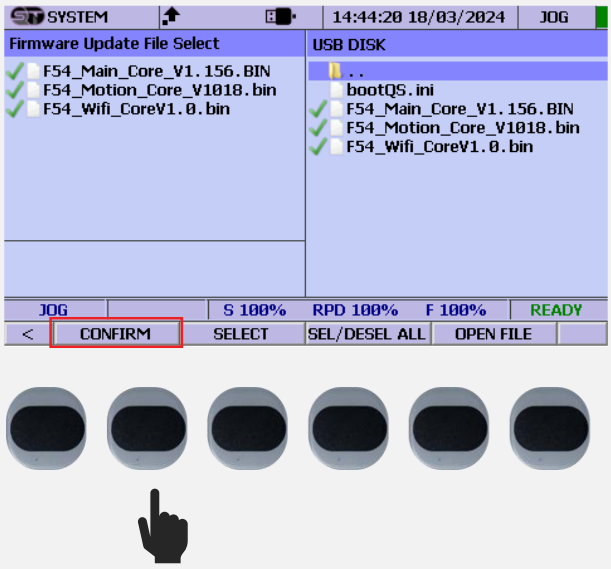
Step 3:

Select FIRMWARE

Step 4:

Select UPDATE

Description images
Model: F54



Instructions
Model: F54

Step 5:

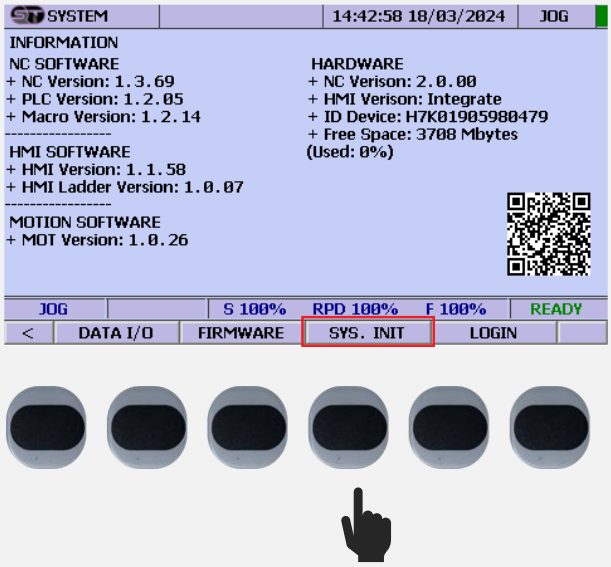
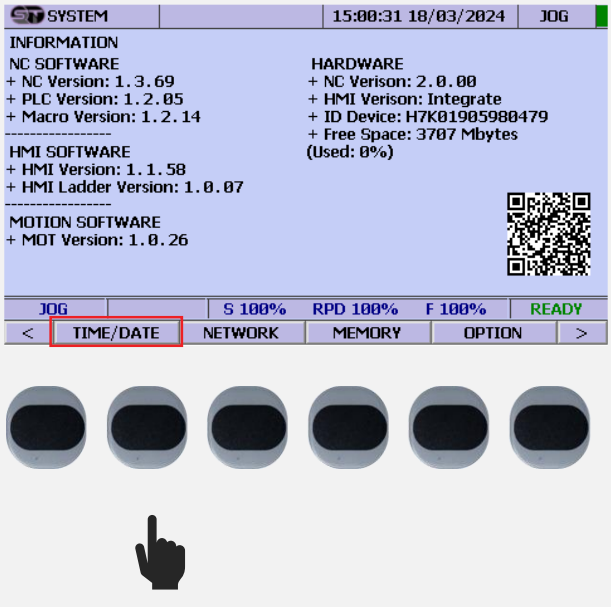
Insert the USB drive into the controller and select the folder containing the firmware. At this point, the controller will automatically recognize the firmware file if it is correctly named and in the proper format. Then, press the CONFIRM button to begin the firmware update process on the controller.

Note:

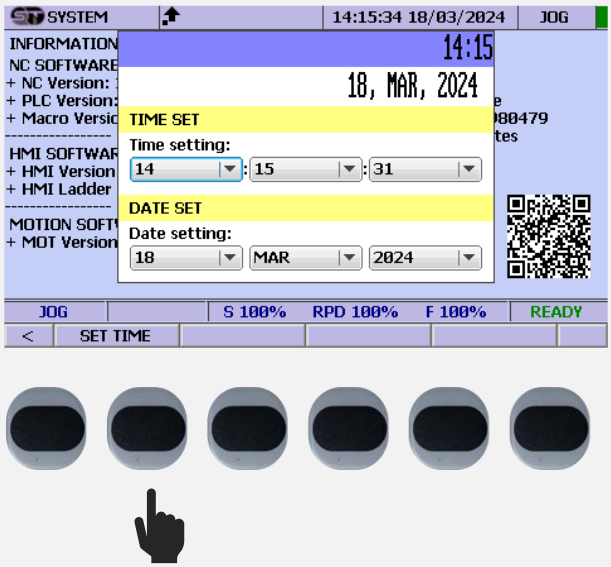
The firmware for the F54 controller is named as follows:

- F54_Main_Core_V1.00.Bin
- F54_Motion_Core_V1061.BIn

4.10.2 Date & Time setting

Description images Model: F54	Instructions Model: F54
	**Step 1:** From SYSTEM screen select SYS.INIT.
	**Step 2:** Select TIME/DATE.

Description images
Model: F54. F86, Astro 6A, Astro 10i



Instructions
Model: F54, F86, Astro 6A, Astro 10i

Step 3:

Set the time value, then press the SET TIME button.

4.11 Transfer data via Wi-Fi network

The F54 controller allows users to transfer data from a computer or laptop to the controller via Wi-Fi by using the “QS Explorer” software.
Access the website www.qstcnc.com to download the software or scan the QR code below to download the software.

Scan this QR code to download “QS Explorer” software.

Scan this QR code to download “QS Editor” software.

(This software is used for editing PLC Ladder)

QS TECHNOLOGY

Instructions:

- Software installation **(4.11.1)**
- Connect device to wi-fi network **(4.11.2)**
- Transfer data via the wi-fi network **(4.11.3)**
- Monitor the operational status of the controller **(4.11.4)**

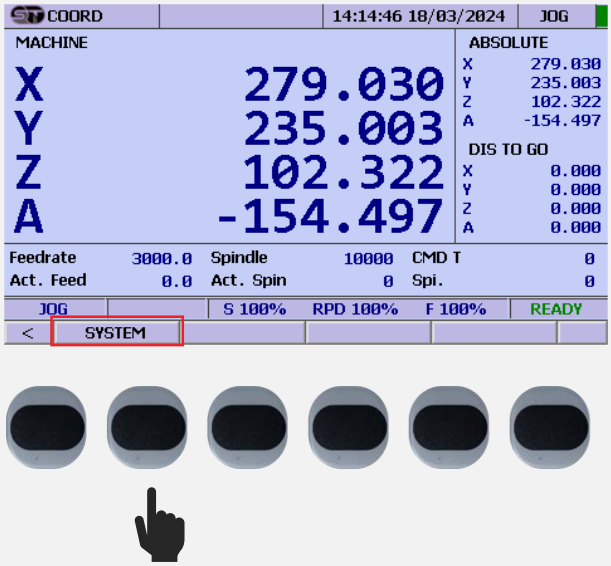
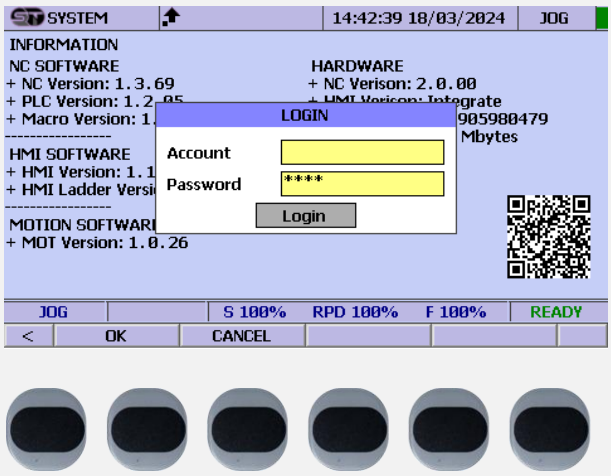
4.11.1 Software installation

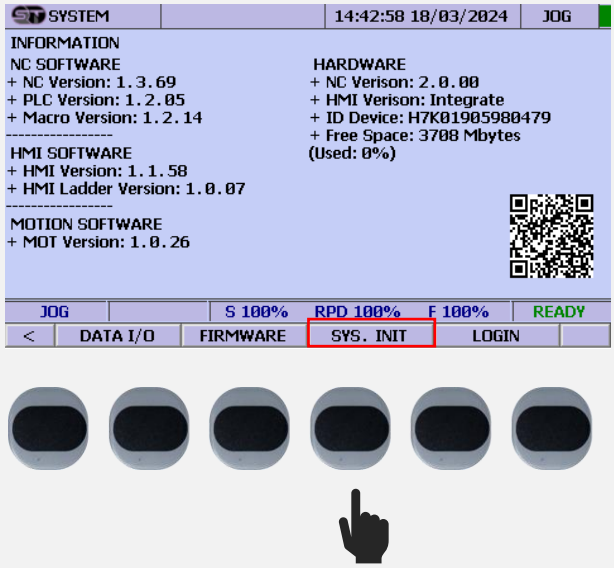
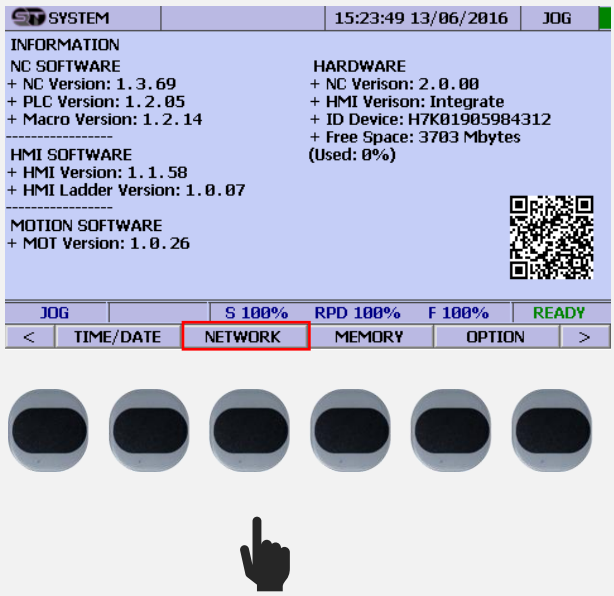
After downloading the software from the website www.qstcnc.com, follow the steps below to install the software.

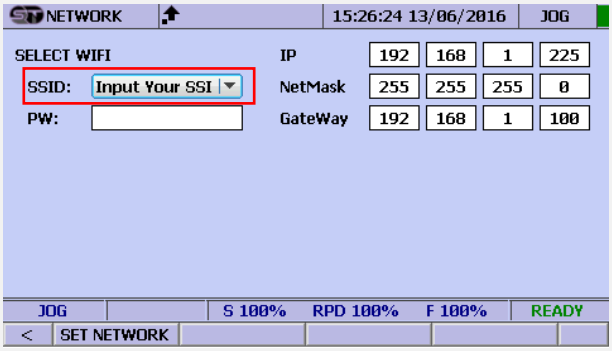
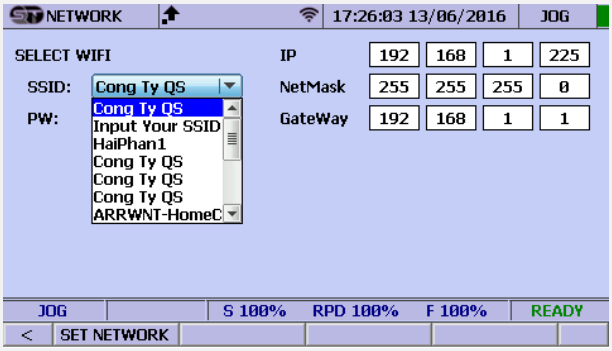


4.11.2 Connect device to wi-fi network

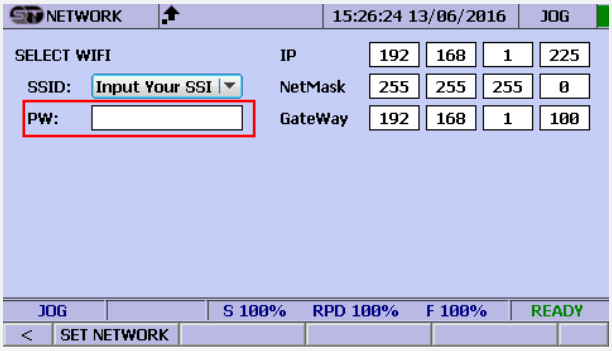
First, the operator needs to attach the antenna to the controller.

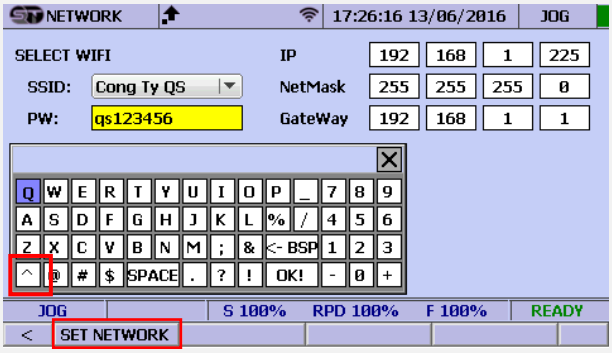
Description images Model: F54	Instructions Model: F54
	**Step 1:** Select SYSTEM.
	**Step 2:** Enter the user password “1415”.

Description images Model: F54	Instructions Model: F54
	**Step 3:** Select SYS.INIT.
	**Step 4:** Selct NETWORK.

Description images Model: F54	Instructions Model: F54
	**Step 5:** Select the SSID option using the arrow keys, then press the ENTER button.
	**Step 6:** Select the wi-fi network.

Description images
Model: F54





Instructions
Model: F54

Step 7:

Enter the wi-fi password.

Step 8:

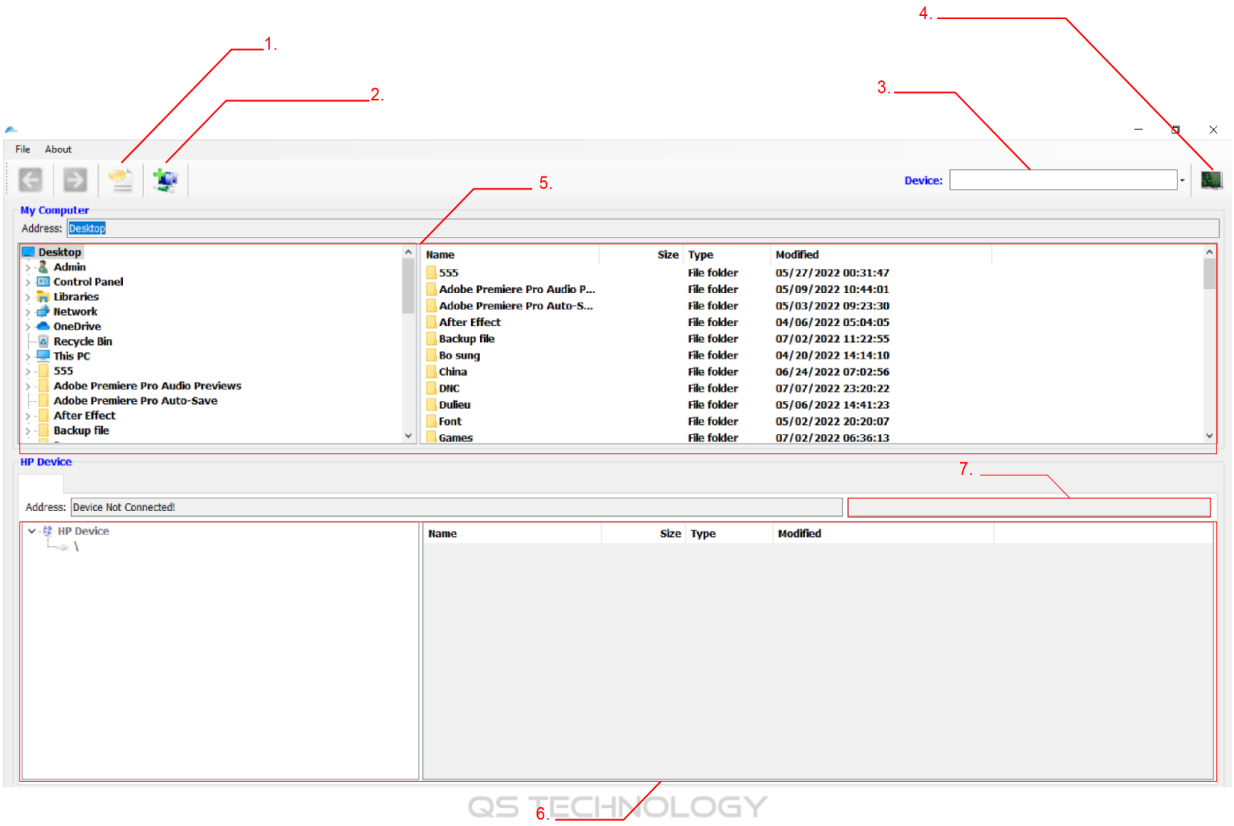
Use the arrow keys on the controller to interact with the on-screen keyboard and enter the Wi-Fi password. After completing this, press the SET NETWORK button. Finally, restart the device.

Note:

Press the button with the red circle around the arrow to type uppercase letters.

4.11.3 Transfer data via wi-fi network

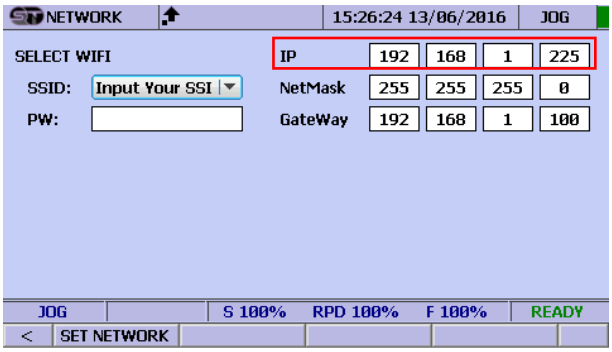
Software Icon:



NO	Name	Function
1	Send MSG	Sending message to controller
2	Add Device	Creat a new device
3	Device List	Device list
4	Get Device Status	Device status
5	Computer	Computer memory area
6	DNC device	Controller memory area
7	Connection status	Connection status

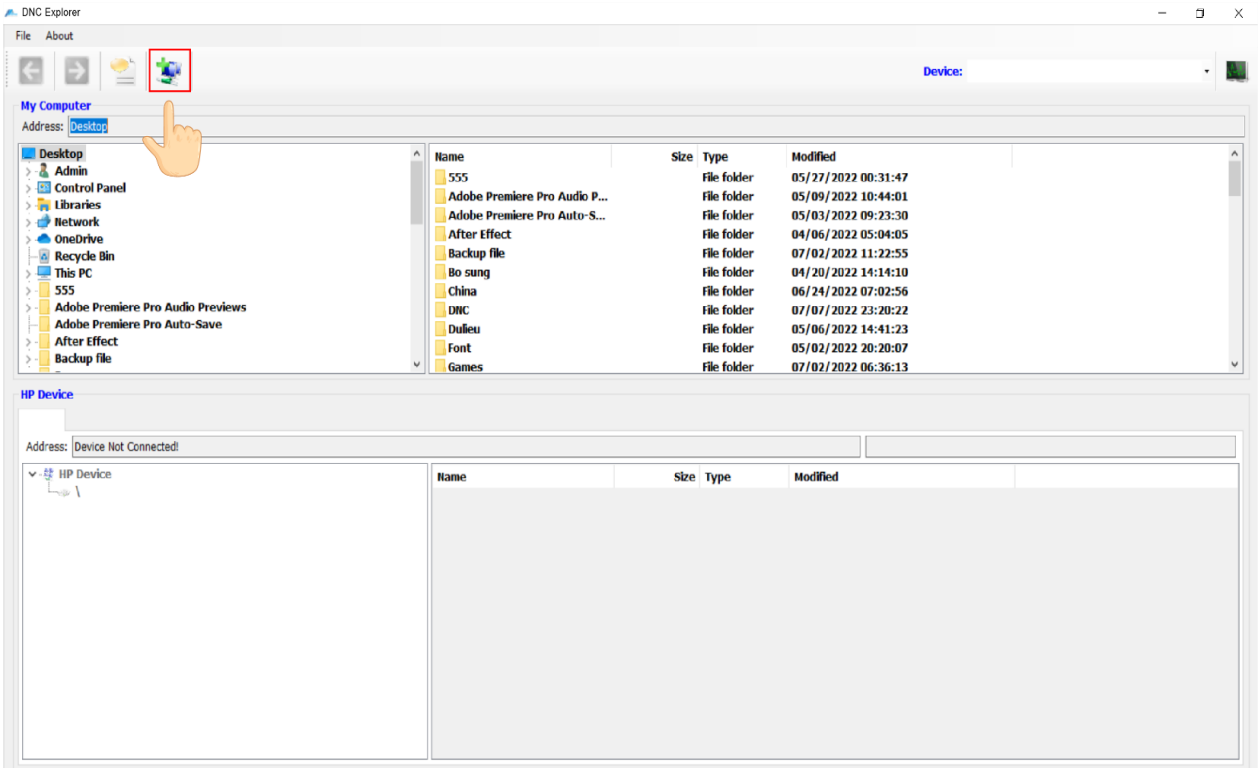
To transfer data from the computer to the controller, the user must first check the controller's IP address on the NETWORK page.

Example: Controller IP address is: 192.168.1.225



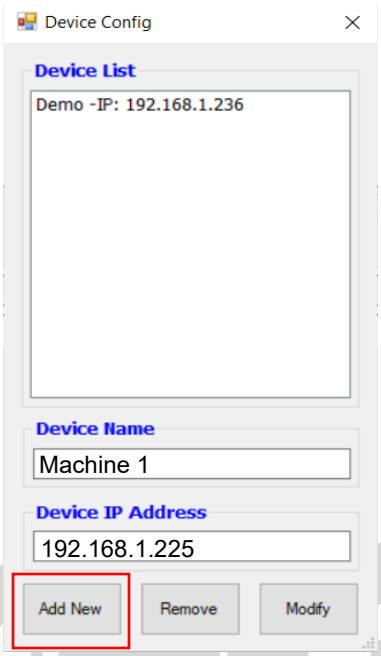
Step 1:

Open the software and select “ADD DEVICE”



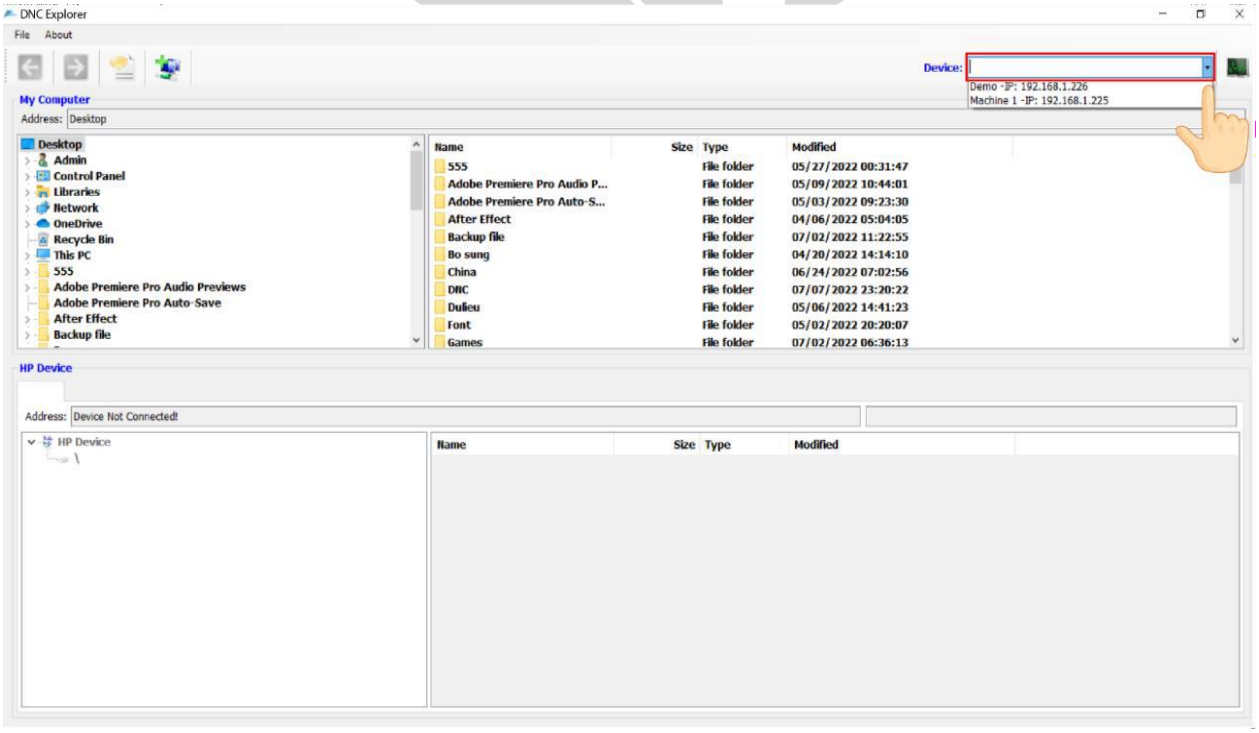
Step 2:

Enter the name and IP address of the controller you want to connect to. Then, press the ADD NEW button to save the device.



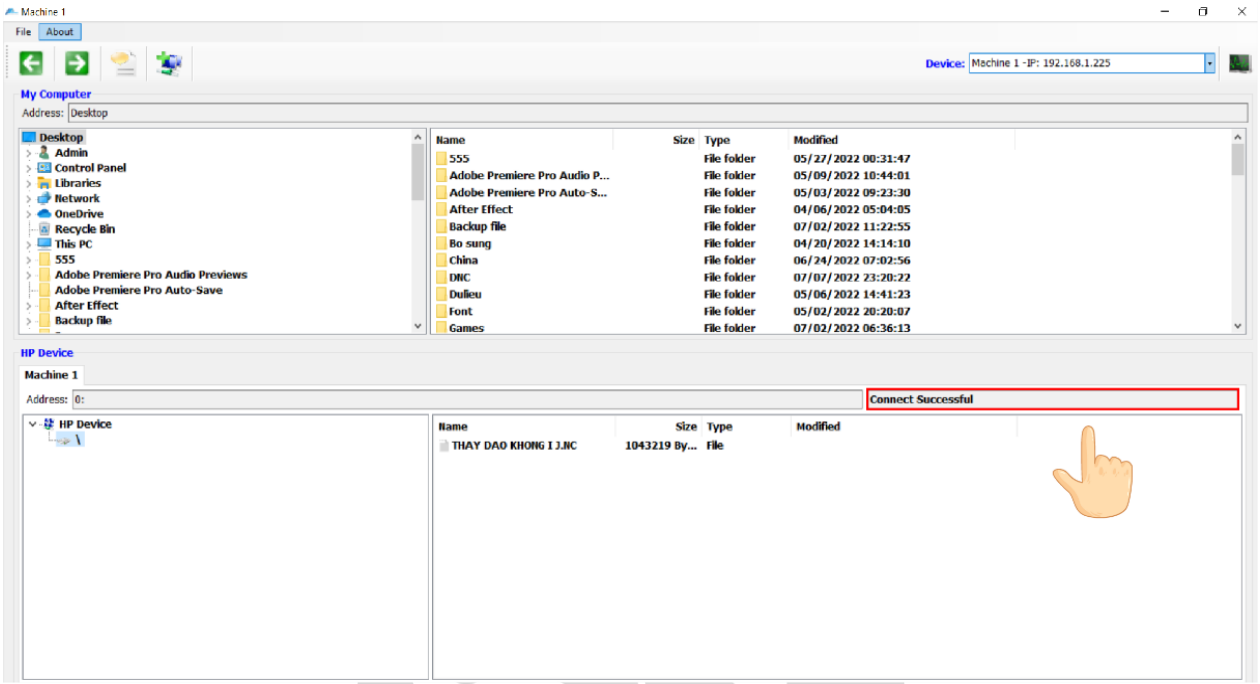
Step 3:

Select the DEVICE LIST box and choose the device you want to connect to.



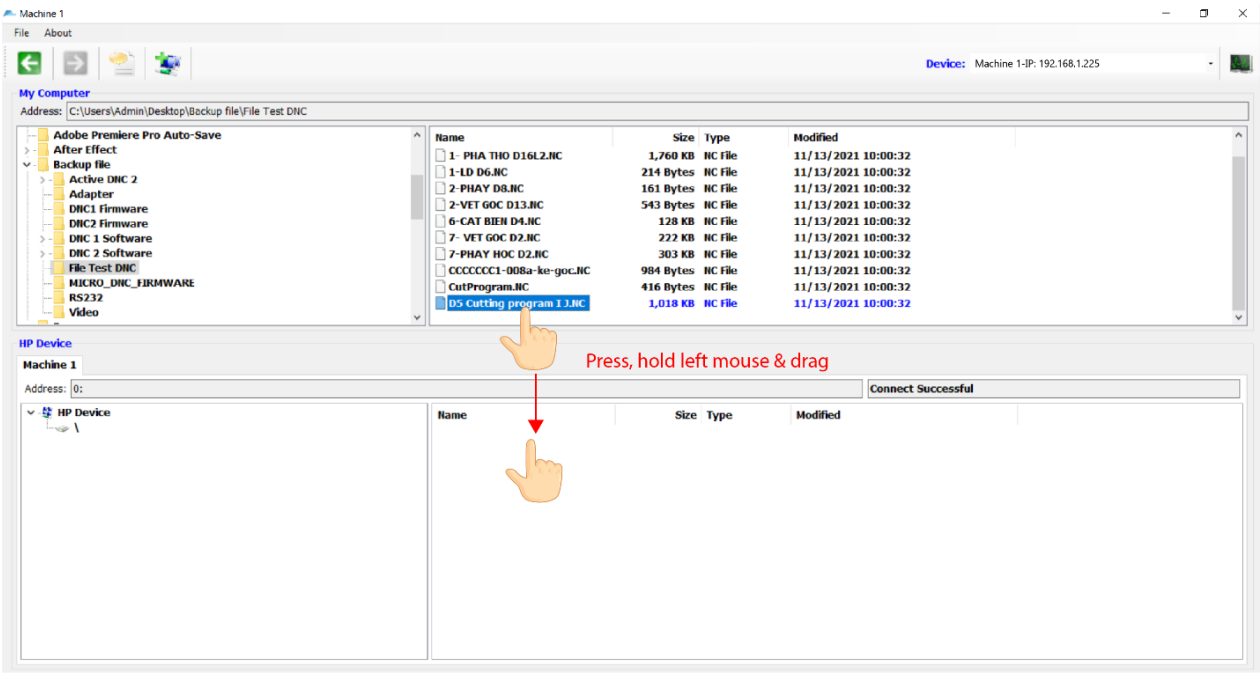
Step 4:

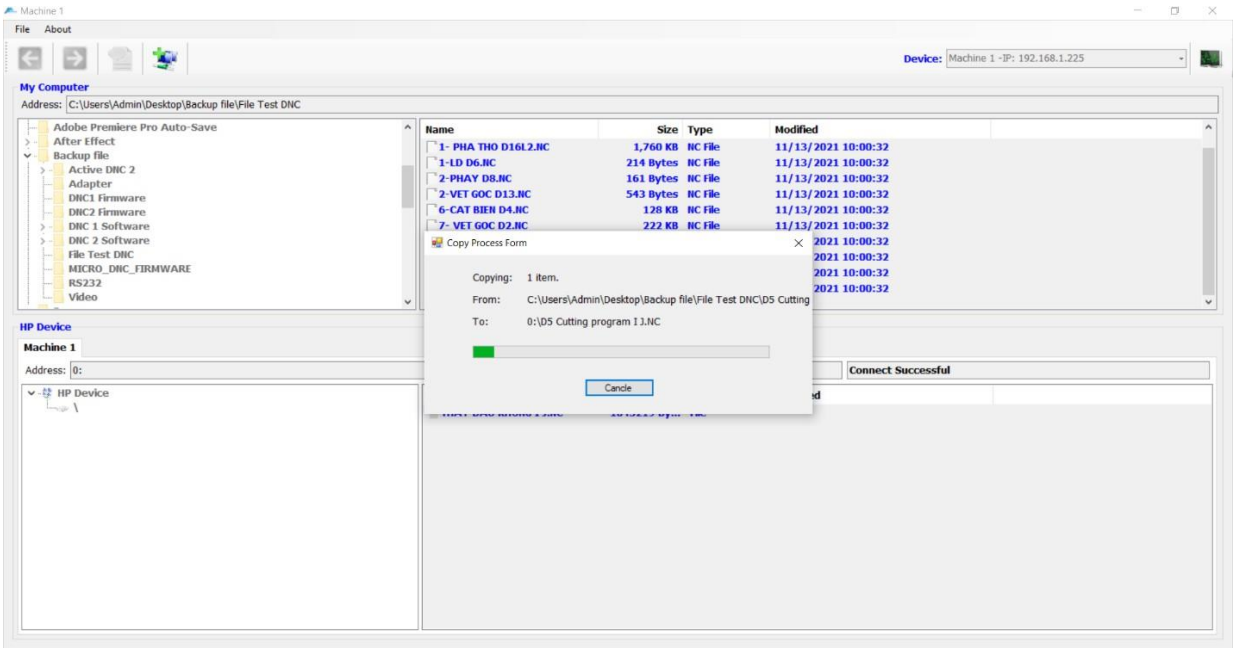
Once the controller has successfully connected to the software, the software will display "Connection Successful".



Step 5:

Select the program from the computer's memory, then click and hold the left mouse button and drag it to the controller's memory below. To retrieve the program from the controller back to the computer, perform the reverse action.

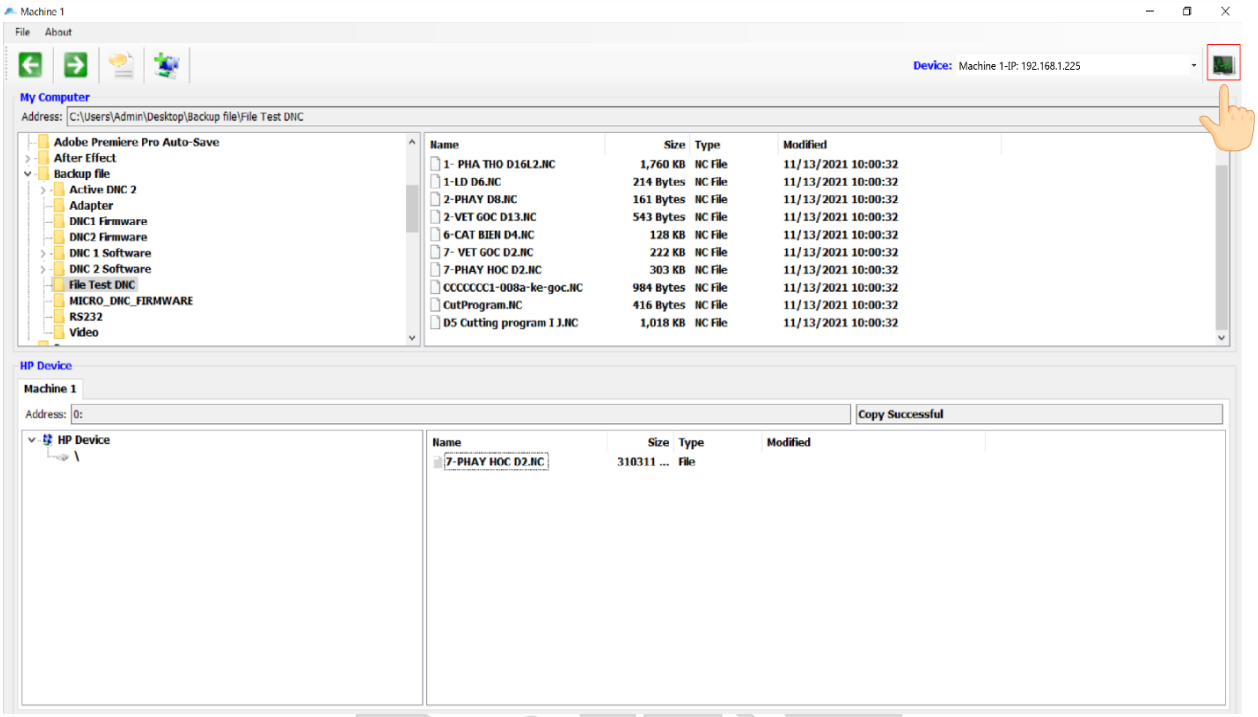




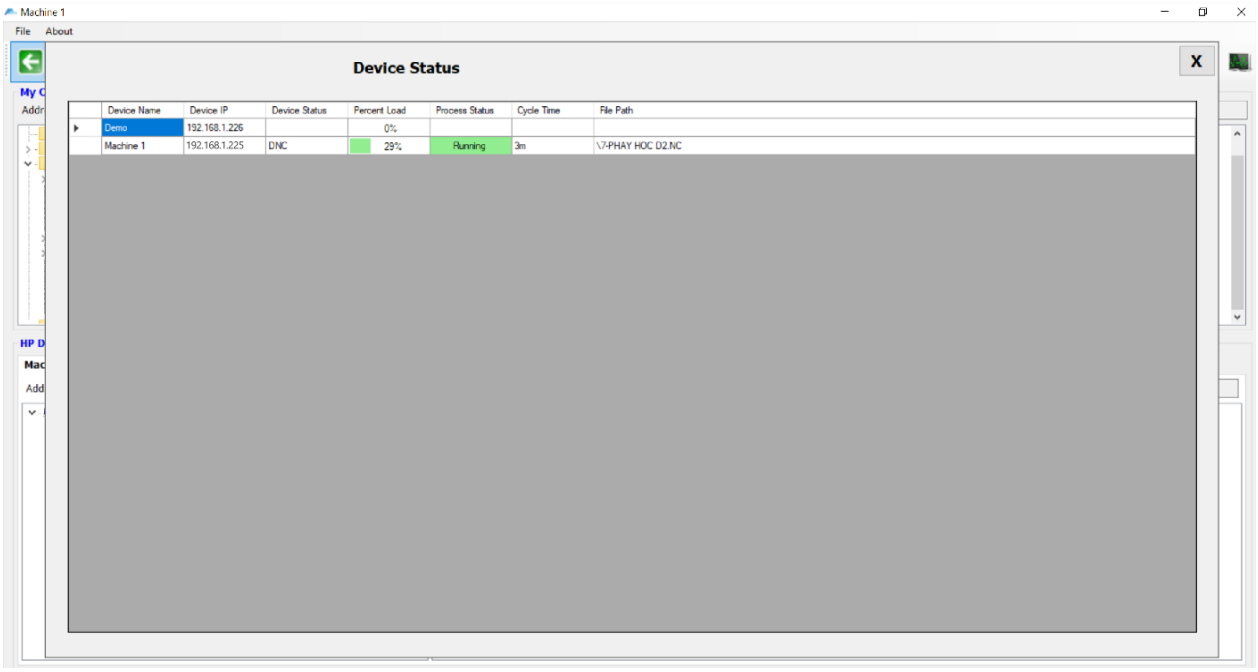
QS TECHNOLOGY

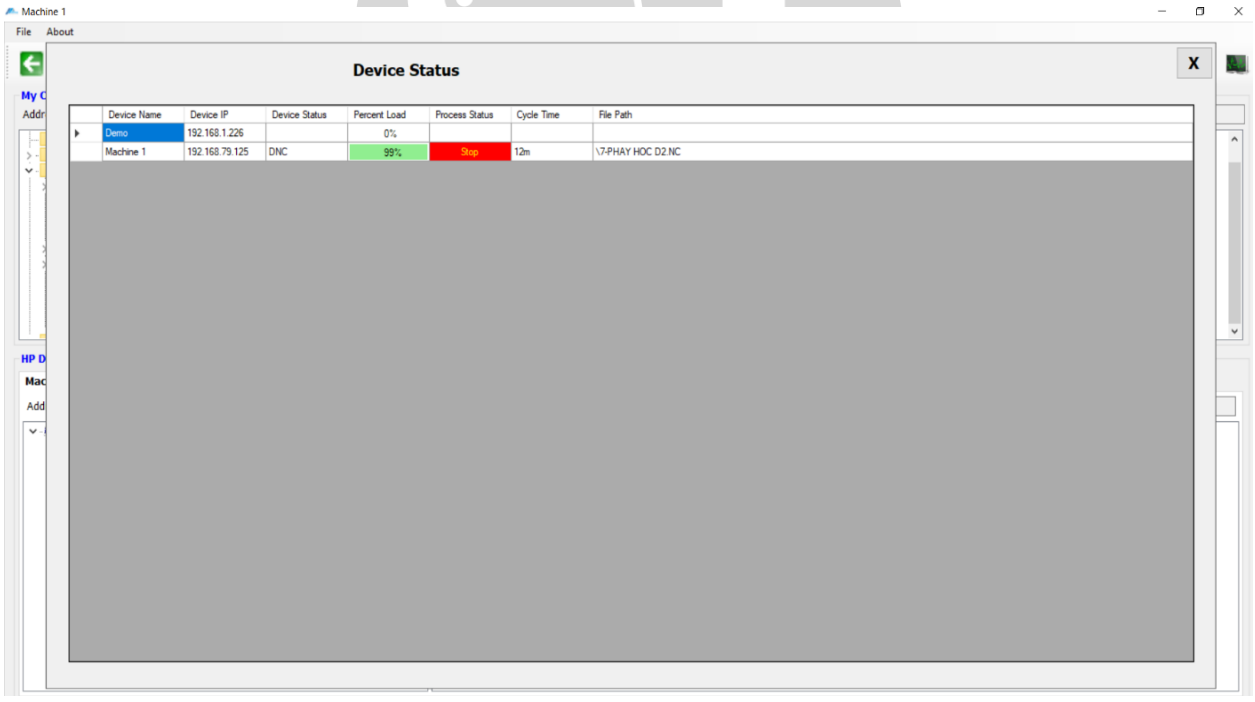
4.11.4 Monitor the operational status of the controller

The user can monitor the operational status of the controller by clicking on the monitor icon in the top-right corner to open the monitoring window.



QS TECHNOLOGY


Machine 1
File
About
Device Status
X
Device Name
Device IP
Device Status
Percent Load
Process Status
Cycle Time
File Path
Demo
192.168.1.226
0%
Machine 1
192.168.1.225
DNC
29%
Running
3m
\7-PHAY HOC D2.NC


Machine 1
File
About
Device Status
X
Device Name
Device IP
Device Status
Percent Load
Process Status
Cycle Time
File Path
Demo
192.168.1.226
0%
Machine 1
192.168.79.125
DNC
99%
Stop
12m
\7-PHAY HOC D2.NC

PART 5 PARAMETERS TABLE

5.1 Axis parameters

Parameter	Description	Work range	Default
#0000 - #0007	Axis enable(AX.ENA)	0:Disable – 1:Enable	0
#0008 - #0015	Axis direction(AX.DIR)	0:Disable – 1:Enable	0
#0016 - #0023	Axis soft limit disable feature(AX.SODI)	0:Disable – 1:Enable	0
#0024 - #0031	Axis home direction(AX.HDR)	0: Forward – 1: Reverse	0
#0032 - #0039	Axis home offset(AX.HDRF)	0:Disable – 1:Enable	0
#0040 - #0047	Axis home fast(AX.HFAS)	0:Disable – 1:Enable	1
#0056 - #0063	MPG direction(MPG.DIR)	0: Forward – 1: Reverse	0
#0088 - #0095	Axis pulse logic(AX.PLG)	0: Low level – 1: High level	0
#0128 - #0135	Encoder direction(ENC.DIR)	0:Disable – 1:Enable	0
#4000	Motion corner velocity control model	0 – 5	0
#4001	Motion corner smooth max level	1 - 100	1
#4002	Motion corner smooth min level	1 - 100	1
#4003	Motion ARC refill buffer percent(%)	0 - 50	0
#4010 - #4017	Axis zero digit visible	0 - 99999	0
#4020 - #4027	Axis homing type	0: DOG on 1: DOG off 2: INDEX	0
#4030	Axis home search dog times	0 – 5	0
#4040 - #4047	Axis name	65 – 90	0
#4050 - #4057	Axis sensor type	0: NU type 1: ENC type 2: RULE type 3: ABS type	0
#4060 - #4067	Axis pulse out	0: Pul/Dir type 1: CW/CCW type 2: A/B type	0
#4070 - #4077	Axis backlash	0: Disable 1: Lin GW type 2: Box GW type	0
#4080 - #4087	Axis alarm sensor	0: Disable 1: NC type 2: NO type	0

#4140 - #4147	Axis type	0: Linear 1: Rotary 1 2: Rotary 2	0
#4150 - #4157	Axis encoder port	0: Disable 1: port 1 2: port 2 3: port 3 4: port 4 5: port 5 6: port 6	0
#4160 - #4167	Axis encoder multiply	0: X1 1: X2 2: X4	0
#8010	Init rapid override (%)		3
#8011	Init federate override (%)		5
#8013	Feedrate default (mm/min)		3000
#8080 - #8087	Axis position loop gain (1/sec)		0
#8090 - #8097	Axis dir signal delay (0.1us)		200
#8150, #8152 #8154	Radius minimum limit federate (um) (1st ,2nd ,3rd)		1000/5000/10000
#8151, #8153 #8155	Radius maximum limit federate (um) (1st ,2nd ,3rd)		1000/3000/6000
#8170, #8172 #8174	Segment length limit federate (um) (1st ,2nd ,3rd)		500/2000/6000
#8171, #8173 #8175	Auto interpolation limit rate (mm/min)		2000/4000/6000
#8180 - #8187	Axis short segment acceleration (mm/s/s)		1000
#8190 - #8197	Axis G00 backlash time (ms)		50
#8200 - #8207	Axis G01 backlash time (ms)		50
#12000 - #12007	Axis pitch mount (BLU)		10000
#12010 - #12017	Axis pulse/mm (BLU)		1000
#12020 - #12027	Axis rapid travel acceleration (m/s/s) G00		1000
#12030 - #12037	Axis cutting acceleration (m/s/s) G01		500
#12040 - #12047	Axis max jog rapid (mm/min)		12000
#12050 - #12057	Max jog federate (mm/min)		6000
#12060 - #12067	Axis max auto rapid (mm/min)		12000
#12070 - #12077	Axis max auto federate (mm/min)		12000
#12080 - #12087	Axis loss pulse check window (BLU)		0
#12090 - #12097	Axis homing federate 1st (mm/min)		2000
#12100 - #12107	Axis homing federate 2nd (mm/min)		200
#12110 - #12117	Axis homing federate 3rd (mm/min)		100
#12120 - #12127	Axis home offset (BLU)		0
#12130 - #12137	Axis MPG acceleration (mm/s/s)		1000
#12140 - #12147	Axis MPG max speed (mm/min)		8000
#12150 - #12157	Axis backlash G00 (BLU)		0
#12160 - #12167	Axis backlash G01 (BLU)		0

#12200 - #12207	Axis negative coordinate of stroke limit 1st (BLU)		0
#12210 - #12217	Axis positive coordinate of stroke limit 1st (BLU)		0
#12220 - #12227	Axis negative coordinate of stroke limit 2nd (BLU)		0
#12230 - #12237	Axis positive coordinate of stroke 2nd (BLU)		0
#12240 - #12247	Axis negative coordinate of stroke limit 3rd (BLU)		0
#12250 - #12257	Axis positive coordinate of stroke 3rd (BLU)		0
#12260 - #12267	Axis negative coordinate of stroke 4th (BLU)		0
#12270 - #12277	Axis positive coordinate of stroke 4th (BLU)		0
#12280 - #12287	Axis reference point 2nd (BLU)		0
#12290 - #12297	Axis reference point 3rd (BLU)		0
#12300 - #12307	Axis reference point 4th (BLU)		0
#12310 - #12317	Axis reference point 5th (BLU)		0
#12320 - #12327	Axis reference point 6th (BLU)		0
#12330 - #12337	Axis jogging acceleration (mm/s/s)		1000
#12340 - #12347	Axis gear (Motor side)		1
#12350 - #12357	Axis gear (Screw side)		1
#12360 - #12367	Axis encoder pulse/round		0

5.2 Spindle parameters

Parameter	Description	Work range	Default
#0104	RES.CODE	0: Default S.code 1: Last S.code	0
#0120 - #0123	Spindle direction (SPI.DIR)	0: Forward – 1: Reverse	0
#4090 - #4093	Spindle mode	0: Disable 1: 0→10V 2: 0→±10V	0
#8020	Init spindle override (%)	0 → 10	5 (= 50%)
#8030, #8032 #8034, #8036	Spindle(1) – Gear (1,2,3,4) (Motor side)	0 – 999999	1
#8031, #8033 #8035, #8037	Spindle(1) – Gear (1,2,3,4) (Screw side)	0 – 999999	1
#8040, #8042 #8044, #8046	Spindle(2) – Gear (1,2,3,4) (Motor side)	0 – 999999	1
#8041, #8043 #8045, #8047	Spindle(2) – Gear (1,2,3,4) (Screw side)	0 – 999999	1

#8050, #8052 #8054, #8056	Spindle(3) – Gear (1,2,3,4) (Motor side)	0 – 999999	1
#8051, #8053 #8055, #8057	Spindle(3) – Gear (1,2,3,4) (Screw side)	0 – 999999	1
#8060, #8062 #8064, #8066	Spindle(4) – Gear (1,2,3,4) (Motor side)	0 – 999999	1
#8061, #8063 #8065, #8067	Spindle(4) – Gear (1,2,3,4) (Screw side)	0 – 999999	1
#8070 - #8073	Spindle (1,2,3,4) float gear motor speed (rpm)	0 – 999999	1
#12170 - #12173	Spindle (1,2,3,4) min speed (rpm)	0 – 999999	10
#12175 - #12178	Spindle (1,2,3,4) max speed (rpm)	0 – 999999	24000
#12180 - #12183	Spindle (1,2,3,4) motor gain analog mode (rpm/volt)	0 – 999999	1
#12185 - #12188	Spindle (1,2,3,4) init speed (rpm)	0 – 999999	1
#12190 - #12193	Spindle (1,2,3,4) acceleration (ms)	0 – 999999	5000
#12195 - #12198	Spindle (1,2,3,4) deceleration (ms)	0 – 999999	5000
#12370 - #12373	Spindle (1,2,3,4) max frequency pulse mode	0 – 999999	200

QS TECHNOLOGY

5.3 General parameters

Parameter	Description	Work range	Default
#0064	G54 default (G54.DEF)	0: Disable – 1: Enable	0
#0065	All position clear (POS.CLR)	0: Forward – 1: Reverse	0
#0066	G04 assignment: X(s) – P(ms). (G04.X/P)	0: G04 X 1: G04 P	0
#0072	MTAD.PRO (Available on touch controller)	0: Disable – 1: Enable	0
#0073	MTAD.CRE (Available on touch controller)	0: Disable – 1: Enable	0
#0074	MTAD.MDI (Available on touch controller)	0: Disable – 1: Enable	0
#0080	JOG.ACC	0: Run by cutting speed 1: Run by jog acceleration	1
#0081	CON.FEED		0

#0096	3DP.EAX		0
#0097	NC.CUT	0: Disable – 1: Enable	0
#0098	3DL.FEED		0
#0112	G99/G98	0: Enable G99 & Disable G98 1: Enable G98 & Disable G99	0
#0136	OPK.DISA (Not available on F54 controller)	0: Disable – 1: Enable	0
#4130	Backup battery	0: Disable – 1: Enable	1
#4131	Reset battery alarm	0: Disable – 1: Enable	0
#4132	Touch panel type (Available on touch controller)		
#4133	LCD panel type (Available on touch controller)		
#4134	Touch panel (Available on touch controller)		
#4135	Speaker	0: Disable – 1: Enable	1
#4136	LCD sleep (min)		15
#4137	Wi-Fi function	0: Disable – 1: Enable	0

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